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**ASSEMBLY-ENHANCED
DIFFERENTIATION OF ISOMERS BY
ION MOBILITY MASS
SPECTROMETRY**

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Assembly-enhanced Differentiation of Isomers by
Ion Mobility Mass Spectrometry

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A thesis submitted in partial fulfilment of the requirements
for the degree of Doctor of Philosophy

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Han Dongqi

ABSTRACT

Isomerism is widely observed in basic biomolecules such as saccharides, amino acids, nucleotides, which are the building blocks of important macromolecules in nature. Isomers are also commonly found in isolated and synthesized products in organic and pharmaceutical studies. The structural differences of isomers can lead to significant differences in their physical and chemical properties, and thus their biological and pharmaceutical activities. Differentiation of isomers is essential in chemistry, biology, pharmaceutical science, medicine, etc. Ion mobility mass spectrometry (IM-MS) is an emerging technique for differentiation of isomers. In this thesis, we report the development and applications of assembly-based approach for differentiation of isomers by IM-MS.

Leucine (Leu), isoleucine (Ile) and allo-isoleucine (allo-Ile) are isomeric amino acids that are difficult to be distinguished by IM-MS due to their very small size difference. In this study, by optimizing the ESI conditions, the types of additive metal ions and their concentration ratios with the amino acids, we have been able to observe higher degrees of assemblies of Leu, Ile or allo-Ile (M) in the presence of copper ion, including $[\text{Cu}_4(\text{M})_6]^{2+}$, $[\text{Cu}_4(\text{M})_7]^+$, $[\text{Cu}_5(\text{M})_8]^{2+}$, $[\text{Cu}_6(\text{M})_{10}]^{2+}$, $[\text{Cu}_7(\text{M})_{12}]^{2+}$, etc. Interestingly, it was found that the difference of the isomer assemblies in drift time and collision cross

sections (CCSs) in IM-MS become larger with the increased number (n) of the assembled amino acids. From n = 1 to 12, the CCS difference (ΔCCS) was increased from 1.6 to 15.2 Å² and from 0.6 to 10.2 Å² between the assemblies of Leu and Ile and between the assemblies of allo-Ile and Ile, respectively, indicating that the isomeric differences could be accumulated and amplified in the assembly process for enhanced discrimination of isomers in IM-MS. The combination of IM-MS, MS/MS and computational modelling have been employed to obtain the structure information of the assemblies and understand their gradually increased differentiation. The preliminary results showed that the assemblies might be formed with the alkyl chains of amino acids protruded outwards and the slight differences between the alkyl chains could be accumulated to eventually cause significant differences in the assemblies. Further investigation on the analogue compounds (e.g., peptides, amino alcohols, keto acids and fatty acids) by IM-MS proved the enhanced differentiation of isomers in the assemblies and supported the structural models.

Monosaccharide and disaccharide isomers including glucose (Glc), fructose (Fru), mannose (Man), galactose (Gal) and maltose (Mal), cellobiose (Cel), isomaltose (IsoMal), lactose (Lac) and sucrose (Suc) with different compositions, connectivity and configurations were selected in this study. With addition of NiCl₂ or ZnCl₂ at the optimal ratio, multimeric saccharide (M), i.e., [Ni(M)_n]⁺²⁺ or [Zn(M)_n]⁺²⁺, could be detected using ESI-MS with n = 1-2 for monosaccharides and n = 1-8 for disaccharides. The IM

separation for the assemblies of disaccharides were found to be enhanced and the increased ΔCCS allowed better differentiation between composition isomers Lac and Gal, between connectivity isomers IsoMal and Mal, and between configuration isomers Mal and Cel.

Propranolol (Prp) is a chiral drug with the S-form more active than the R-form. Chiral phenylalanine (Phe) was chosen as the chiral selector to form complexes with Prp in the presence of copper ion. IM study showed that the ΔCCS for the Prp enantiomers was 0 in the dimeric complex ion $[\text{Cu}(\text{Phe})(\text{Prp})]^+$ and increased to $6.41\text{\AA}^2$ for the trimeric complex ion $[(\text{Cu})_2(\text{Phe})_2(\text{Prp})]^+$, indicating the enhanced differentiation for the enantiomers in higher assembly. Further study and the structural interpretation of the complexes for the enantiomer differentiation are still ongoing.

In summary, isomers including Leu, Ile and allo-Ile, Glc, Fru, Man, Gal, Mal, Cel, IsoMal, Lac and Suc, S- and R-Prp, which are constitutional, configurational, and chiral isomers, respectively, were investigated for their differentiations by IM-MS in this project. With the presence of metal ions and optimized ESI conditions, various multimeric assemblies of isomers were observed in ESI-MS. Investigation by IM-MS revealed that, in general, higher assembly could significantly enhance the differentiation of isomers. Structural investigation suggested that the initial isomeric differences could be accumulated in the assembly and lead to significant differences in higher assemblies of the isomers. This study proposes a new strategy for differentiation of isomers and provides new insights into the structures and applications of molecular assemblies in the gas phase.

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Journal papers

1. Han, D.-Q.; Yao, Z.-P., Chiral mass spectrometry: An overview. *Trac-Trends in Analytical Chemistry* **2020**, *123*, 115763.
2. Wu, Q.; Wang, J.-Y.; Han, D.-Q.; Yao, Z.-P., Recent advances in differentiation of isomers by ion mobility mass spectrometry. *Trac-Trends in Analytical Chemistry* **2020**, *124*, 115801.
3. Han, D.-Q.; Yao, Z.-P., Assembly-enhanced differentiation of leucine, isoleucine and allo-isoleucine by ion mobility mass spectrometry. (Under preparation)

Conference papers

1. Han, D.-Q. Assembly-enhanced differentiation of leucine, isoleucine and allo-isoleucine by ion mobility mass spectrometry. 8th Asia Oceania Mass Spectrum Conference: Young Scientist Forum and Main Conference Session 9D: FAIMS, IMS and DMS Developments in Mass Spectrometry, Macau, 3-8 Jan 2020. (Oral presentation)
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List of abbreviations

Abbreviation	Full form
AA	Amino acids
Ala	Alanine
allo-Ile	allo-Isoleucine
BCCA	Branched-chain amino acid
$\alpha/ \beta/\gamma$ -CD	$\alpha/ \beta/\gamma$ -Cyclodextrins
CCS	Collision cross section
CD	Circular dichroism
Cel	Cellobiose
CID	Collision induced dissociation
CR	Chiral recognition ratio method
DFT	Density functional theory
DTIM-MS	Drift tube ion mobility mass spectrometry
EHSS	Exact hard-sphere scattering
ESI	Electrospray ionization
Fru	Fructose
FT-ICT	Fourier transform-ion cyclotron resonance
Gal	Galactose

Glc	Glucose
Gly	Glycine
HECD	Hot electron capture dissociation
Ile	Isoleucine
IM-MS	Ion mobility mass spectrometry
Isomal	Isomaltose
KM	Kinetic method
Lac	Lactose
Leu	Leucine
Mal	Maltose
MALDI	Matrix-assisted laser desorption/ionization
Man	Mannose
MD	Molecular dynamics
MUSD	Maple syrup urine disease
MS	Mass spectrometry
MS/MS	Tandem mass spectrometry
NMR	Nuclear magnetic resonance
Phe	Phenylalanine
Pro	Proline
Prp	Propranolol

Suc	Sucrose
TIMS	Trap ion mobility mass spectrometry
TOF	Time of flight
Trp	Tryptophan
TWIM-MS	Travelling wave ion mobility mass spectrometry
Tyr	Tyrosine
t_d	Drift time
UV	Ultraviolet
V_w	Van der Waals volumes

Chapter 1: Introduction

1.1 Isomers

Isomers are chemical species with the same formula but different structures. There are two major types of isomers, constitutional isomers that have different orders of atom connection (e.g., functional isomers and positional isomers), and stereoisomers that have the same order of atom connection but different geometries (e.g., *cis-trans* isomers and enantiomers).¹ Isomers exist widely in nature and products of chemical synthesis.²⁻

³ Many basic molecules, such as amino acids, nucleotides, fatty acids and monosaccharides, have isomers. As they are the building blocks of important biomolecules such as proteins, nucleic acids, lipids and polysaccharides, the isomers can significantly affect the biological functions of the resulting biomolecules.⁴ For examples, the isomers of polysaccharides or glycans with different linkages or compositions are associated with the selectivity of microbial fermentation;⁵ our ability to taste is regulated by the chiral receptors in our mouths that act as "sense", and for the two enantiomers of amino acids such as leucine, histidine and tyrosine, their L-forms are bitter, whereas their D-forms are sweet;⁶ Arachidonic acid a precursor of lipid mediators, whose natural form is *cis*-form, is easily isomerized to *trans* isomers that could induce selective apoptosis and specific, species-independent cytotoxicity on retinal microvessels;⁷ the L-17mer of Zintevir shows better anti-HIV activity than its

D-17mer.⁸ Isomeric biomolecules in human body can also be aging or disease related.⁹⁻
¹⁰ For example, 67%, 65% and 73% of disease-associated microRNA are structural isomers in the disease models for colorectal cancer, malignant ovarian cancer and epithelial ovarian cancer, respectively.¹¹ Lactosylceramide (LacCer) and its isomer galabiosylceramide (Ga₂) were analyzed as biomarkers together and showed high levels in fabry patients. However, it was reported that LacCer might not be an alpha-galactosidase related biomarker because its level in urine could be negligible compared to the Ga₂.⁴

Isomerism is commonly observed in organic chemistry and pharmaceutical science as isomers coexist commonly in isolated and synthesized products.²⁻³ For instance, oligosaccharides isolated from human milk or Lotus seeds contain many stereoisomers.¹²⁻¹³ Chemical synthesis can yield isomers and racemates from intermediates to products, such as Omura's intermediate and madindoline or felodipine.¹⁴⁻¹⁵ As stereoselectivity plays an important role in the human body especially for biosynthesis of drug targets, it is not surprising that isomeric property affects drug affinity and binding. Therefore, isomers of drugs may have different effects on the receptors and biochemical pathways, which can be reflected in different metabolic rates and biological effects and toxicology. For instance, thalidomide was used to alleviate morning sickness in pregnant women in 1950s.¹⁶ Shortly after the drug

was sold, thousands of infants were born with phocomelia, and it was that its R-enantiomer is an effective sedative while the S-enantiomer may cause teratogen formation. Other examples include cisplatin, a widely used anti-tumour agent, versus its clinically ineffective isomer transplatin,¹⁷ and Hinokiresinol, whose trans-form shows significantly decreased neuronal injury ability, compared with cis-form with the stereo-specific augmentation of Cu/Zn-SOD activity.¹⁸

Therefore, investigation of individual isomers is required for mechanism exploration or quality monitoring in many cases. Although stereoselective synthesis and purification are widely applied, especially for synthesis design and pharmaceutical production,¹⁹ still many isomers could not be isolated or synthesized with high purity and analyzed individually.²⁰⁻²¹ Differentiation of isomers is thus essential in chemistry, biology, pharmaceutical science, medicine, environmental science, food science, etc.

1.2 Techniques for differentiation of isomers

Various techniques have been developed and applied for differentiation of isomers, including circular dichroism (CD), nuclear magnetic resonance (NMR), X-ray crystallography, chromatography, capillary electrophoresis (CE) and mass spectrometry (MS).²²⁻²³ CD can be used to distinguish optical isomers, and NMR and X-ray crystallography can provide detailed structural information of analytes. However,

these methods are generally time-consuming, require relatively large amounts of samples, and typically cannot be used for analysis of mixture samples. Chromatography and CE are commonly used separation techniques for analysis of isomers.²⁴⁻²⁵ Although coupling with MS improves the qualification capability of these separation techniques, differentiation by these techniques is still a challenging task for many isomers and separation time is another concern in many cases.

Compared to other techniques, MS has significant advantages in terms of speed, sensitivity and specificity. MS can also offer a unique gas-phase environment to probe molecular recognitions without the interferences of solvents or other mediators, thus allowing a better understanding on fundamental mechanism of molecular recognitions.²⁶⁻²⁸ Because isomers show the same mass-to-charge (m/z) value in the mass spectra, and typically cannot be directly distinguished in conventional MS. The differentiation of isomers by MS is usually achieved by tandem MS (MS/MS) where fragmentation patterns enabled the study of energy differences of isomers.²⁹⁻³³ For example, MS/MS was used for discrimination of isomeric PCR products to allow determination of adjacent nucleotides and sequence variations in larger nucleic acids.³⁴ In the structural study of flavonoids glycoside isomers, differences were observed in the relative abundances of the Y_1 or Y_0 ions resulting from the loss of the one or two *C*-glycosyl moieties between the di-*O*-glycosyl and *O*-diglycosides or di-*C,O*-glycosides

and *C,O*-diglycosides. For *O*-diglycosides, the linkage could also be determined by comparing the relative abundances of the Y_1 and Y_0 , with $Y < Y_0$ indicative of 1-2 linkages and conversely indicative of 1-6 connections.³⁵ Multi-stage fragmentation MS has also been applied in isomer recognition. For example, the differentiation of isomeric steroid hormone metabolites (cis versus trans) was achieved at MS^4 by comparing the relative intensities of certain ion species.³⁶ Positional isomers from purified glycoconjugate fraction were easily characterized and differentiated using the automatic MS^n .³⁷

1.3 Differentiation of isomers by IM-MS

In recent years, ion mobility mass spectrometry (IM-MS) has merged as a very promising tool for isomer differentiation.³⁸⁻⁴¹ Ion mobility (IM) can separate gas-phase ions based on their charge, size and shape, according to their differential mobility through a buffer gas in an electric field.⁴² The coupling of IM with MS can provide an additional dimension to resolve isobaric ions that are indistinguishable by MS alone. IM-MS can also offer structural and molecular dynamics information of the ions based on the collision cross section (CCS) obtained,^{18, 38} which is a rotationally-averaged value represents the effective area for the interaction between an individual ion and the neutral gas through which it is travelling. The CCS could be divided into theoretical

CCS (CCS_{theo})⁴³ and experimental CCS (CCS_{exp}),⁴⁴ the latter one is calculated from drift time (t_d) obtained from IM, such as Mason–Schamp equation in drift-tube IM (DTIM) or calibrants in travelling wave IM (TWIM).⁴⁵⁻⁴⁶ Typically, ions with larger collision cross section (CCS) values have lower mobility and thus need a longer drift time to travel through the ion mobility cell. For example, in dihydroindenofluorene family who has five positional isomers, anti-geometry and syn-geometry with same para linkage showed longer and shorter drift times in IM, with CCS values as 179 and 157 Å², respectively.⁴⁷ However, the CCS value is only affected by size but also the shape factor, such as the geometry. For example, the poly(ethylene glycol) with linear geometry travelled drift tube using a longer time than the star poly(ethylene glycol).⁴⁸ Interestingly, even the concaveness, a very minor element of shape, could also influence the detected CCS, the surface of the ion with more dents has larger drift time represented in IM than the ones have smoother surface.⁴⁹

IM-MS thus has been applied for analysis of several isomers such as glycans,^{29, 50} lipids,³¹ peptides,⁵¹ proteins⁵² and others.^{47, 53-57} For example, by using field asymmetric IM-MS, about 75% of the four major types of glycerol- and phospho-lipids isomers could be separated and differentiated including sn, chain length, double bond position, and cis/trans isomers.⁵⁸ By introducing the chiral selector *N*-*tert*-butoxycarbonyl-O-benzylserine to form diastereomeric dimers, amino acid enantiomers were allowed to

be better separated in IM.⁵⁹ The D-AA containing peptides were well resolved with L-AA containing peptides with resolution up to 1.9 by trapped IM-MS (TIMS).⁶⁰ The metallocupramolecular isomers with same shape as trigonal prismatic cages but different in the orientations were distinguished by IM, performing as the trans form with CCS as 572 Å² had longer drift time than the cis form with CCS as 503 Å².⁶¹

Recently, combination of IM-MS with other techniques has been proposed for differentiation of isomers.⁶²⁻⁶³ For example, the subtle difference in macromolecules such as glycan is difficult to be detected, while IM-CID-IM-MS, the fragmentation of certain ions could be achieved before and after the ions go into the IM, thus the drift time of precursor ions and fragment ions, the fragmentation pattern could be combined for better identification of carbohydrate isomers.⁶⁴⁻⁶⁵

Metals also play an important role on the isomer differentiation by IM. Five topoisomers of lasso peptides could be poorly separated by TIMS-MS due to their slight CCS differences, when with the addition of metal ions, the separation was improved.⁶⁶ The sodium ion adducts of lacto-*N*-fucopentaose isomers showed better separation results than lithium ion adducts of lacto-*N*-fucopentaose isomers, however, not all the metal ions worked on improving the separation, in contrary, the IM separation of some metal adducts of trisaccharide isomers even poorer than the separation of trisaccharide

isomers alone.⁶⁷

1.4 Previous findings and our assumptions

Differentiation of isomers has been the research effort for a long time.^{30, 68-70} In our previous work, we observed the general trend that binuclear copper bound trimers allowed better chiral discrimination of R- and S-Naproxen by MS/MS than mononuclear copper bound dimers; and that binuclear copper bound tetramers allowed better separation of chiral amino acids in IM spectra than mononuclear copper bound dimers and trimers.^{30, 70} For example, with chiral tryptophan (Trp) as the selective reagent for chiral differentiation of tyrosine (Tyr) enantiomers, the Δ CCSs between the assemblies of Tyr enantiomers were 0.74, 2.40 and 3.40 Å² for dimeric complex ion [Cu(Tyr)(Trp) – 2H + Na]⁺, trimeric complex ion [Cu(Tyr)(Trp)₂ – 3H + 2Na]⁺ and tetrameric complex ion [Cu₂(Tyr)(Trp)₃ – 4H + Na]⁺, respectively. Some studies from other groups also supported this observation.^{55, 71-74} For example, Domalain *et al.* reported⁷² the gradually improved differentiation of the *cis*- and *trans-steroid* isomers from monomers to dimers and trimers.

Assembly is a process for association of individual species, typically allowing disordered components in a system to form organized structures or patterns as a

consequence of specific, local interactions among the components. For biomolecules, assembly is the inherent ability to assemble from their units through random interactions of molecules and then form larger shapes,⁷⁵ and is a common process in nature.⁷⁶⁻⁷⁸ Serine (Ser) octamers, a famous self-assemblies in gas phase for its homochiral preference even in the racemic mixtures, leading an suggestion that the origin of homochirality in life.⁷⁹ When the assembled units are isomers, the stereo-structural difference was shown that could affect the final assembly results. For example, the chirality of amino acid could affect the handedness of self-assembled nanofibrils where the L-form units resulted the left-hand twisting helix whereas the D-form units resulted the right-hand twisting helix.⁸⁰ The slight difference between cyanostilbene-based positional isomers had a huge influence on the luminescent properties of their self-assembled materials.⁸¹ However, to our knowledge, the assembly-induced differentiation of isomers by MS were only limited to a few individual cases and very low degree of assembly. A systematic study on the effects of assembly on differentiation of isomers by MS, with the extensions to larger assemblies and various isomer analytes, would be highly desirable.

1.5 Objectives of the PhD project

This PhD project aims to conduct assembly-based approach for differentiation of isomers by IM-MS and MS/MS. The study is focused on but not limited to the following four parts:

(1) Selection of representative isomer samples:

Amino acid isomers: Leu, Ile and allo-Ile and analogue compounds (e.g., peptides, amino alcohols, keto acids and fatty acids) with differences in alkyl chains; saccharide isomers: monosaccharides (glucose, mannose, fructose and galactose) and disaccharides (maltose, cellobiose, isomaltose, lactose and sucrose) with differences in compositions, connectivity, configurations; Chiral drugs: propranolol, a chiral drug with one chiral center.

(2) Optimization for detection of assemblies under MS:

The ESI parameters including voltage, gas flowrate, temperature, etc. and other experimental factors such as metal ions, pH, solvents and concentration that are related to the molecular assembly were optimized for detection of multimeric complex ions.

(3) Study on assembly-driven differentiation of isomers by IM-MS and MS/MS:

The differentiation of isomers by IM-MS was evaluated and compared using drift time and CCSs at different multimerization stages.

The differentiation of isomers by MS/MS was carried out for chiral drug by

comparing the chiral recognition (CR) values at different multimerization stages.

(4) Structural investigation of assemblies:

The structural information of the assemblies was obtained by the combination of IM-MS, MS/MS and computational modelling.

1.6 Outline of this thesis

In Chapter 1, the background of isomers as well as various techniques for differentiation of isomer are introduced, with emphasis on the fundamental and applications on differentiation of isomers by IM-MS and MS/MS.

In Chapter 2, Leu, Ile, allo-Ile and analogue compounds were studied to understand how assembly could affect the differentiation of the isomers. The optimized conditions were studied and employed for detection of higher degrees of assemblies. Assembly ions of isomers upon different assembly degrees were studied in terms of drift times and CCSs in IM-MS under the optimized conditions. A general trend of larger differences of the isomer assemblies in drift time and CCSs in IM-MS with the increased number (n) of the assembled amino acids was observed. The combination of IM-MS, MS/MS and computational modelling were employed to obtain the structure information of the assemblies and understand their gradually increased differentiation. The results showed that the assemblies might be formed with the alkyl chains of amino acids protruded outwards and the slight differences between the alkyl chains could be accumulated to eventually cause significant differences in the assemblies.

In Chapter 3, monosaccharide and disaccharide isomers were studied to understand whether assembly could also enhance the differentiation of the saccharide isomers,

including the composition, connectivity and configuration isomers. The assembly conditions were optimized, and the assembly ions of isomers upon different assembly degrees were studied in terms of drift times and CCSs in IM-MS under the optimized conditions. The IM separations of the assemblies were found to be enhanced with the assembled molecules, allowing the Δ CCS to increase. The related studies proposed that the structures of the complexes might be metal centered with the saccharides surrounded and the difference of the configuration, linkage and monomer unit might cause the diameter or surface concaveness differences of the assemblies and thus their enhanced differentiations.

In Chapter 4, chiral drug propranolol was studied to understand whether assembly could enhance differentiation of chiral drug enantiomers. Several types of chiral selectors were introduced and evaluated. The results showed that the sizes and structures of chiral selectors had important effects on the detection of the complex ions, chiral differentiation in MS/MS and IM-MS study.

In Chapter 5, the results of the whole research were summarized and discussed. Some general conclusions were obtained. The prospects of this study were also discussed in this chapter.

Chapter 2: Differentiation of leucine, isoleucine and allo-isoleucine, and analogue isomers by IM-MS

2.1 Introduction

Amino acids (AAs) are building blocks of proteins and play critical roles in living organisms. Leucine (Leu), Isoleucine (Ile) and allo-Isoleucine (allo-Ile) (Figure 2-1), isomers in branched-chain amino acid (BCAA), are prone to occurrence of isomerized invert in peptides and proteins, giving different biological functions.⁸²⁻⁸⁴ For examples, when an Ile residue in acetyl-CoA carboxylase is replaced by Leu, the herbicide-sensitive grass becomes herbicide-resistant;⁸⁵ the second residue of the peptides from skin secretions of Bombinas species was found to either L-Ile or D-allo-Ile, while the latter one had a stronger antibiotic activity and a higher rate of killing against the microorganisms;⁸³ Ile with sec-butyl side chain tends to be more α -helix destabilizing in peptides than Leu with isobutyl side chain due to its sterically imposing branches.⁸⁶ For measurement of BCAA in blood for diagnosis of maple syrup urine disease (MSUD),⁸⁷⁻⁸⁸ allo-Ile, the metabolism byproduct of MSUD, is more pathognomonic than Leu and Ile, since the latter ones are diet-related and can cause false-positive results.⁸⁹⁻⁹⁰

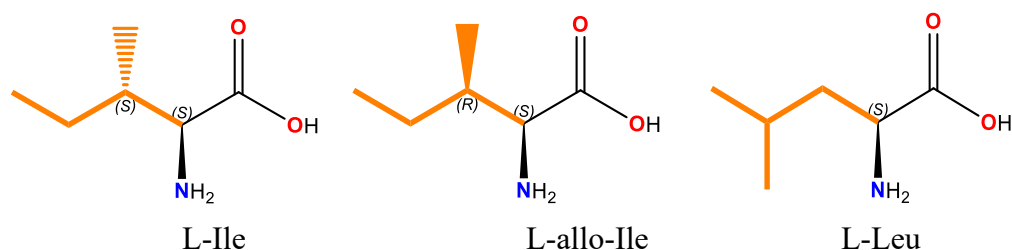
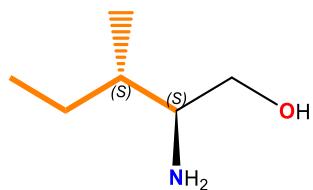


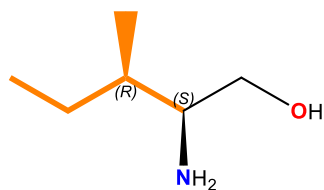
Figure 2-1. Structures of Leu, Ile and allo-Ile.

Several methods have been developed for differentiation of Leu, Ile and allo-Ile.⁹¹ A study by Devin and coauthors achieved separation of Leu, Ile and allo-Ile by using LC and combined MS/MS to establish an accurate measurement of allo-Ile in blood examples of MSUD patients.⁹⁰ Wang et al. reported the discrimination of Leu and Ile by comparing the intensities of the fragment ions using Matrix-assisted laser desorption/ionization time of flight MS (MALDI-TOF/TOF-MS).⁹² A peptide containing Ile residue was distinguished from one containing Leu residue by comparing the fragmentation using hot electron capture dissociation (HECD) on a Fourier transform-ion cyclotron resonance (FT-ICT) mass spectrometer for characterizing hemoglobin disease.⁹³ IM-MS is an emerging technique for differentiation of isomers and can offer unique structural information based on the measured CCSs. However, the differentiation of Leu, Ile and allo-Ile by IM-MS is still difficult due to their slight CCS difference of less than $2 \text{ \AA}^2$.⁸⁴

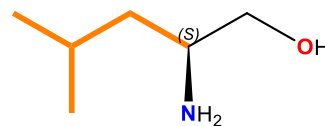
In this chapter, we aim to explore the effects of assembly on the differentiation of Leu, Ile and allo-Ile in IM-MS. Other analogue isomers (e.g., amino alcohols, keto acids and fatty acids) with branched alkyl side chains and their homo di-, tri-, tetrapeptides (Figures 2-2 and 2-3) were also selected for comparative and extended study. The solution conditions and ESI parameters were optimized for detection of higher degrees of assemblies, and the effect of assembly on isomer differentiation was studied by TWIM-MS in terms of drift time and CCS. IM-MS, MS/MS combined with computational modelling were applied to gain the comprehensive structures of assemblies, and the mechanism for the assembly-enhanced differentiation was discussed.



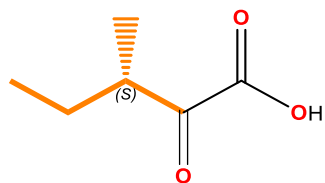
(S)-Isoleucinol



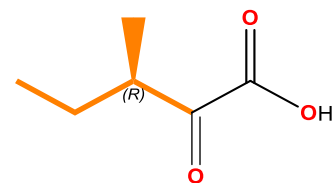
(R)-Isoleucinol



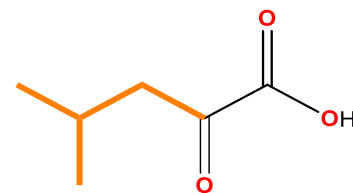
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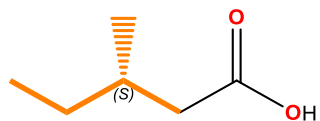
(S)-3-Methyl-2-oxovaleric acid



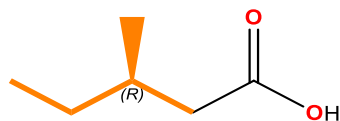
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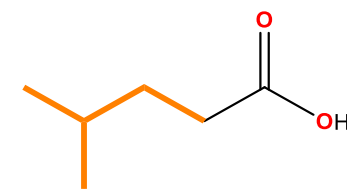
4-Methyl-2-oxovaleric acid



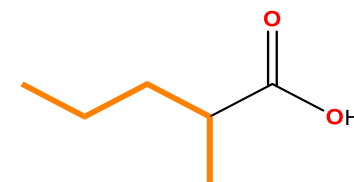
(S)-3-Methyl-pentanoic acid



(R)-3-Methyl-pentanoic acid

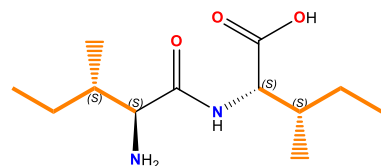


4-Methylvaleric acid

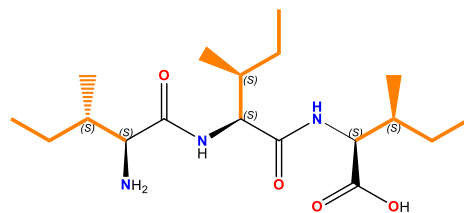


2-Methylvaleric acid

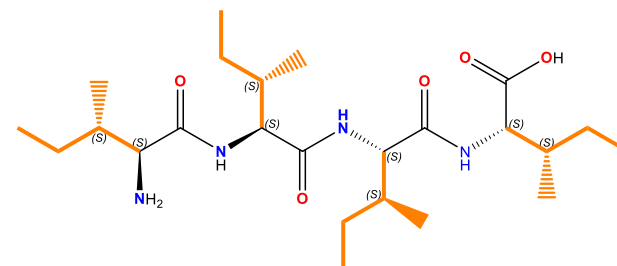
Figure 2-2. Structures of analogue isomers.



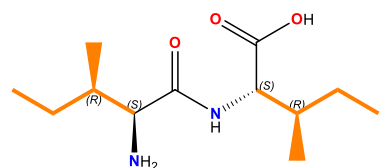
L-Ile-L-Ile



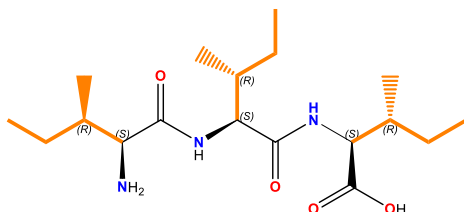
L-Ile-L-Ile-L-Ile



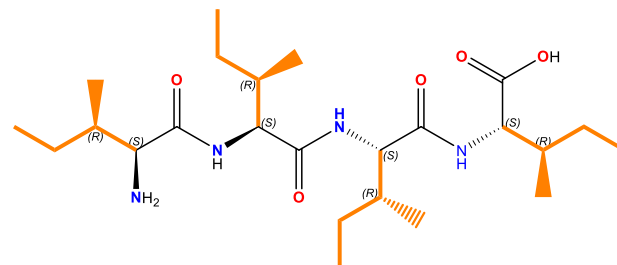
L-Ile-L-Ile-L-Ile-L-Ile



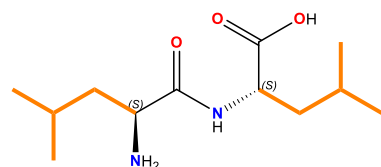
L-allo-Ile-L-allo-Ile



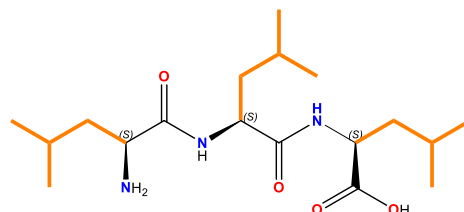
L-allo-Ile-L-allo-Ile-L-allo-Ile



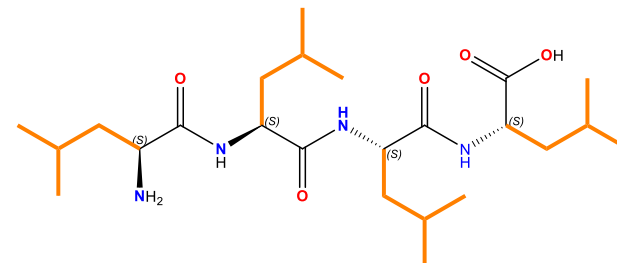
L-allo-Ile-L-allo-Ile-L-allo-Ile-L-allo-Ile



L-Leu-L-Leu



L-Leu-L-Leu-L-Leu



L-Leu-L-Leu-L-Leu-L-Leu

Figure 2-3. Structures of di-, tri- and tetrapeptides of Ile, allo-Ile and Leu.

2.2 Experimental

2.2.1 Materials

The optically pure chiral amino acids (L-Leu, D-Leu, L-Ile, D-Ile, L-allo-Ile and D-allo-Ile) and analogue compounds ((S)-Isoleucinol, (R)-Isoleucinol, (S)-Leucinol, (S)-3-Methyl-2-oxovaleric acid, (R)-3-Methyl-2-oxovaleric acid, 4-Methyl-2-oxovaleric acid, (S)-3-Methyl-pentanoic acid, (R)-3-Methyl-pentanoic acid, 4-Methylvaleric acid and 2-Methylvaleric acid) were purchased from Sigma-Aldrich Co (St. Louis, MO). The dipeptides (L-Leu-L-Leu, L-Ile-L-Ile and L-allo-Ile-L-allo-Ile) and tripeptides (L-Leu-L-Leu-L-Leu, L-Ile-L-Ile-L-Ile and L-allo-Ile-L-allo-Ile-L-allo-Ile) were purchased from GL Biochem Ltd (Shanghai, China). The dipeptides (L-Gly-L-Gly, L-Ala-L-Ala, L-Pro-L-Pro, L-Phe-L-Phe), tripeptides (L-Gly-L-Gly-L-Gly, L-Ala-L-Ala-L-Ala, L-Pro-L-Pro-L-Pro, L-Phe-L-Phe-L-Phe) and tetrapeptides (L-Leu-L-Leu-L-Leu-L-Leu, L-Ile-L-Ile-L-Ile-L-Ile, L-allo-Ile-L-allo-Ile-L-allo-Ile-L-allo-Ile, L-Gly-L-Gly-L-Gly-L-Gly, L-Ala-L-Ala-L-Ala-L-Ala, L-Pro-L-Pro-L-Pro-L-Pro and L-Phe-L-Phe-L-Phe-L-Phe) samples were purchased from Synpeptide Co Ltd (Nanjing, China). The metal salts ($\text{Cu}(\text{CH}_3\text{COOH})_2$, CuCl_2 , ZnCl_2 , NiCl_2 , MgCl_2 and LiNO_3 , NaNO_3 , KNO_3) were purchased from Aladdin (Shanghai, China), and the HPLC grade methanol was purchased from BCR International Trading Co Ltd. (California, USA). Ultrapure water was produced by a MilliQ device (Millipore, Milford, MA, USA). All

the chemicals were used directly without further purification.

2.2.2 Sample preparation

For IM-MS experiments, amino acids, peptides, analogue compounds and metal salts were dissolved separately in water to achieve a concentration of 5 mM. The sample solutions used for the MS analysis were prepared by mixing the stock solutions, and then diluting with 50:50 water/methanol, to obtain a series of final solutions containing 50 μ M metal salts with 5 μ M, 25 μ M, 50 μ M, 100 μ M, 250 μ M, 500 μ M or 1000 μ M amino acids, peptides or analogue compounds.

For NMR and CD experiments, amino acids (L-Leu, D-Leu, L-Ile, D-Ile, L-allo-Ile and D-allo-Ile) were dissolved separately in water and D₂O to achieve a concentration of 5 mM and 10 mM. Copper acetate salt was dissolved separately in water and D₂O to achieve a concentration of 5 mM and 50 mM. Acetic acid was diluted with D₂O, to obtain a final solution of 5mM. The sample solution used for the CD analysis was prepared by mixing the stock solutions and then diluting with water to obtain a final solution containing 400 μ M amino acids and 100 μ M, 400 μ M, 800 μ M copper acetate salt, to obtain final ratios of 0, 4:1, 4:4, 4:8 between amino acids and metal salt. The sample solution used for the NMR analysis was prepared by removing 0.5 ml 10 mM stock solutions in D₂O into NMR tubes and adding 0, 0.02 ml, 0.06 ml, 0.1 ml 50 mM

copper acetate in D₂O to obtain final ratios of 5:0, 5:1, 5:3, 5:5 between amino acids and metal salt, respectively.

2.2.3 Mass spectrometry

IM-MS and MS/MS experiments were performed using a Synapt G2 mass spectrometer (Waters, Manchester, UK). Electrospray ionization (ESI) conditions were as follows: capillary voltage, 3 kV; sampling cone, 40 V; extraction cone, 5 V; source temperature, 100 °C; desolvation temperature, 350°C; cone gas, 50 L h⁻¹; desolvation gas, 500 L h⁻¹. All the experiments were performed in the positive ion mode. The sample solutions were directly introduced to the mass spectrometer through the ESI source at a flow rate of 20 μL min⁻¹. All the spectral data represented the combined 50 scans in the direct infusion experiments. For the TWIM experiments, the drift gas was N₂, the traveling wave height was set at 40 V, and traveling wave velocity was chosen as 800 m s⁻¹. For MS/MS analysis, the transfer collision energy (CE) was set at 5, 10, 15, 20 and 25 eV to observe the fragmentation pattern. The mass spectra acquired were analyzed by the Driftscope program. The CCS values of the investigated complexes were calculated by using polyalanine ions with a concentration of 10 mg mL⁻¹ to plot a calibration curve as described by the literature,⁴⁵ and the CCS values of polyalanine were obtained from the literature.⁹⁴ The mass spectra acquired were analyzed by the Drift scope program.

2.2.4 Circular dichroism

CD spectra were recorded on the Jasco J-810 spectropolarimeter using a 1 mm cell at room temperature. Scans were made from 190 to 350 nm with a speed of 200 nm/min.

2.2.5 Nuclear magnetic resonance

^1H NMR spectra were recorded using a Bruker AV500 (500 MHz) spectrometer. Some of the ^1H spectra were referenced using a residual solvent peak, DMSO- d_6 (δ 2.50).

2.2.6 Computational details

Structures of the most promising copper complexes were studied using unrestricted B3LYP with a 6-31+ G(d) basis set. All geometries were optimized using the Gaussian 09 program package. Theoretical collision cross sections were evaluated using the exact hard-sphere scattering (EHSS) method from the structures from Gaussian optimization calculations.

2.3 Results and discussion

2.3.1 Optimization for detection of assembly ions

ESI is a soft ionization technique that could effectively generate assembly ions of amino acids.³⁰ In this study, the ESI parameters including voltage, gas flowrate, temperature,

etc. and other experimental factors such as metal ions, pH, solvents and concentration that are related to the molecular assembly⁹⁵⁻⁹⁶ were optimized for detection of multimeric complex ions of Leu, Ile, allo-Ile, analogue compounds and peptides.

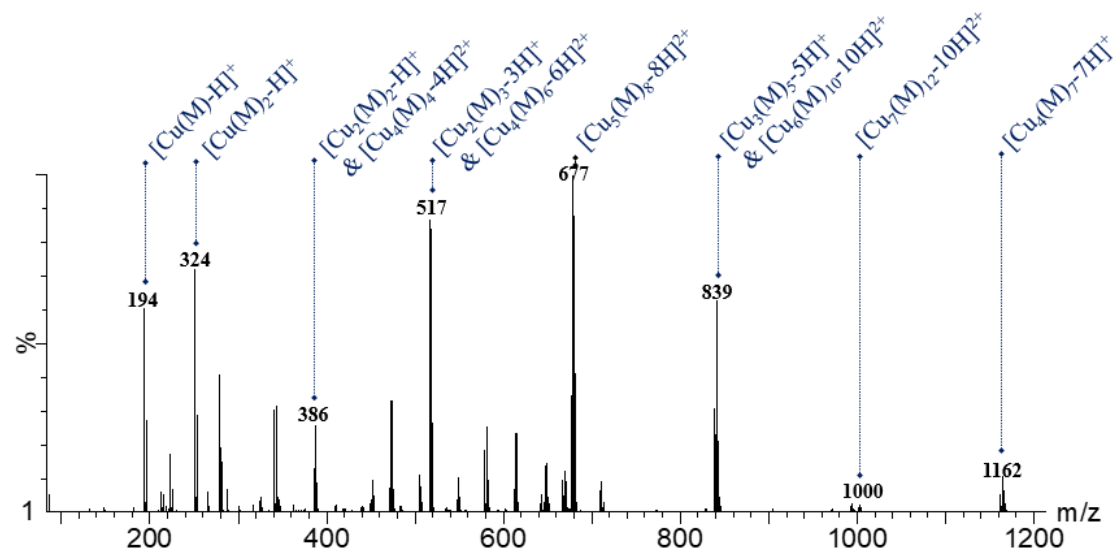


Figure2-4. ESI-MS spectrum of L-Ile with addition of $\text{Cu}(\text{CH}_3\text{COO})_2$ at a concentration ratio of Ile: Metal=10:1.

Figure 2-4 is a typical mass spectrum obtained for Ile in the presence of copper acetate, showing detection of various multimeric complex ions, including the hepta-copper bound dodecameric ion of Ile. The compositions of the complex ions (see Tables 2-1 to 2-7 for the list of the identified complex ions) were assigned by the combination of accurate masses, isotopic patterns and MS/MS spectra as discussed in section 2.3.2.

The additive metal ions were found to be one of the key factors affecting the assembly. For example, as shown in Figure 2-5, comparing the mass spectra of the AA solutions

without or with addition of metal, addition of metal ions was found to be crucial for detection of higher degree assembly ions. Metal ions including Cu^{2+} , Zn^{2+} , Mg^{2+} and Ni^{2+} were attempted in this study, and addition of Cu^{2+} , especially from $\text{Cu}(\text{COOCH}_3)_2$, could significantly promote the assembly and allow detection of self-assembled dodecamer of the AAs, such as $[(\text{M})_6 + (\text{Cu}^{\text{II}})_4 - 6\text{H}]^{2+}$ (m/z 517.1456), $[(\text{M})_7 + (\text{Cu}^{\text{II}})_4 - 7\text{H}]^+$ (m/z 1162.3887), $[(\text{M})_8 + (\text{Cu}^{\text{II}})_5 - 8\text{H}]^{2+}$ (m/z 680.6962), $[(\text{M})_{10} + (\text{Cu}^{\text{II}})_6 - 10\text{H}]^{2+}$ (m/z 841.2603). The largest complex ions detected were hexameric ions $[(\text{M})_6 + (\text{Mg}^{\text{II}})_4 - 6\text{H}]^+$ (m/z 438.2422) and $[(\text{M})_6 + (\text{Cu}^{\text{II}})_4 - 6\text{H}]^{2+}$ (m/z 517.1349) with the addition of MgCl_2 and CuCl_2 at concentration ratio 1:1, respectively, and tetrameric ions such as $[(\text{M})_4 + (\text{Ni}^{\text{II}})_2 - 3\text{H}]^+$ (m/z 637.2119) and $[(\text{M})_4 + (\text{Zn}^{\text{II}})_2 - 3\text{H}]^+$ (m/z 651.2167) with the addition of NiCl_2 and ZnCl_2 at concentration ratio 1:1, respectively. The further optimized conditions in our study enabled detection of these large assembly ions that have not been reported before.

For dipeptides, as shown in Figure 2-6, the largest complex ions detected were hexameric ions such as $[(\text{M})_6 + (\text{Mg}^{\text{II}})_4 - 6\text{H}]^{2+}$ (m/z 778.0106), $[(\text{M})_6 + (\text{Cu}^{\text{II}})_4 - 6\text{H}]^{2+}$ (m/z 918.3234) and $[(\text{M})_6 + (\text{Ni}^{\text{II}})_4 - 2\text{H}]^+$ (m/z 845.3804), trimeric ions such as $[(\text{M})_3 + (\text{Zn}^{\text{II}})_2 + 2\text{H}_2\text{O} - 3\text{H}]^+$ (m/z 897.3707), with the addition of MgCl_2 , CuCl_2 , NiCl_2 and ZnCl_2 , respectively. Addition of $\text{Cu}(\text{COOCH}_3)_2$ again could allow detection of the highest assemblies such as $[(\text{M})_8 + (\text{Cu}^{\text{II}})_8 - 14\text{H}]^{2+}$ (m/z 1223.3738). These complex

ions of the amino acids and peptides might involve the coordination bonding, i.e., donation of pair of electrons from the nitrogen or oxygen atoms in amino acids to the transition metals,⁹⁷⁻⁹⁹ while the presence of Cl⁻ might suppress the ESI signals, causing detection of lower degrees of assembly ions with ZnCl₂, MgCl₂, CuCl₂ and NiCl₂ as compared with Cu(COOCH₃)₂.¹⁰⁰⁻¹⁰¹

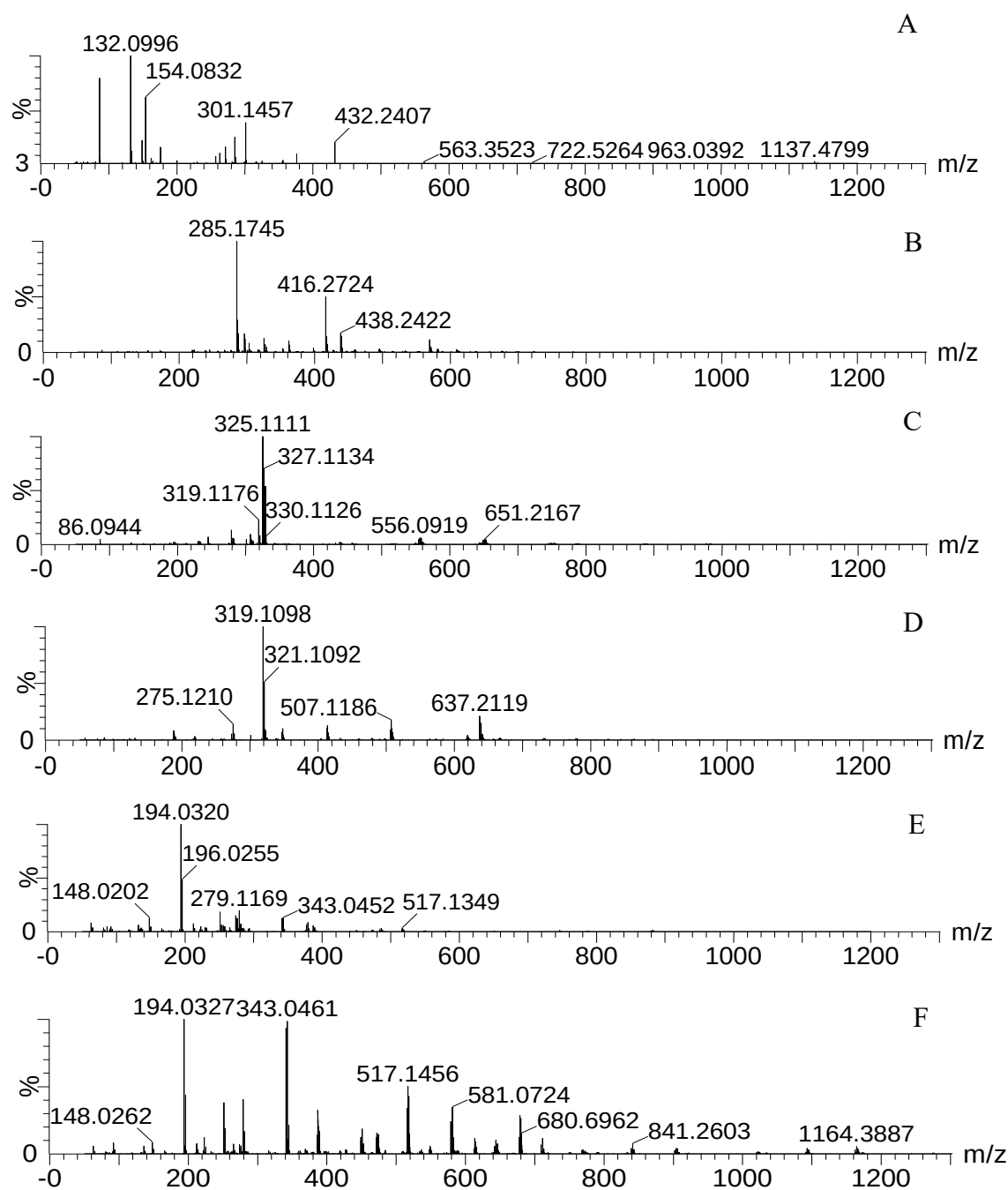


Figure 2-5. ESI-MS spectra for L-Ile without (A) and with addition of MgCl_2 (B), ZnCl_2 (C), NiCl_2 (D), CuCl_2 (E), $\text{Cu}(\text{COOCH}_3)_2$ (F) at a concentration ratio of Ile: Metal=1:1.

The assignments of major peaks are summarized in Table 2-1.

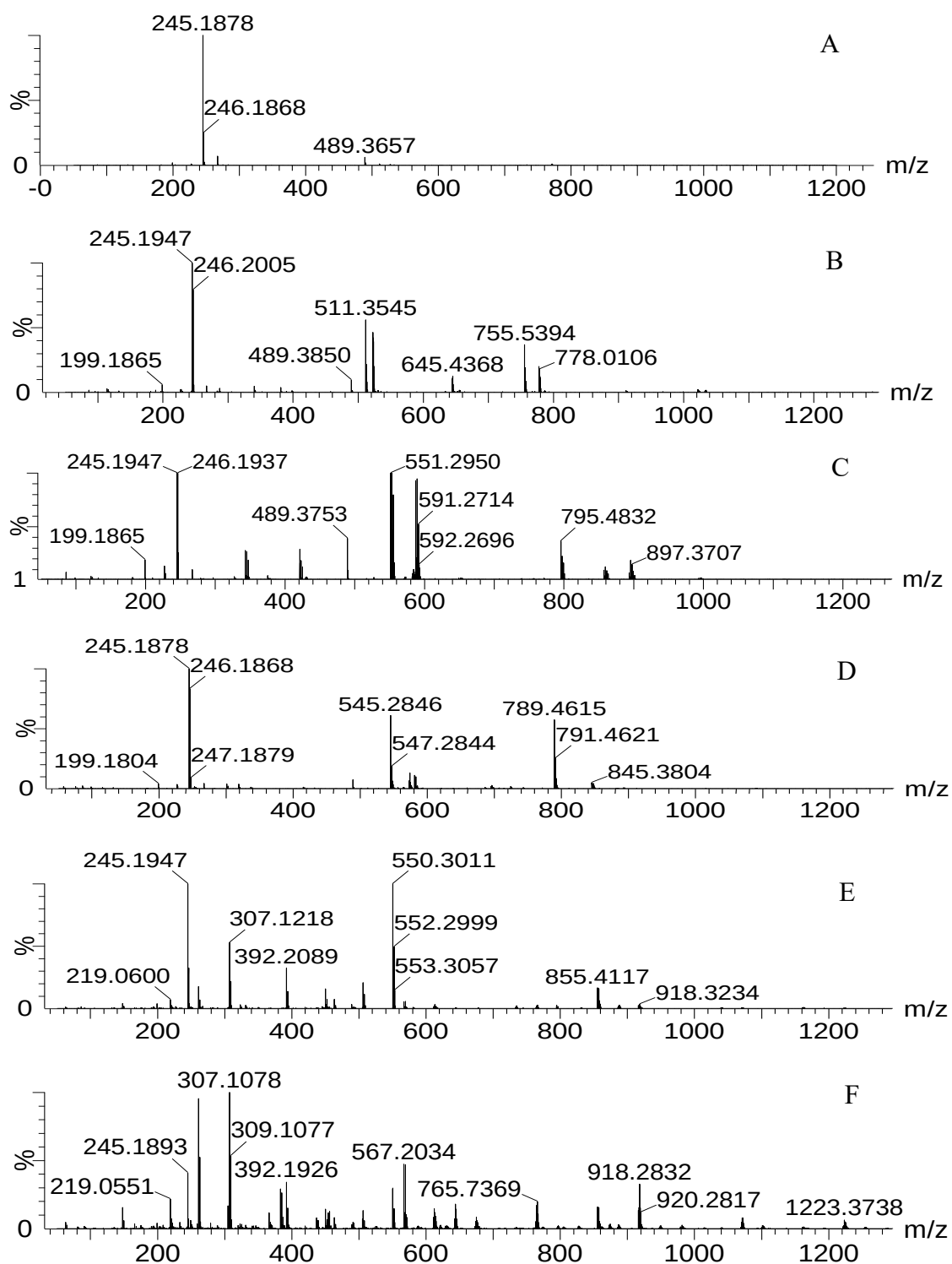


Figure 2-6. ESI-MS spectra for L-Leu-L-Leu without (A) and with addition of MgCl₂ (B), ZnCl₂ (C), NiCl₂ (D), CuCl₂ (E), Cu(COOCH₃)₂ (F) at concentration ratio of dipeptide: Metal=1:1. The assignments of major peaks are summarized in Table 2-2.

Concentration ratio of the analytes to metal ions, i.e., 10:1, 8:1, 5:1, 2:1, 1:1 or 1:10, was another important factor for the self-assembly. For AAs, as shown in Figure 2-7, with Cu^{2+} to Ile at a 1:10 molar ratio, higher degree assembly ions of Ile, such as m/z 517.1357, 678.7045, 841.2603, 1001.8007 and 1164.3887, corresponding to $[(\text{M})_6 + (\text{Cu}^{\text{II}})_4 - 6\text{H}]^{2+}$, $[(\text{M})_8 + (\text{Cu}^{\text{II}})_5 - 8\text{H}]^{2+}$, $[(\text{M})_{10} + (\text{Cu}^{\text{II}})_6 - 10\text{H}]^{2+}$, $[(\text{M})_{12} + (\text{Cu}^{\text{II}})_7 - 12\text{H}]^{2+}$ and $[(\text{M})_7 + (\text{Cu}^{\text{II}})_4 - 7\text{H}]^+$, showed stronger signals than with a 1:1 molar ratio, indicating the higher molar ratio of Ile enabled better signals of larger assemblies to be detected. As shown in Figure 2-8, the increased ratio of Ile/copper in general favored detection of the assembly ions with higher degrees.¹⁰²⁻¹⁰³

For dipeptides, however, as shown in Figure 2-9, compared with the 10:1 ratio, the equal molar ratio of dipeptides to the copper ion enabled better detection of larger assemblies, such as $[(\text{M})_6 + (\text{Cu}^{\text{II}})_6 - 10\text{H}]^{2+}$ (m/z 918.8234) and $[(\text{M})_8 + (\text{Cu}^{\text{II}})_8 - 14\text{H}]^{2+}$ (m/z 1223.4215). For Keto acids, the lower molar ratio of analyte to the copper ion was also better for detection of larger complex ions such as $[(\text{M})_5 + (\text{Cu}^{\text{II}})_3 - 5\text{H}]^+$ (m/z 766.1961), $[(\text{M})_6 + (\text{Cu}^{\text{II}})_4 + 2\text{H}_2\text{O} - 4\text{H}]^+$ (m/z 891.1712) and $[(\text{M})_7 + (\text{Cu}^{\text{II}})_4 + \text{H}_2\text{O} - 2\text{H}]^+$ (m/z 975.2316) for (S)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid (Figure 2-10). These indicated that the concentration ratio was an important factor for the assembly and should be considered carefully for different

analytes.

In sum, with addition of copper ion, assemblies up to hepta-copper bound dodecamers of Leu, Ile and allo-Ile, octa-copper bound octamers of the dipeptides, tetra-copper bound tetramers of the tripeptides, di-copper bound dimers of the tetrapeptides, hexa-copper bound octamers of (S)-3-methyl-2-oxovaleric acid, (R)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid, tetra-copper bound octamers of (S)-3-methyl-pentanoic acid, (R)-3-methyl-pentanoic acid, 4-methylvaleric acid and 2-methylvaleric acid, di-copper bound dimers of (S)-isoleucinol, (R)-isoleucinol and (S)-leucinol were detected by ESI-MS for the first time.

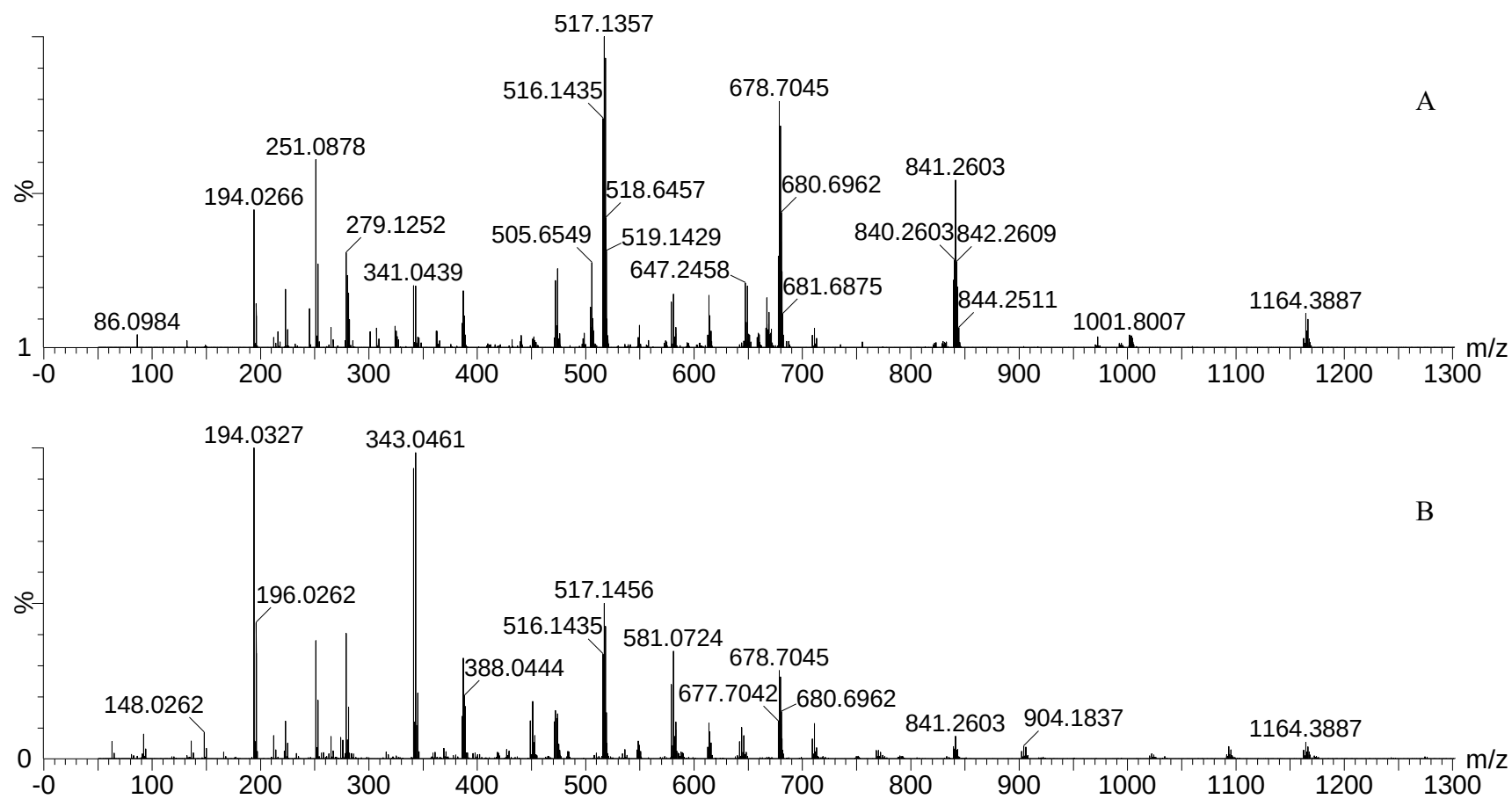


Figure 2-7. ESI-MS spectra for L-Ile with addition of $\text{Cu}(\text{COOCH}_3)_2$ at the concentration ratios of Ile: Metal=10:1 (A) and 1:1 (B). The assignments of major peaks are summarized in Table 2-1.

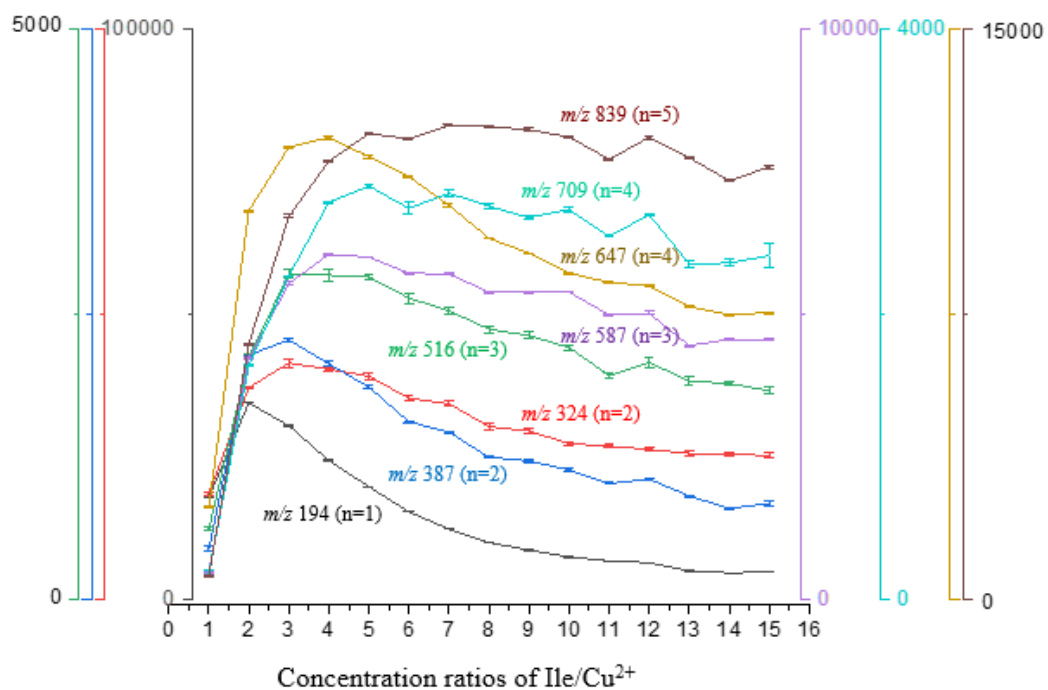


Figure 2-8. Plots of intensities of different oligomeric ions against the concentration ratio of Ile/Cu²⁺.

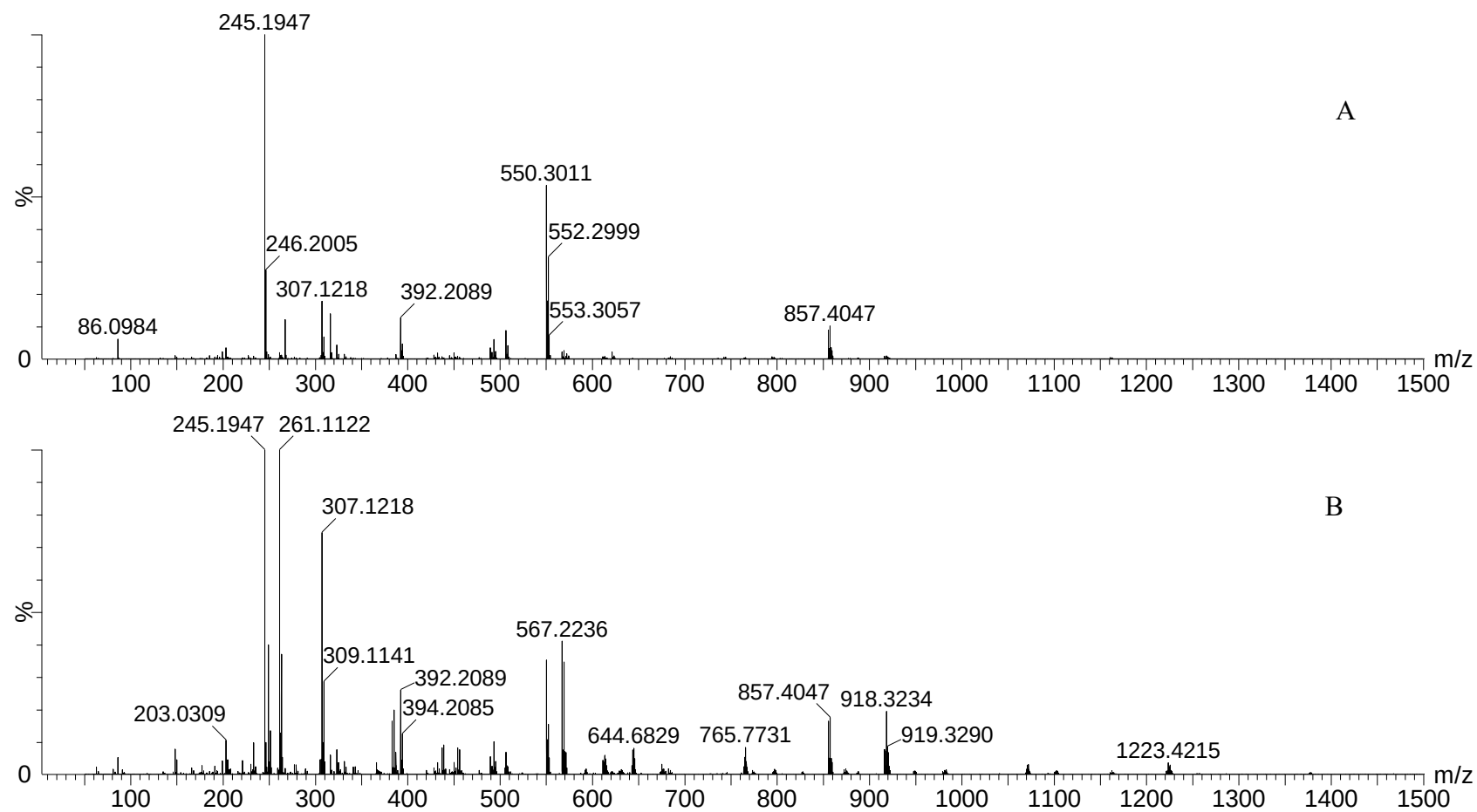


Figure 2-9. ESI-MS spectra for L-Leu-L-Leu with addition of $\text{Cu}(\text{COOCH}_3)_2$ at the concentration ratios of dipeptide: Metal=10:1 (A) and 1:1 (B). The assignments of major peaks are summarized in Table 2-2.

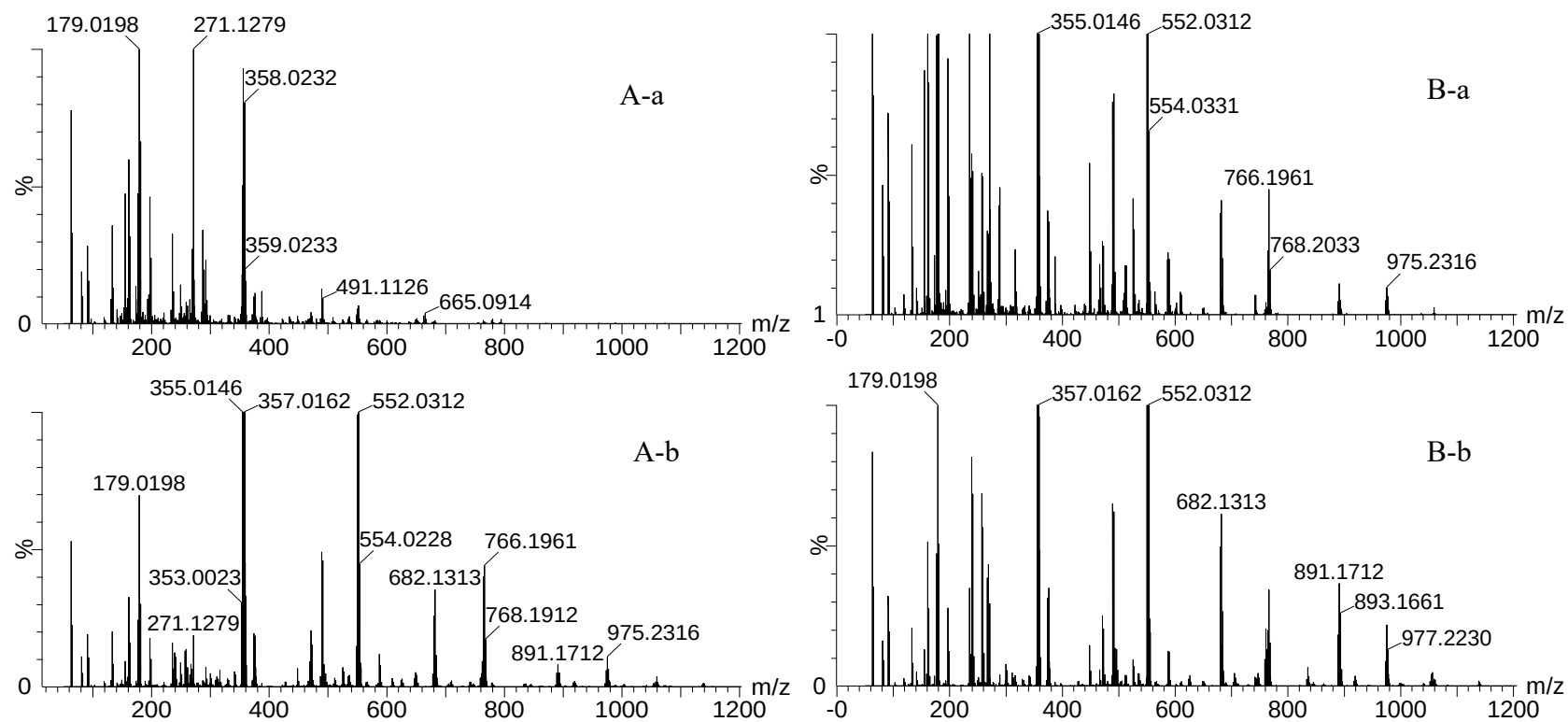


Figure 2-10. ESI-MS spectra for (S)-3-methyl-2-oxovaleric acid (A) and 4-methyl-2-oxovaleric acid (B) with addition of $\text{Cu}(\text{COOCH}_3)_2$ at the concentration ratios of Keto acids: Metal=10:1 (-a) and 10:5 (-b). The assignments of major peaks are summarized in Table 2-5.

2.3.2 Identification of assembly ions

The compositions of the complex ions were identified based on their accurate monoisotopic masses, isotopic patterns, fragmentation patterns of MS/MS spectra and measurements of isobaric peaks by IM. The identified complex ions of AAs, peptides and analogue compounds were listed in Tables 2-1 to 2-7.

Complex ions containing transition metals have special isotopic distributions, and the isotopic patterns of the complex ions are informative for determination of their compositions. For example, as shown in Figure 2-11, the complex ions with different numbers of copper ion showed different isotopic distribution patterns, allowing determination of the Cu number in the complex ions.

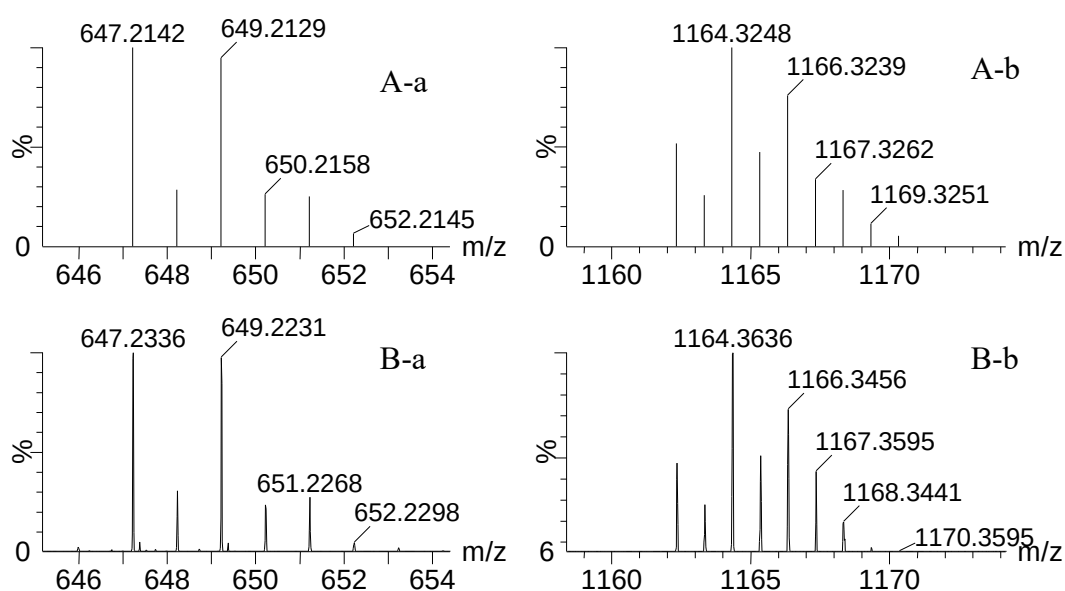


Figure 2-11. Theoretical mass spectra (A) and experimental mass spectra (B) of $[(M)_4 + (Cu^{II})_2 - 3H]^+$ (-a) and $[(M)_7 + (Cu^{II})_4 - 7H]^+$ (-b).

Some mass peaks contained different complex ions with the same m/z value, and such isobaric ions could be separated in IM-MS, typically due to their different charged states. For example, as shown in Figure 2-12, isobaric ions such as $[\text{Cu}_2(\text{M})_2 - \text{H}]^+$ and $[\text{Cu}_4(\text{M})_4 - 4\text{H}]^{2+}$ (m/z 387), $[\text{Cu}_2(\text{M})_3 - 3\text{H}]^+$ and $[\text{Cu}_4(\text{M})_6 - 6\text{H}]^{2+}$ (m/z 517), and $[\text{Cu}_3(\text{M})_5 - 5\text{H}]^+$ and $[\text{Cu}_6(\text{M})_{10} - 10\text{H}]^{2+}$ (m/z 839) could be separated by IM-MS and further identified. These separated singly charged ions and doubly charged ions allowed grouping for the further study.

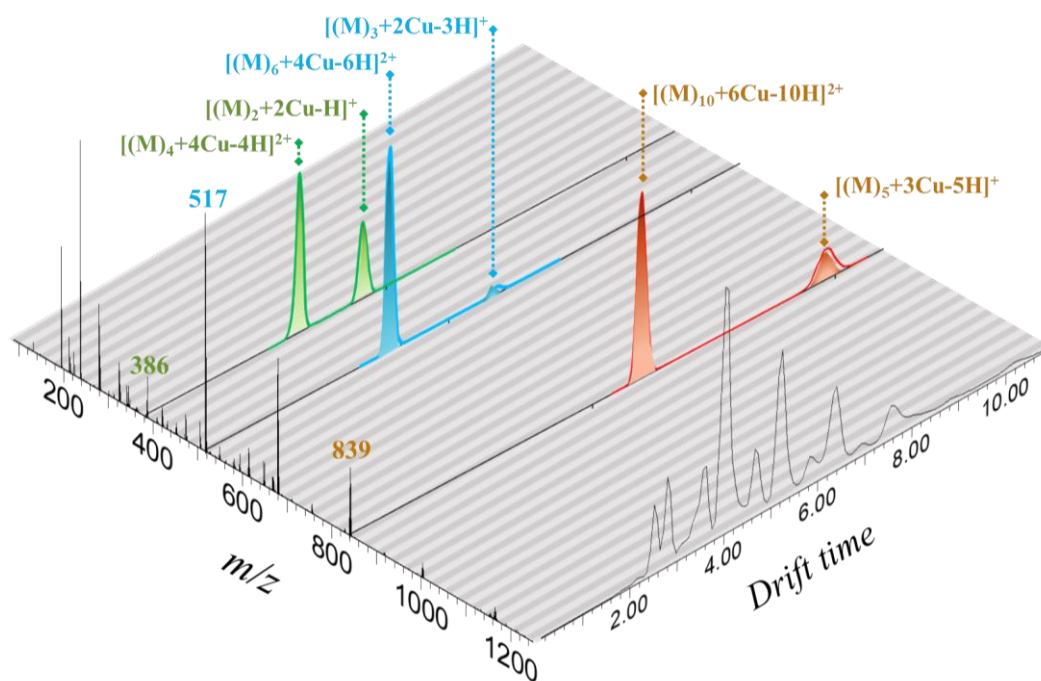


Figure 2-12. The 3D plot of m/z versus drift time for L-Ile with addition of $\text{Cu}(\text{COOCH}_3)_2$ at a concentration ratio of 20:1, which allowed separation of isobaric peaks at m/z 387, m/z 516 and m/z 839.

Fragmentation by MS/MS combined with IM was also applied to further confirm the identification. Take AAs as an example, it was found that for singly charged ions, the assembly ions gradually lost one M (Ile, Leu or allo-Ile) or one Cu^{2+} with the increasing collision energy (Figure 2-13). Precursor ion m/z 1164 $[(\text{M})_7 + (\text{Cu}^{\text{II}})_4 - 7\text{H}]^+$ produced product ions m/z 904 $[(\text{M})_5 + (\text{Cu}^{\text{II}})_4 - 7\text{H}]^+$, m/z 841 $[(\text{M})_5 + (\text{Cu}^{\text{II}})_3 - 5\text{H}]^+$, m/z 711 $[(\text{M})_4 + (\text{Cu}^{\text{II}})_2 + \text{Cu}^{\text{I}} - 4\text{H}]^+$ and m/z 581 $[(\text{M})_3 + \text{Cu}^{\text{II}} + (\text{Cu}^{\text{I}})_2 - 3\text{H}]^+$, precursor ion m/z 711 $[(\text{M})_4 + (\text{Cu}^{\text{II}})_2 + \text{Cu}^{\text{I}} - 4\text{H}]^+$ produced product ion m/z 581 $[(\text{M})_3 + \text{Cu}^{\text{II}} + (\text{Cu}^{\text{I}})_2 - 3\text{H}]^+$, and precursor ion m/z 581 $[(\text{M})_3 + \text{Cu}^{\text{II}} + (\text{Cu}^{\text{I}})_2 - 3\text{H}]^+$ or 647 $[(\text{M})_4 + (\text{Cu}^{\text{II}})_2 - 3\text{H}]^+$ produced product ion m/z 517 $[(\text{M})_3 + (\text{Cu}^{\text{II}})_2 - 3\text{H}]^+$. It is notable that product ion such as m/z 579 $[(\text{M})_3 + \text{Cu}^{\text{II}} + (\text{Cu}^{\text{I}})_2 - 3\text{H}]^+$ from precursor assembly ion m/z 709 $[(\text{M})_4 + (\text{Cu}^{\text{II}})_2 + \text{Cu}^{\text{I}} - 4\text{H}]^+$ showed the same drift time as the directly detected assembly ion m/z 579 $[(\text{M})_3 + \text{Cu}^{\text{II}} + (\text{Cu}^{\text{I}})_2 - 3\text{H}]^+$, suggesting that they have the same structure and the larger assemblies might be formed from the smaller ones.

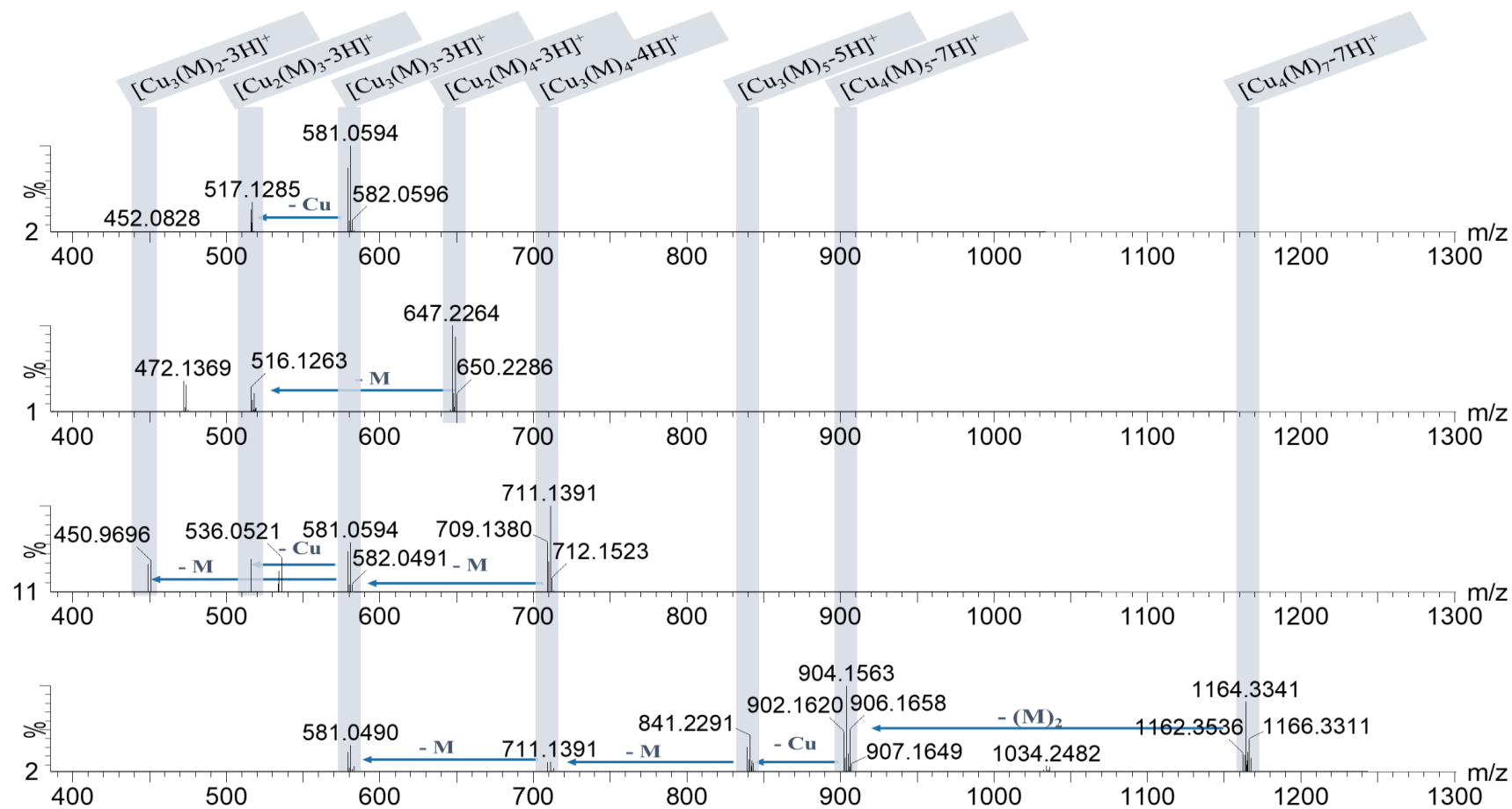


Figure 2-13. MS/MS spectra of typical singly charged oligomers on m/z 579, m/z 647, m/z 709 and m/z 1162 under the collision energy (CE) of 15 eV.

The patterns in the doubly charged groups were different. Major doubly charged assembly ions were selected as precursor ions to be activated with increasing collision voltages and separated in the TWIMS cell. As mass selection in CID was based on m/z value, singly charged ions and doubly charged ions with the same m/z values were trapped and dissociated together. By combining the results in Figure 2-14, we could remove the interference of the fragments from the singly charged ions. It could be clearly seen that the product ion of m/z 1002 $[(M)_{12} + (Cu^{II})_7 - 12H]^{2+}$ was m/z 841 $[(M)_{10} + (Cu^{II})_6 - 5H]^{2+}$, the product ion of m/z 841 $[(M)_{10} + (Cu^{II})_6 - 5H]^{2+}$ was m/z 678 $[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$, and the product ion of m/z 678 $[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$ was m/z 517 $[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$. Unlike in the singly charged group, in the doubly charged group, the major loss fragment was m/z 161, $Cu(M)_2$, one of the most stable basic blocks in any states. The drift time of product ions, such as m/z 678 $[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$ from precursor assembly ion m/z 841 $[(M)_{10} + (Cu^{II})_6 - 5H]^{2+}$, showed the same time with m/z 678 $[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$ formed at single stage MS. The results were interesting and probably provided a clue for explanation of the extremely complicated assembly process in gas phase.

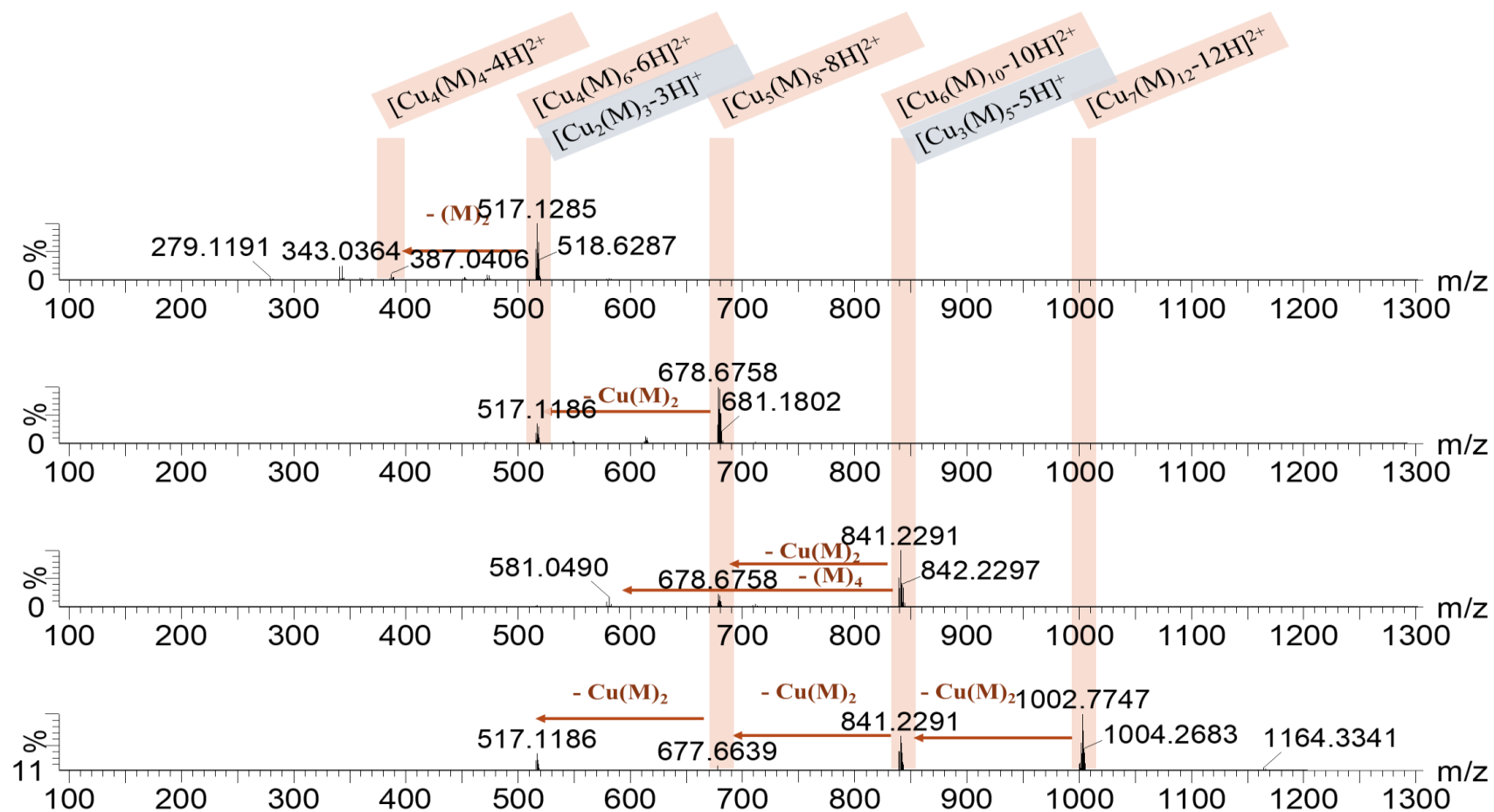


Figure 2-14. Dissociation pathway of typical doubly charged oligomers by mass selected CID/IMS (m/z 579, $t_d=6\text{ms}$; m/z 709, $t_d=8\text{ms}$).

Table 2-1. Assembly ions of L-Leu, L-Ile and L-allo-Ile (M) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[M + H]^+$	$C_6H_{14}NO_2$	132.1025	132.0997	-0.0028
$[M + Cu^I]^+$	$C_6H_{13}O_2NCu$	194.0242	194.0215	-0.0027
$[(M)_2 + Cu^{II} - H]^+$	$C_{12}H_{25}N_2O_4Cu$	324.1110	324.1147	0.0037
$[(M)_2 + (Cu^I)_2 - H]^+$	$C_{12}H_{25}N_2O_4Cu_2$	387.0406	387.1138	0.0732
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$C_{18}H_{36}N_3O_6Cu_2$	516.1196	516.1184	-0.0012
$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	$C_{18}H_{36}N_3O_6Cu_3$	579.0492	579.0678	0.0186
$[(M)_4 + (Cu^{II})_2 - 3H]^+$	$C_{24}H_{49}N_4O_8Cu_2$	647.2142	647.2192	0.0050
$[(M)_4 + (Cu^{II})_2 + Cu^I - 4H]^+$	$C_{24}H_{48}N_4O_8Cu_3$	709.1360	709.2733	0.1373
$[(M)_4 + (Cu^{II})_2 + (Cu^I)_2 - 4H]^{2+}$	$C_{24}H_{48}N_4O_8Cu_4$	386.0328	386.0379	0.0051
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	$C_{30}H_{60}N_5O_{10}Cu_3$	839.2228	839.2333	0.0105
$[(M)_6 + (Cu^{II})_3 + Na - 6H]^+$	$C_{36}H_{72}N_6O_{12}Cu_3Na$	992.2994	992.2747	-0.0247
$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	$C_{36}H_{72}N_6O_{12}Cu_4$	516.1196	516.1184	-0.0012
$[(M)_7 + (Cu^{II})_4 - 7H]^+$	$C_{42}H_{84}N_7O_{14}Cu_4$	1162.3260	1162.2460	-0.0800
$[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$	$C_{48}H_{96}N_8O_{16}Cu_5$	677.6712	677.6772	0.0060
$[(M)_{10} + (Cu^{II})_6 - 10H]^{2+}$	$C_{60}H_{120}N_{10}O_{20}Cu_6$	839.2228	839.2333	0.0105
$[(M)_{12} + (Cu^{II})_6 + 2Na - 18H]^{2+}$	$C_{72}H_{138}N_{12}O_{24}Cu_6Na_2$	989.2759	989.3353	0.0594
$[(M)_{12} + (Cu^{II})_7 - 12H]^{2+}$	$C_{72}H_{144}N_{12}O_{24}Cu_7$	1000.7744	1000.8061	0.0317
$[(M) + Zn^I]^+$	$C_6H_{12}NO_2Zn$	194.0159	194.0145	-0.0014
$[(M)_2 + Zn^{II} - H]^+$	$C_{12}H_{25}N_2O_4Zn$	325.1106	325.1111	0.0005
$[(M)_3 + (Zn^{II})_2 - 3H]^+$	$C_{18}H_{36}N_3O_6Zn_2$	518.1187	518.1288	0.0101

$[(M)_4 + (Zn^{II})_2 - 3H]^+$	$C_{24}H_{49}N_4O_8Zn_2$	649.2133	649.2131	-0.0002
$[(M)_6 + (Zn^{II})_4 - 6H]^{2+}$	$C_{36}H_{72}N_6O_{12}Zn_4$	518.1187	518.1288	0.0101
$[(M)_8 + (Zn^{II})_4 - 6H]^{2+}$	$C_{48}H_{98}N_8O_{16}Zn_4$	649.2133	649.2131	-0.0002
$[(M) + Ni^{II} - H]^+$	$C_6H_{12}NO_2Ni$	188.0222	188.0175	-0.0047
$[(M)_2 + Ni^{II} - H]^+$	$C_{12}H_{25}N_2O_4Ni$	319.1168	319.1098	-0.0070
$[(M)_3 + (Ni^{II})_2 - 3H]^+$	$C_{18}H_{36}N_3O_6Ni_2$	506.1311	506.1164	-0.0147
$[(M)_4 + (Ni^{II})_2 - 3H]^+$	$C_{24}H_{49}N_4O_8Ni_2$	637.2257	637.2119	-0.0138
$[(M)_4 + (Ni^{II})_3 - 4H]^{2+}$	$C_{24}H_{48}N_4O_8Ni_3$	347.0766	347.0682	-0.0084
$[(M)_5 + (Ni^{II})_3 - 5H]^+$	$C_{30}H_{60}N_5O_{10}Ni_3$	824.2401	824.245	0.0049
$[(M)_5 + (Ni^{II})_3 - 4H]^{2+}$	$C_{30}H_{61}N_5O_{10}Ni_3$	412.6239	412.611	-0.0129
$[(M)_6 + (Ni^{II})_4 - 6H]^{2+}$	$C_{36}H_{72}N_6O_{12}Ni_4$	506.1311	506.1164	-0.0147
$[(M)_9 + (Ni^{II})_5 - 9H]^{2+}$	$C_{54}H_{108}N_9O_{18}Ni_5$	730.2290	730.1030	-0.1260
$[(M)_{10} + (Ni^{II})_6 - 10H]^+$	$C_{60}H_{120}N_{10}O_{20}Ni_6$	824.2401	824.2424	0.0023
$[(M) + Mg^{II} - H]^+$	$C_6H_{12}NO_2Mg$	154.0718	154.0724	0.0006
$[(M)_2 + Mg^I - H]^+$	$C_{12}H_{25}N_2O_4Mg$	285.1665	285.1745	0.0080
$[(M)_3 + Mg^{II} - H]^+$	$C_{18}H_{38}N_3O_6Mg$	416.2611	416.2724	0.0113
$[(M)_3 + (Mg^{II})_2 - 3H]^+$	$C_{18}H_{36}N_3O_6Mg_2$	438.2305	438.2422	0.0117
$[(M)_4 + (Mg^{II})_2 - 3H]^+$	$C_{24}H_{49}N_4O_8Mg_2$	569.3251	569.3466	0.0215
$[(M)_4 + (Mg^{II})_2 - 2H]^{2+}$	$C_{24}H_{50}N_4O_8Mg_2$	285.1665	285.1745	0.0080
$[(M)_5 + (Mg^{II})_3 - 5H]^+$	$C_{30}H_{60}N_5O_{10}Mg_3$	722.3892	722.3974	0.0082
$[(M)_6 + (Mg^{II})_4 - 6H]^+$	$C_{36}H_{72}N_6O_{12}Mg_4$	438.2305	438.2422	0.0117
$[(M)_8 + (Mg^{II})_5 - 8H]^+$	$C_{48}H_{96}N_8O_{16}Mg_5$	580.3098	580.3256	0.0158
$[(M)_{10} + (Mg^{II})_6 - 10H]^+$	$C_{60}H_{120}N_{10}O_{20}Mg_6$	722.3892	722.3974	0.0082

Table 2-2. Assembly ions of dipeptides (Leu-Leu, Ile-Ile and allo-Ile) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[(M) + Zn^{II} + 2H_2O - H]^+$	$C_{12}H_{27}N_2O_5Zn$	343.1211	343.0869	-0.0342
$[M + (Zn^{II})_2 + 3H_2O - 5H]^+$	$C_{12}H_{25}N_2O_6Zn_2$	421.0296	421.0958	0.0662
$[(M)_2 + Zn^{II} - H]^+$	$C_{24}H_{47}N_4O_6Zn$	551.2787	551.2950	0.0163
$[(M)_2 + Zn^{II} + 2H_2O - H]^+$	$C_{24}H_{51}N_4O_8Zn$	587.2998	587.2658	-0.0340
$[(M)_2 + (Zn^{II})_2 + 2H_2O - 3H]^+$	$C_{24}H_{49}N_4O_8Zn_2$	649.2133	649.1797	-0.0336
$[(M)_2 + (Zn^{II})_2 + 4H_2O - 3H]^+$	$C_{24}H_{53}N_4O_{10}Zn_2$	685.2345	685.1682	-0.0663
$[(M)_3 + Zn^{II} - H]^+$	$C_{36}H_{71}N_6O_9Zn$	795.4574	795.4708	0.0134
$[(M)_3 + (Zn^{II})_2 - 3H]^+$	$C_{36}H_{69}N_6O_9Zn_2$	857.3709	857.3918	0.0209
$[(M)_3 + (Zn^{II})_2 + 2H_2O - 3H]^+$	$C_{36}H_{73}N_6O_{11}Zn_2$	893.392	893.3735	-0.0185
$[(M)_3 + (Zn^{II})_3 + 4H_2O - 5H]^+$	$C_{36}H_{75}N_6O_{13}Zn_3$	991.3266	991.2722	-0.0544
$[(M) + Ni^{II} - H]^+$	$C_{12}H_{23}N_2O_3Ni$	301.1062	301.1078	0.0016
$[(M)_2 + Ni^{II} - H]^+$	$C_{24}H_{47}N_4O_6Ni$	545.2849	545.2846	-0.0003
$[(M)_2 + Ni^I + H_2O]^+$	$C_{24}H_{50}N_4O_7Ni$	564.3033	564.2542	-0.0491
$[(M)_2 + Ni^{II} + 2H_2O - H]^+$	$C_{24}H_{51}N_4O_8Ni$	581.3060	581.2618	-0.0442
$[(M)_3 + Ni^{II} - H]^+$	$C_{36}H_{71}N_6O_9Ni$	789.4636	789.4615	-0.0021
$[(M)_3 + (Ni^I)_2 - H]^+$	$C_{36}H_{69}N_6O_9Ni_2$	845.3833	845.3677	-0.0156
$[(M)_4 + (Ni^I)_2 + 2H_2O]^{2+}$	$C_{48}H_{100}N_8O_{14}Ni_2$	564.3033	564.2542	-0.0491
$[(M)_6 + (Ni^I)_4 - 2H]^+$	$C_{72}H_{138}N_{12}O_{18}Ni_4$	845.3833	845.3350	-0.0483
$[(M) + Mg^{II} - H]^+$	$C_{12}H_{23}N_2O_3Mg$	267.1559	267.1626	0.0067
$[(M)_2 + Mg^{II} - H]^+$	$C_{24}H_{47}N_4O_6Mg$	511.3346	511.3545	0.0199
$[(M)_3 + (Mg^{II})_2 - 3H]^+$	$C_{36}H_{69}N_6O_9Mg_2$	777.4827	777.5114	0.0287
$[(M)_4 + (Mg^{II})_2 - 2H]^{2+}$	$C_{48}H_{94}N_8O_{12}Mg_2$	511.3346	511.3545	0.0199

$[(M)_4 + (Mg^{II})_3 - 4H]^{2+}$	$C_{48}H_{92}N_8O_{12}Mg_3$	522.3193	522.3405	0.0212
$[(M)_5 + (Mg^{II})_3 - 4H]^{2+}$	$C_{60}H_{116}N_{10}O_{15}Mg_3$	644.4086	644.4280	0.0194
$[(M)_6 + (Mg^{II})_4 - 6H]^{2+}$	$C_{72}H_{138}N_{12}O_{18}Mg_4$	777.4827	777.5114	0.0287
$[(M)_7 + (Mg^{II})_5 - 8H]^{2+}$	$C_{84}H_{160}N_{14}O_{21}Mg_5$	911.5567	911.6008	0.0441
$[(M)_8 + (Mg^{II})_5 - 8H]^{2+}$	$C_{96}H_{184}N_{16}O_{24}Mg_5$	1032.6461	1032.6674	0.0213
$[(M) + Cu^I]^+$	$C_{12}H_{24}N_2O_3Cu$	307.1083	307.1218	0.0135
$[(M)_2 + Cu^{II} - H]^+$	$C_{24}H_{47}N_4O_6Cu$	550.2792	550.3011	0.0219
$[(M)_2 + Cu^{II} + H_2O - 2H]^+$	$C_{24}H_{48}N_4O_7Cu$	567.2819	567.2236	-0.0583
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	$C_{24}H_{45}N_4O_6Cu_2$	611.1931	611.2122	0.0191
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$C_{36}H_{69}N_6O_9Cu_2$	855.3718	855.3989	0.0271
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	$C_{36}H_{67}N_6O_9Cu_3$	916.2858	916.3269	0.0411
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	$C_{48}H_{89}N_8O_{12}Cu_4$	1221.3784	1221.4373	0.0589
$[(M)_4 + (Cu^{II})_4 - 6H]^{2+}$	$C_{48}H_{90}N_8O_{12}Cu_4$	611.1931	611.2122	0.0191
$[(M)_4 + (Cu^{II})_5 - 6H]^{2+}$	$C_{48}H_{90}N_8O_{12}Cu_5$	642.6579	642.6893	0.0314
$[(M)_5 + (Cu^{II})_5 - 8H]^{2+}$	$C_{60}H_{112}N_{10}O_{15}Cu_5$	763.7394	763.7812	0.0418
$[(M)_6 + (Cu^{II})_6 - 10H]^+$	$C_{72}H_{134}N_{12}O_{18}Cu_6$	916.2858	916.3269	0.0411
$[(M)_7 + (Cu^{II})_7 - 12H]^+$	$C_{84}H_{156}N_{14}O_{21}Cu_7$	1068.8320	1068.8776	0.0456
$[(M)_8 + (Cu^{II})_8 - 14H]^+$	$C_{96}H_{178}N_{16}O_{24}Cu_8$	1221.3784	1221.4373	0.0589

Table 2-3. Assembly ions of tripeptides (Leu-Leu-Leu, Ile-Ile-Ile and allo-Ile-allo-Ile-allo-Ile) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[(M) + H]^+$	$C_{18}H_{36}N_3O_4$	358.4961	358.5718	0.0757
$[(M) + Cu^{II} - H]^+$	$C_{18}H_{34}N_3O_4Cu$	420.0263	420.1976	0.1713
$[(M)_2 + Cu^{II} - H]^+$	$C_{36}H_{69}N_6O_8Cu$	777.5145	777.6287	0.1142
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	$C_{36}H_{67}N_6O_8Cu_2$	839.0446	839.0989	0.0543
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	$C_{54}H_{103}N_9O_{12}Cu_3$	1258.0629	1258.1101	0.0472
$[(M)_4 + (Cu^{II})_4 - 6H]^{2+}$	$C_{72}H_{137}N_{12}O_{16}Cu_4$	791.0018	791.0239	0.0221

Table 2-4. Assembly ions of tetrapeptides (Leu-Leu-Leu-Leu, Ile-Ile-Ile-Ile and allo-Ile-allo-Ile-allo-Ile-allo-Ile) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[(M) + H]^+$	$C_{24}H_{47}N_4O_5$	471.3546	471.3586	0.0040
$[(M) + Cu^I - H]^+$	$C_{24}H_{45}N_4O_5Cu$	532.2686	532.2737	0.0051
$[(M)_2 + Cu^{II} - H]^+$	$C_{48}H_{91}N_8O_{10}Cu$	1002.6154	1002.6237	0.0083
$[(M)_2 + (Cu^{II})_2 - 2H]^{2+}$	$C_{48}H_{90}N_8O_{10}Cu_2$	532.2686	532.2737	0.0051
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	$C_{48}H_{89}N_8O_{10}Cu_2$	1063.5294	1063.5411	0.0117

Table 2-5. Assembly ions of keto acids ((S)-3-methyl-2-oxovaleric acid, (R)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[(M) + Cu^I]^+$	$C_6H_{10}O_3Cu$	192.6799	192.9968	0.3169
$[(M)_2 + (Cu^I)_3 - 2H]^+$	$C_{12}H_{18}O_6Cu_3$	446.8991	446.9178	0.0187
$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	$C_{18}H_{27}O_9Cu_3$	575.9543	575.9783	0.0240
$[(M)_3 + (Cu^I)_4 - 3H]^+$	$C_{18}H_{27}O_9Cu_4$	638.8839	638.9089	0.0250
$[(M)_4 + (Cu^{II})_2 + (Cu^I)_2 - 4H]^{2+}$	$C_{24}H_{36}O_{12}Cu_4$	383.9695	383.9843	0.0148
$[(M)_4 + (Cu^{II})_2 + Cu^I - 4H]^+$	$C_{24}H_{36}O_{12}Cu_3$	705.0095	705.0320	0.0225
$[(M)_8 + (Cu^{II})_4 + (Cu^I)_2 - 8H]^{2+}$	$C_{48}H_{74}O_{24}Cu_6$	705.0095	705.0320	0.0225

Table 2-6. Assembly ions of fatty acids ((S)-3-methyl-pentanoic acid, (R)-3-methyl-pentanoic acid, 4-methylvaleric acid and 2-methylvaleric acid) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[(M) + Cu^I]^+$	$C_6H_{12}O_2Cu$	179.0133	179.0198	0.0065
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	$C_{12}H_{21}O_4Cu_2$	355.0032	355.0229	0.0197
$[(M)_3 + (Cu^{II})_3 + H_2O - 5H]^+$	$C_{18}H_{33}O_7Cu_3$	550.0114	550.0330	0.0216
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	$C_{30}H_{55}O_{10}Cu_3$	764.1683	764.2063	0.0380
$[(M)_6 + (Cu^{II})_4 + 2H_2O - 4H]^+$	$C_{36}H_{72}O_{14}Cu_4$	887.1361	887.1618	0.0257
$[(M)_7 + (Cu^{II})_4 + H_2O - 2H]^+$	$C_{42}H_{84}O_{15}Cu_4$	971.2956	971.2141	-0.0815
$[(M)_8 + (Cu^{II})_4 - 2H]^+$	$C_{48}H_{94}O_{16}Cu_4$	1052.4312	1052.1343	-0.2969

Table 2-7. Assembly ions of amino alcohols ((S)-isoleucinol, (R)-isoleucinol and (S)-leucinol) observed in the ESI-MS spectra.

Composition	Formula	Mass (Da)		
		Theoretical	Measured	Error
$[(M) + H]^+$	$C_6H_{16}NOCu$	118.1973	118.1266	-0.0707
$[(M) + 2H_2O + (Cu^I)_2 - H]^+$	$C_6H_{19}NO_2Cu_2$	279.3040	279.1534	-0.1506
$[(M) + 3H_2O + (Cu^I)_2 - H]^+$	$C_6H_{21}NO_3Cu_2$	297.3193	297.2635	-0.0558
$[(M)_2 + Cu^{II} + 2H_2O - 5H]^+$	$C_{12}H_{31}N_2O_4Cu$	328.9157	329.2868	0.3711
$[(M)_2 + H_2O + Cu^{II} + Cu^I - 2H]^+$	$C_{12}H_{32}N_2O_3Cu_2$	377.4702	377.2573	-0.2129

2.3.3 The effect of assembly on differentiation of Leu, Ile and allo-Ile

The drift time of complex ions of Leu, Ile and allo-Ile were measured and their CCSs were calculated (Table 2-8). It was found that higher assembly led to larger IM separations between Leu and Ile, allo-Ile and Ile, Leu and allo-Ile. As shown in Figure 2-15, the drift time difference was increased from 0 to 0.9 ms, 0.1 to 0.5 ms and 0 to 0.4 ms between the assemblies of Leu and Ile, the assemblies of allo-Ile and Ile, and the assemblies of Leu and allo-Ile, respectively. To describe the discrimination, the peak-to-peak resolution (R_{p-p}) was calculated using equation as described in the literature,³⁰ where t_{d1} and t_{d2} , and W_{b1} and W_{b2} are the drift times and peak widths of the two separated ions.

$$R_{p-p} = \frac{2(t_{d2} - t_{d1})}{W_{b1} + W_{b2}} \quad (1)$$

The R_{p-p} values for the assembly ions of allo-Ile and Ile were 0, 0, 0.234 and 0.401 for $[(M) + Cu^I]^+$, $[(M)_2 + Cu^{II} - H]^+$, $[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$ and $[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$, respectively, indicating the improved differentiation of isomers with the higher assemblies.

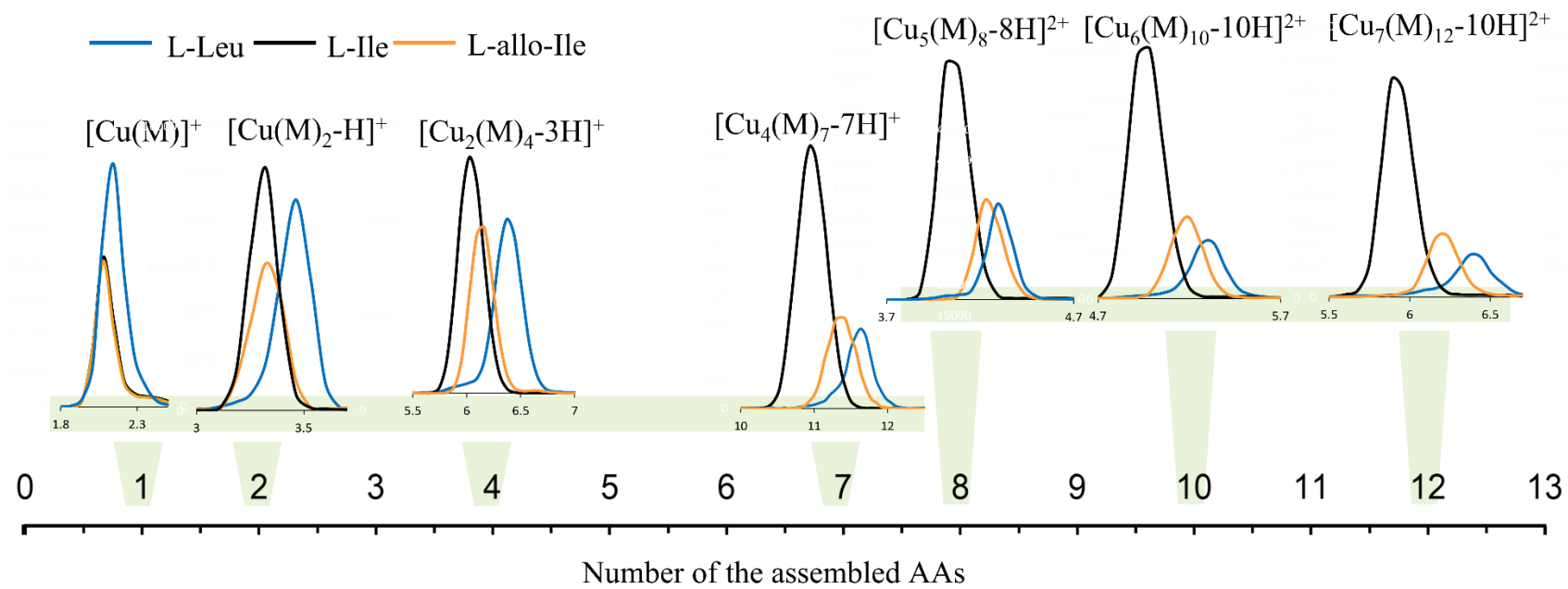


Figure 2-15. The effect of self-assembly on the differences of drift time among L-Leu, L-Ile and L-allo-Ile as observed in IM-MS.

The CCS values of complex ions at the same assembly degree were compared between Leu and Ile, allo-Ile and Ile, as shown in Table 2-8 and Figure 2-16. It was found that the difference of the isomer assemblies in CCSs became larger with the increased number (n) of the assembled amino acids. Note that the Δ CCS before any assembly, or initial Δ CCS, for Leu vs. Ile was about 1.6 Å²; for allo-Ile vs. Ile was about 0.6 Å²; and for L-form vs. D-form was 0 Å² in N₂, respectively.⁸⁴ From n = 1 to 12, the CCS difference (Δ CCS) was increased from 1.6 to 15.2 Å² and from 0.6 to 10.2 Å² between the assemblies of Leu and Ile and between the assemblies of allo-Ile and Ile, respectively. The Δ CCSs between isomers with other additive metal ions were also compared. As shown in Table 2-8, from n = 1 to 8, 10 and 10 for ZnCl₂, MgCl₂ and NiCl₂ respectively, the Δ CCSs between the assemblies of Leu and Ile were increased from 1.6 to 9.5, 14.0 and 18.6 Å², and the Δ CCS between the assemblies of allo-Ile and Ile were increased from 0.6 to 4.7, 9.1 and 9.1 Å². The results showed that copper ions allowed the best discrimination.

As shown in Figure 2-16, the slope of Δ CCS against the assembly number showed the trend: Leu vs. Ile > allo-Ile vs. Ile > Leu vs. allo-Ile > L-form vs. D-form, which is the same for differences between the individual isomers, indicating that the enhanced differences were probably originated from the accumulation of initial structural

differences of the AAs. For the isomers with a larger initial ΔCCS , the assembly number n needed for a significant differentiation could be smaller, and vice versa. If $\Delta\text{CCS} > 4 \text{ \AA}^2$ is set to indicate the differentiation of isomers based on the measurement errors of 0.11 ms for drift time and thus $2 \text{ \AA}^2$ for CCS, it could be achieved with $n = 3$ for Leu vs. Ile, and $n = 8$ for allo-Ile vs Ile could exceed $4 \text{ \AA}^2$. This further supported the notion that the accumulation of intrinsic differences of the assembly units led to the enhanced differentiation of isomers by assembly.

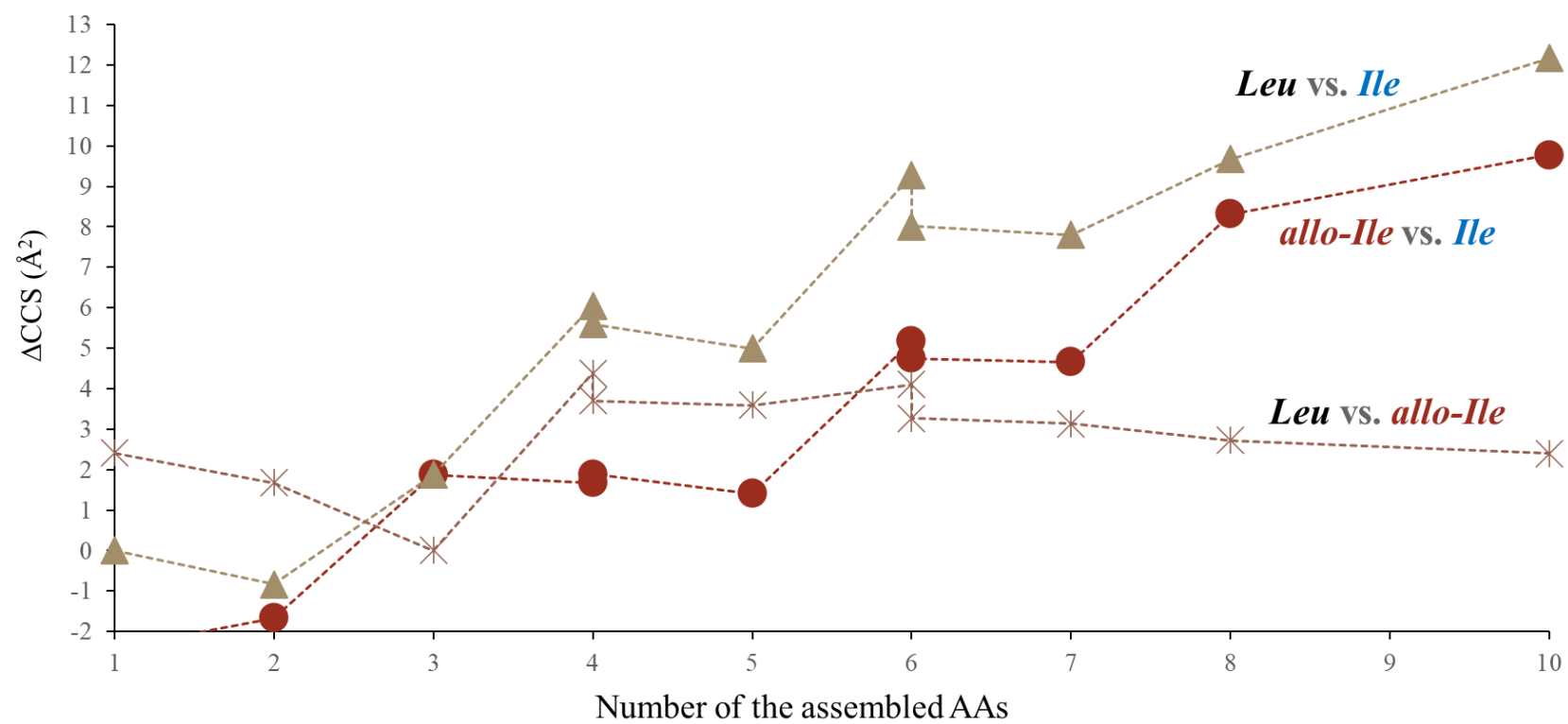


Figure 2-16. The effect of self-assembly on ΔCCS between Leu vs. Ile, *allo-Ile* vs. Ile and Leu vs. *allo-Ile*.

Table 2-8. Summary of the measured CCSs and t_d for copper bound assembly ions of L-Leu, L-Ile and L-allo-Ile (n = 3 intraday measurements).

Composition	Leu				Ile				allo-Ile			
	L		D		L		D		L		D	
	t_d [ms]	CCS _{exp} (Ω_{He}) [Å ²]	t_d [ms]	CCS _{exp} (Ω_{He}) [Å ²]	t_d [ms]	CCS _{exp} (Ω_{He}) [Å ²]	t_d [ms]	CCS _{exp} (Ω_{He}) [Å ²]	t_d [ms]	CCS _{exp} (Ω_{He}) [Å ²]	t_d [ms]	CCS _{exp} (Ω_{He}) [Å ²]
[(M) + H] ⁺	2.43	66.2	2.43	66.2	2.43	66.2	2.43	66.2	2.43	66.2	2.43	66.2
[(M) + Cu ^I] ⁺	2.76	72.1	2.76	72.1	2.76	72.1	2.76	72.1	2.76	72.1	2.76	72.1
[(M) ₂ + Cu ^{II} – H] ⁺	4.52	110.0	4.52	110.0	4.30	105.1	4.30	105.1	4.41	107.5	4.41	107.5
[(M) ₂ + (Cu ^I) ₂ – H] ⁺	4.96	118.8	4.96	118.8	4.85	116.4	4.85	116.4	4.96	118.8	4.96	118.8
[(M) ₃ + (Cu ^{II}) ₂ – 3H] ⁺	6.84	157.7	6.84±0.11	156.5±0.9	6.39	148.2	6.39	148.2	6.50	150.5	6.50	150.5
[(M) ₄ + (Cu ^{II}) ₂ – 3H] ⁺	7.83	184.0	7.83±0.11	182.9±0.9	7.72	175.0	7.83	175.0	7.83	177.3	7.83±0.11	178.4±0.9
[(M) ₄ + (Cu ^{II}) ₂ + Cu ^I – 4H] ⁺	8.49±0.11	189.3±0.9	8.60	192.6	8.05	181.4	8.16	183.7	8.27	185.9	8.27±0.11	187.0±0.9
[(M) ₄ + (Cu ^{II}) ₂ + (Cu ^I) ₂ – 4H] ²⁺	3.64	176.3	3.64	176.3	3.53	171.4	3.53	171.4	3.53	171.4	3.53	171.4
[(M) ₅ + (Cu ^{II}) ₃ – 5H] ⁺	10.80	236.1	10.80±0.11	235.0±1.1	10.36	227.4	10.36±0.11	226.3±1.2	10.46	228.5	10.47	229.6
[(M) ₆ + (Cu ^{II}) ₃ + Na – 6H] ⁺	12.90±0.1	275.7±1.1	13.01	278.8	12.02±0.11	258.4±1.1	12.02±0.11	258.4±1.3	12.46	268.1	12.46	268.1
[(M) ₆ + (Cu ^{II}) ₄ – 6H] ²⁺	4.41	208.8	4.41	208.8	4.08±0.11	196.9±1.2	4.08±0.11	196.9±1.2	4.30	204.1	4.30	204.1
[(M) ₇ + (Cu ^{II}) ₄ – 7H] ⁺	14.66	310.0	14.66	310.0	13.67	291.0	13.67±0.11	293.1±1.2	14.11	299.5	14.11	299.5
[(M) ₈ + (Cu ^{II}) ₅ – 8H] ²⁺	5.62	259.4	5.62	259.4	5.29	245.5	5.29	245.5	5.51	254.7	5.51	254.7
[(M) ₁₀ + (Cu ^{II}) ₆ – 10H] ²⁺	6.84	309.3	6.84±0.11	307.1±2.0	6.39±0.11	293.0±1.9	6.39±0.11	293.0±2.0	6.73	304.8	6.73	304.8
[(M) ₁₂ + (Cu ^{II}) ₆ + 2Na – 18H] ²⁺	8.05	358.4	8.05	358.4	7.5±0.11	338.3±2.0	7.5±0.11	338.3±2.0	7.83	349.5	7.83	349.5
[(M) ₁₂ + (Cu ^{II}) ₇ – 12H] ²⁺	8.05	358.1	8.05	358.1	7.61	340.2	7.61	340.2	7.94	353.6	7.94	353.6
[(M) + Zn ^I] ⁺	2.54	66.9	2.54	66.9	2.54	66.9	2.54	66.9	2.54	66.9	2.54	66.9
[(M) ₂ + Zn ^{II} – H] ⁺	4.52	109.9	4.63	112.4	4.41	107.5	4.41	107.5	4.41	107.5	4.41	107.5

$[(M)_3 + (Zn^{II})_2 - 3H]^+$	6.50	150.5	6.50	150.5	6.28±0.11	145.9±0.8	6.28	145.9	6.28	145.9	6.28	145.9
$[(M)_4 + (Zn^{II})_2 - 3H]^+$	8.27	186.3	8.38	188.5	8.05	181.8	8.16	184.0	8.16	184.0	8.16	184.0
$[(M)_6 + (Zn^{II})_4 - 6H]^{2+}$	4.41	208.8	4.41	208.8	4.19	199.3	4.19	199.3	4.3±0.07	204.0±0.8	4.30	204.0
$[(M) + Ni^{II} - H]^+$	2.43	64.4	2.43	64.4	2.43	64.4	2.43	64.4	2.43	64.4	2.43	64.4
$[(M)_2 + Ni^{II} - H]^+$	4.41	107.6	4.41	107.6	4.3	105.2	4.3	105.2	4.30	105.2	4.3	105.2
$[(M)_3 + (Ni^{II})_2 - 3H]^+$	6.61	152.9	6.61	152.9	6.28	146.0	6.28	146.0	6.28	146.0	6.28	146.0
$[(M)_4 + (Ni^{II})_2 - 3H]^+$	8.38	188.6	8.38	188.6	8.05	181.8	8.05	181.8	8.05	181.8	8.05	181.8
$[(M)_4 + (Ni^{II})_3 - 4H]^{2+}$	3.31	162.0	3.31	162.0	3.20	157.1	3.20	157.1	3.20	157.1	3.20	157.1
$[(M)_5 + (Ni^{II})_3 - 5H]^+$	10.47	229.7	10.58	231.8	10.14	223.1	10.14	223.1	10.25	225.3	10.25	225.3
$[(M)_5 + (Ni^{II})_3 - 4H]^{2+}$	3.86	185.7	3.86	185.7	3.75	180.9	3.75	180.9	3.75	180.9	3.75	180.9
$[(M)_6 + (Ni^{II})_4 - 6H]^{2+}$	4.30	204.1	4.30	204.1	4.08	194.6	4.08	194.6	4.19	199.4	4.19	199.4
$[(M)_9 + (Ni^{II})_5 - 9H]^{2+}$	5.73	263.7	5.84	268.3	5.40	249.9	5.4±0.07	249.9±0.9	5.62	259.1	5.62	259.1
$[(M)_{10} + (Ni^{II})_6 - 10H]^+$	6.84	309.4	6.84	309.4	6.39	290.8	6.39	290.8	6.61	299.9	6.61	299.9
$[(M) + Mg^{II} - H]^+$	2.76	73.4	2.76	73.4	2.76	73.4	2.76	73.4	2.76	73.4	2.76	73.4
$[(M)_2 + Mg^{II} - H]^+$	4.63	113.0	4.74	115.5	4.52	110.6	4.52	110.6	4.52	110.6	4.52	110.6
$[(M)_3 + Mg^{II} - H]^+$	6.17	144.5	6.17	144.5	6.06	142.2	6.06	142.2	6.06	142.2	6.06	142.2
$[(M)_3 + (Mg^{II})_2 - 3H]^+$	6.61	153.6	6.61	153.6	6.28	146.6	6.39	149.0	6.39	149.0	6.39	149.0
$[(M)_4 + (Mg^{II})_2 - 3H]^+$	8.49	191.4	8.49±0.07	191.4±1.2	8.16	184.6	8.16	184.6	8.27	186.9	8.27	186.9
$[(M)_4 + (Mg^{II})_2 - 2H]^{2+}$	3.31	162.8	3.31	162.8	3.20	157.9	3.20	157.9	3.31	162.8	3.31	162.8
$[(M)_5 + (Mg^{II})_3 - 5H]^+$	10.47	230.2	10.47	230.2	9.92	219.2	10.03	221.4	10.14	223.6	10.14	223.6
$[(M)_6 + (Mg^{II})_4 - 6H]^+$	4.30	204.6	4.30	204.6	4.08	195.1	4.08	195.1	4.19	199.9	4.19	199.9
$[(M)_8 + (Mg^{II})_5 - 8H]^+$	5.62	259.9	5.62	259.9	5.18	241.3	5.18	241.3	5.40	250.6	5.40	250.6
$[(M)_{10} + (Mg^{II})_6 - 10H]^+$	6.73	305.3	6.73	305.3	6.39	291.3	6.39	291.3	6.61	300.4	6.61	300.4

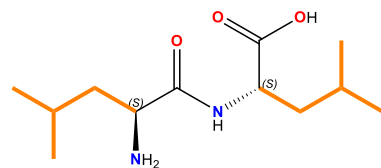
2.3.4 Investigation on the structures of the assembly ions

Recent IM-MS on multimerization of several polymers and amyloid- β peptide^{48, 104-107, 29} revealed two types of assembly mode for the shapes of assembly products, i.e., isotropic assembly in which the final assemblies are of spherical shape or near-spherical shape, and linear assembly in which the final assemblies are of fibrillar, helical, cylindrical or other anisotropic shape,^{105, 108} and the assembly mode could be described by $\Omega_n = \Omega_1 n^{\text{power}}$, where Ω_1 embodies the added mean CCS volume at adding a monomer unit and n is multimerization degree. For example, in one of the studies¹⁰⁹ with all the CCSs upon n fitted, a parameter power close to 2/3 was obtained, indicating that the assemblies grew to spherical shapes or near-spherical shapes, i.e., isotropic assembly. If a parameter power close to 1 is obtained after the fitting, an assembly into anisotropic shape like fibril, helices or cylinder is indicated.

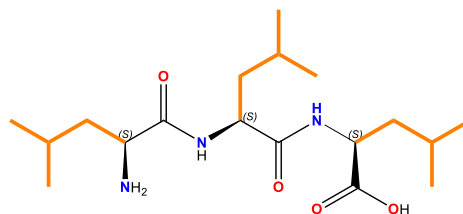
2.3.4.1 Prediction model for the shapes of the assembly ions

The presence of amino ($-\text{NH}_2$) and carboxyl ($-\text{COOH}$) groups makes long peptides or short peptide assemblies prone to form linear assemblies as fibrillar or helical shapes in solutions.^{80, 109} In our study, the assembly was driven by metal ions under ESI to form the complex ions in gas phase, and the assembly mode might be different from those in solutions. Thus, we established a prediction model according to the equation mentioned above to predict the shapes of the assemblies. For establishment of the prediction model,

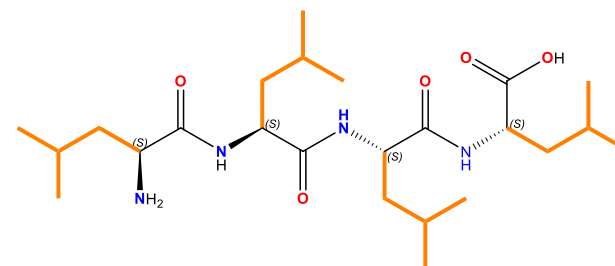
CCSs of linear and cyclic peptides were used for the training. These peptides (e.g., Figures 2-17 to 2-23) were designed to contain only one kind of AAs with or without rigid chain, i.e., L-Phe, L-Leu, L-Ile, L-allo-Ile, L-Pro, L-Ala and L-Gly, and thus the increment residue, i.e., unit, could be consistently obtained and compared in these linear and cyclic peptides. The theoretical CCS values from the perfect linear and cyclic conformations were calculated and summarized in Table 2-9. The fitting equation for the linear peptides was a linear function, $y=ax + b$, and the equation for cyclic peptides was an exponential function, $y=cx^{0.66}$, with a , b and c as the constants. After comparing the constants and theoretical CCSs of the amino acids and Δ_{unit} (for AA minus H_2O), we noticed that the values of constant c for cyclic peptides and a for linear peptides were close to the CCS of additive unit (AA minus H_2O) and the values of constant b for linear peptides were close to the CCS of H_2O loss in peptide bond formation.



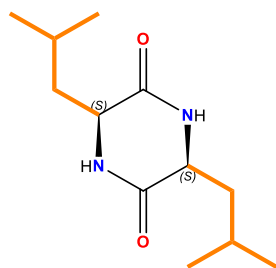
L-Leu-L-Leu (linear)



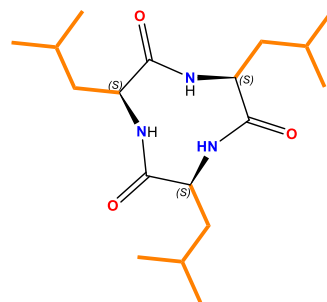
L-Leu-L-Leu-L-Leu (linear)



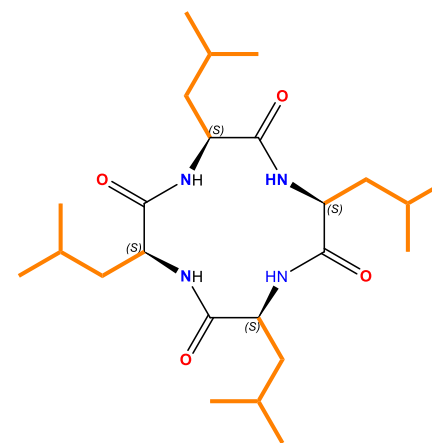
L-Leu-L-Leu-L-Leu-L-Leu (linear)



L-Leu-L-Leu (cyclic)

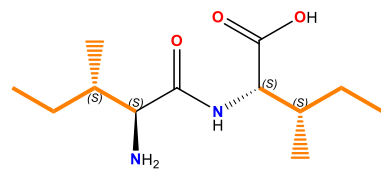


L-Leu-L-Leu-L-Leu (cyclic)

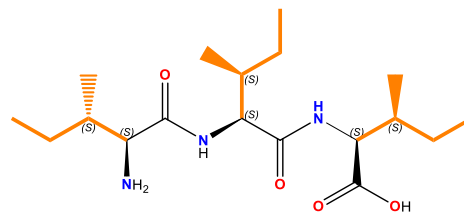


L-Leu-L-Leu-L-Leu-L-Leu (cyclic)

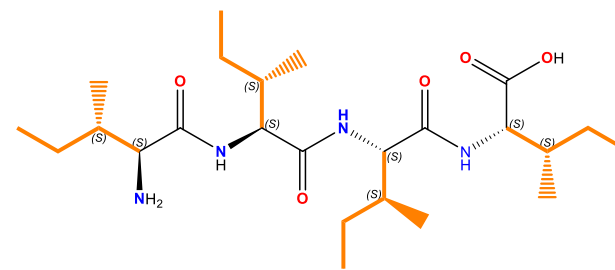
Figure 2-17. Linear and cyclic di-, tri- and tetra-peptides of L-Leu.



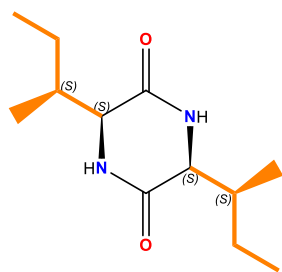
L-Ile-L-Ile (linear)



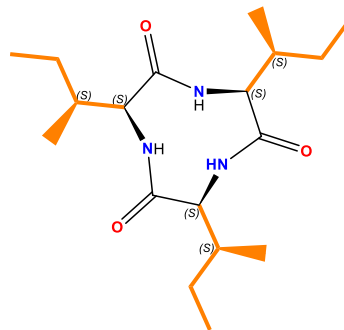
L-Ile-L-Ile-L-Ile (linear)



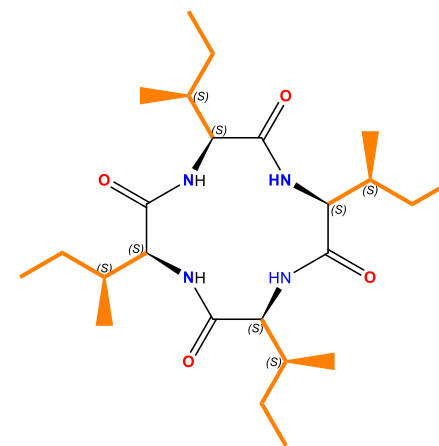
L-Ile-L-Ile-L-Ile-L-Ile (linear)



L-Ile-L-Ile (cyclic)

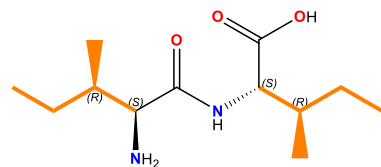


L-Ile-L-Ile-L-Ile (cyclic)

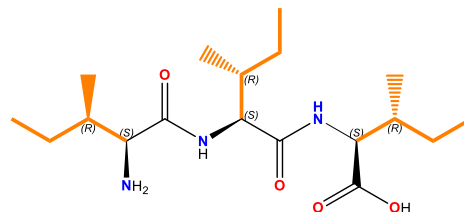


L-Ile-L-Ile-L-Ile-L-Ile (cyclic)

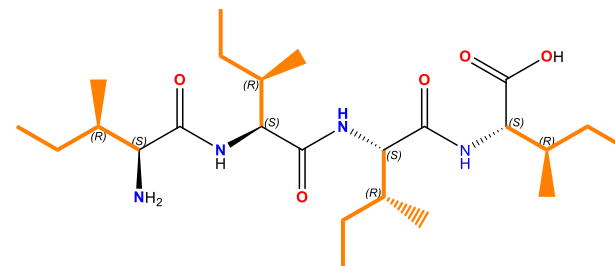
Figure 2-18. Linear and cyclic di-, tri- and tetra-peptides of L-Ile.



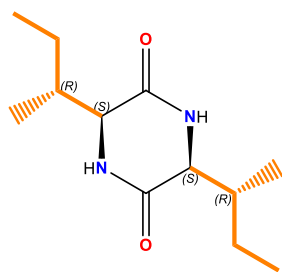
L-allo-Ile-L-allo-Ile (linear)



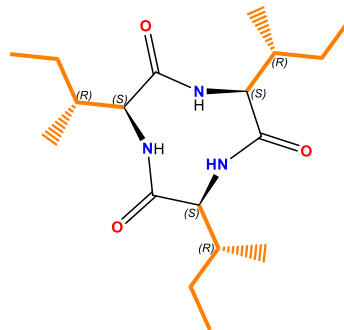
L-allo-Ile-L-allo-Ile-L-allo-Ile (linear)



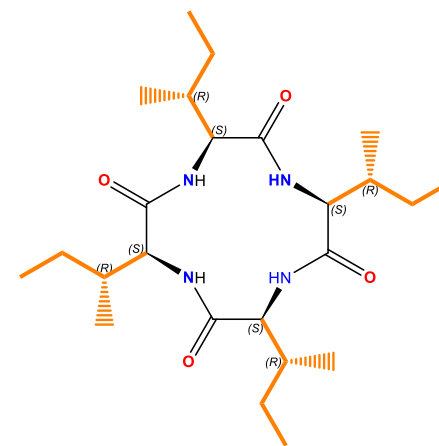
L-allo-Ile-L-allo-Ile-L-allo-Ile-L-allo-Ile (linear)



L-allo-Ile-L-allo-Ile (cyclic)

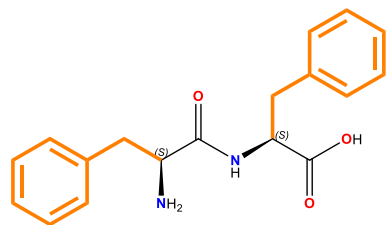


L-allo-Ile-L-allo-Ile-L-allo-Ile (cyclic)

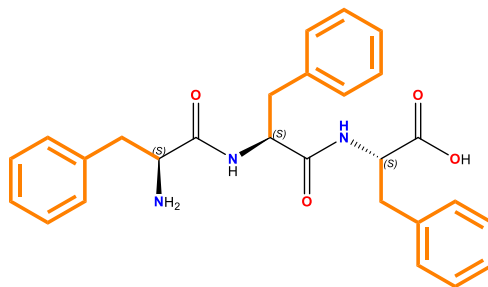


L-allo-Ile-L-allo-Ile-L-allo-Ile-L-allo-Ile (cyclic)

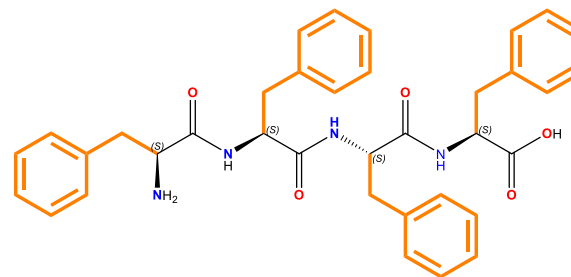
Figure 2-19. Linear and cyclic di-, tri- and tetra-peptides of L-allo-Ile.



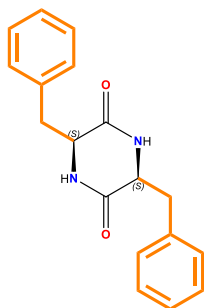
L-Phe-L-Phe (linear)



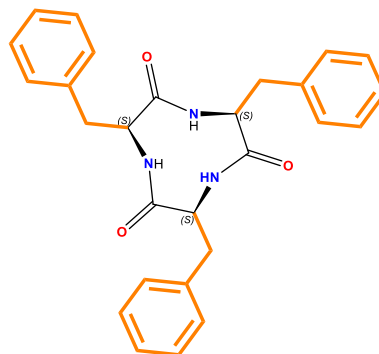
L-Phe-L-Phe-L-Phe (linear)



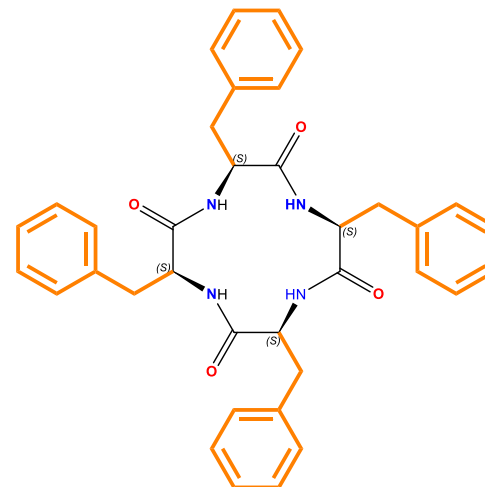
L-Phe-L-Phe-L-Phe-L-Phe (linear)



L-Phe-L-Phe (cyclic)

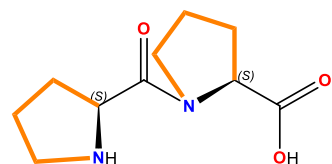


L-Phe-L-Phe-L-Phe (cyclic)

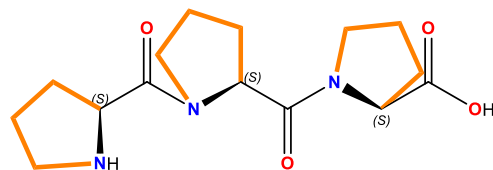


L-Phe-L-Phe-L-Phe-L-Phe (cyclic)

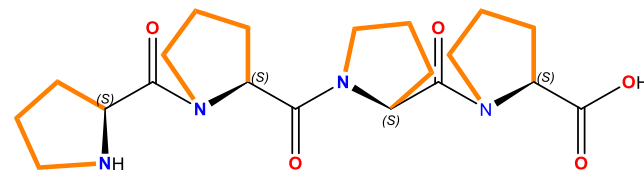
Figure 2-20. Linear and cyclic di-, tri- and tetra-peptides of L-Phe.



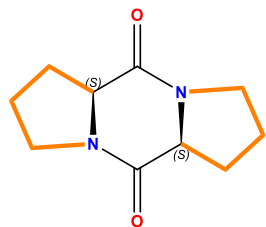
L-Pro-L-Pro (linear)



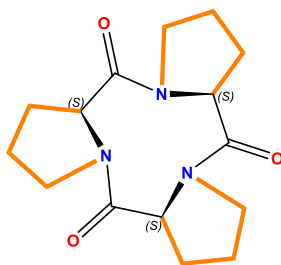
L-Pro-L-Pro-L-Pro (linear)



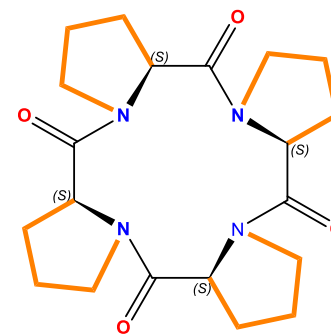
L-Pro-L-Pro-L-Pro-L-Pro (linear)



L-Pro-L-Pro (cyclic)

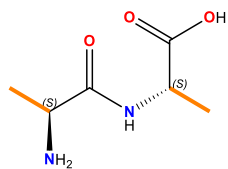


L-Pro-L-Pro-L-Pro (cyclic)

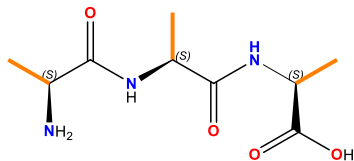


L-Pro-L-Pro-L-Pro-L-Pro (cyclic)

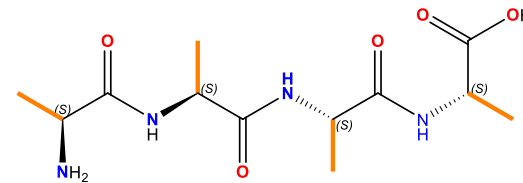
Figure 2-21. Linear and cyclic di-, tri- and tetra-peptides of L-Pro.



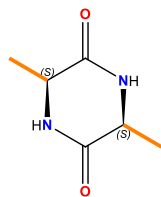
L-Ala-L-Ala (linear)



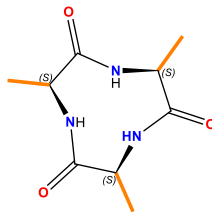
L-Ala-L-Ala-L-Ala (linear)



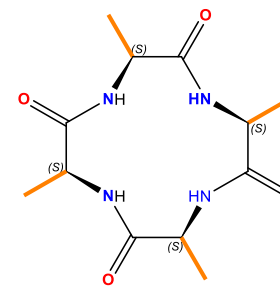
L-Ala-L-Ala-L-Ala-L-Ala (linear)



L-Ala-L-Ala (cyclic)

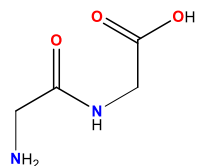


L-Ala-L-Ala-L-Ala (cyclic)

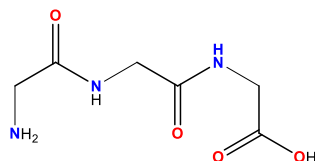


L-Ala-L-Ala-L-Ala-L-Ala (cyclic)

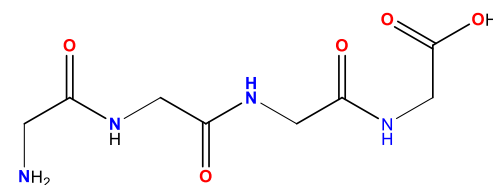
Figure 2-22. Linear and cyclic di-, tri- and tetra-peptides of L-Ala



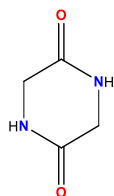
L-Gly-L-Gly (linear)



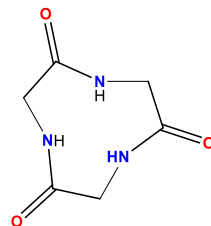
L-Gly-L-Gly-L-Gly (linear)



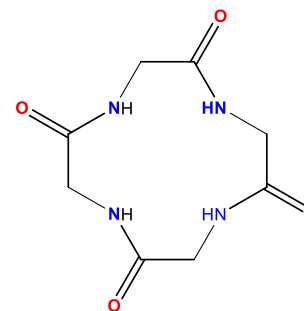
L-Gly-L-Gly-L-Gly-L-Gly (linear)



L-Gly-L-Gly (cyclic)



L-Gly-L-Gly-L-Gly (cyclic)



L-Gly-L-Gly-L-Gly-L-Gly (cyclic)

Figure 2-23. Linear and cyclic di-, tri- and tetra-peptides of L-Gly

Table 2-9. Theoretical CCSs and fitting equations for linear and cyclic peptides.

Amino acid in peptides	Number of AAs	Theoretical CCS		Unit (AA- H ₂ O)	Linear peptide $y=ax+b$	Cyclic peptide $y=cx^{0.66}$
		Linear	Cyclic			
L-Gly	1	40.3		C ₂ H ₃ NO	$y=19.66x+20.53$	$y=32.59x^{0.66}$
	2	59.8	49.9			
	3	79.4	63.6			
	4	99.2	77.8			
	5	118.8	89.9			
	6	138.5	106.7			
	7	158.2	115.6			
	8	177.9	138.3			
L-Ala	1	45.7		C ₃ H ₅ NO	$y=23.85x+22.31$	$y=40.34x^{0.66}$
	2	70.1	60.2			
	3	94.1	79.6			
	4	118.2	100.0			
	5	141.7	112.9			
	6	165.4	133.8			
	7	189.2	144.2			
	8	213.0	165.3			
L-Pro	1	52.4		C ₅ H ₇ NO	$y=28.19x+24.68$	$y=46.27x^{0.66}$
	2	81.3	70.8			
	3	109.6	95.1			
	4	137.5	111.0			
	5	165.7	131.6			
	6	193.9	150.0			
	7	221.9	166.1			
	8	250.1	189.9			
L-Phe	1	70.1		C ₉ H ₉ NO	$y=39.84x+31.75$	$y=75.82x^{0.66}$
	2	116.5	108.6			
	3	149.0	148.5			

	4	192.6	184.9			
	5	224.3	220.4			
	6	276.6	253.7			
	7	304.6	270.9			
	8	354.5	307.4			
L-Leu	1	61.9		C ₆ H ₁₁ NO	y=35.51x+28.26	y=64.76x ^{0.66}
	2	99.4	92.8			
	3	136.1	127.4			
	4	171.4	160.2			
	5	206.4	185.4			
	6	241.7	216.2			
	7	275.7	234.7			
	8	311.8	260.2			
L-Ile	1	60.8			y=34.89x+27.91	y=61.51x ^{0.66}
	2	98.6	90.2			
	3	133.1	123.3			
	4	168.3	151.6			
	5	203.1	175.8			
	6	237.7	203.8			
	7	272.0	220.7			
	8	305.9	249.0			
L-allo-Ile	1	60.7			y=35.19x+27.72	y=61.79x ^{0.66}
	2	98.7	90.5			
	3	134.3	122.8			
	4	169.6	150.5			
	5	204.4	175.3			
	6	239.3	204.2			
	7	273.4	221.3			
	8	308.3	253.8			

Interestingly, it could be clearly observed that the order of a and c was the same as the order for the size of the residue side chain, i.e., L-Phe > L-Leu > L-Ile and L-allo-Ile > L-Pro > L-Ala > L-Gly. Meantime, with the increasing degree of the assembly, the linear assembly showed larger and larger CCS than the corresponding cyclic assembly, as shown in Figure 2-24, allowing clear differentiation of the shapes with higher assembly. Accordingly, in this study, two standard assembly trends were included in the prediction model: standard linear assembly and standard isotropic assembly.

The constants c and a of the standard trends were set as the CCSs of $[M]^+$ or $[M + H]^+$ and $[Cu(M)]^+$ which were the main additive units in our assembly. As shown in Figure 2-25, the experimental CCSs of all the assemblies were plotted and compared with the standard trend lines. It is clearly showed that most of the data points are lower than the isotropic assembly lines, even lower than the unit that is equal to $[M]^+$ or $[M + H]^+$. The assemblies of Ala, Pro, Leu, Ile, allo-Ile and Phe were more prone to isotropic assembly, especially as compared with the assemblies of Gly. These results indicated that in general the assemblies tended to be isotropic, and the shapes of the detected assembly ions were compact spherical or near-spherical shapes.

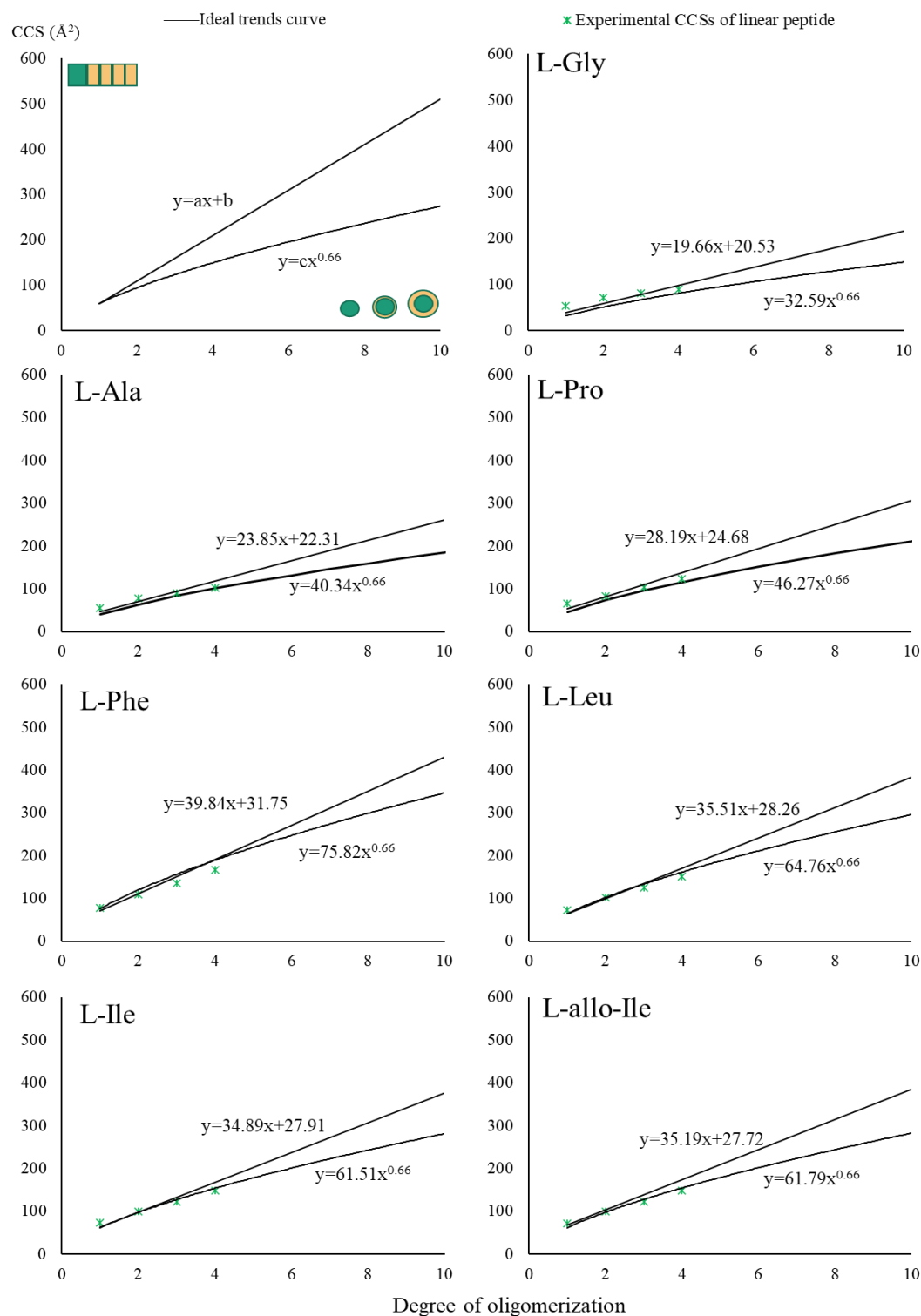


Figure 2-24. Plots of the linear and cyclic growth trend curve of theoretical CCSs for the perfect liner and cyclic peptides. Green dots are for the experimental CCSs of the linear peptides.

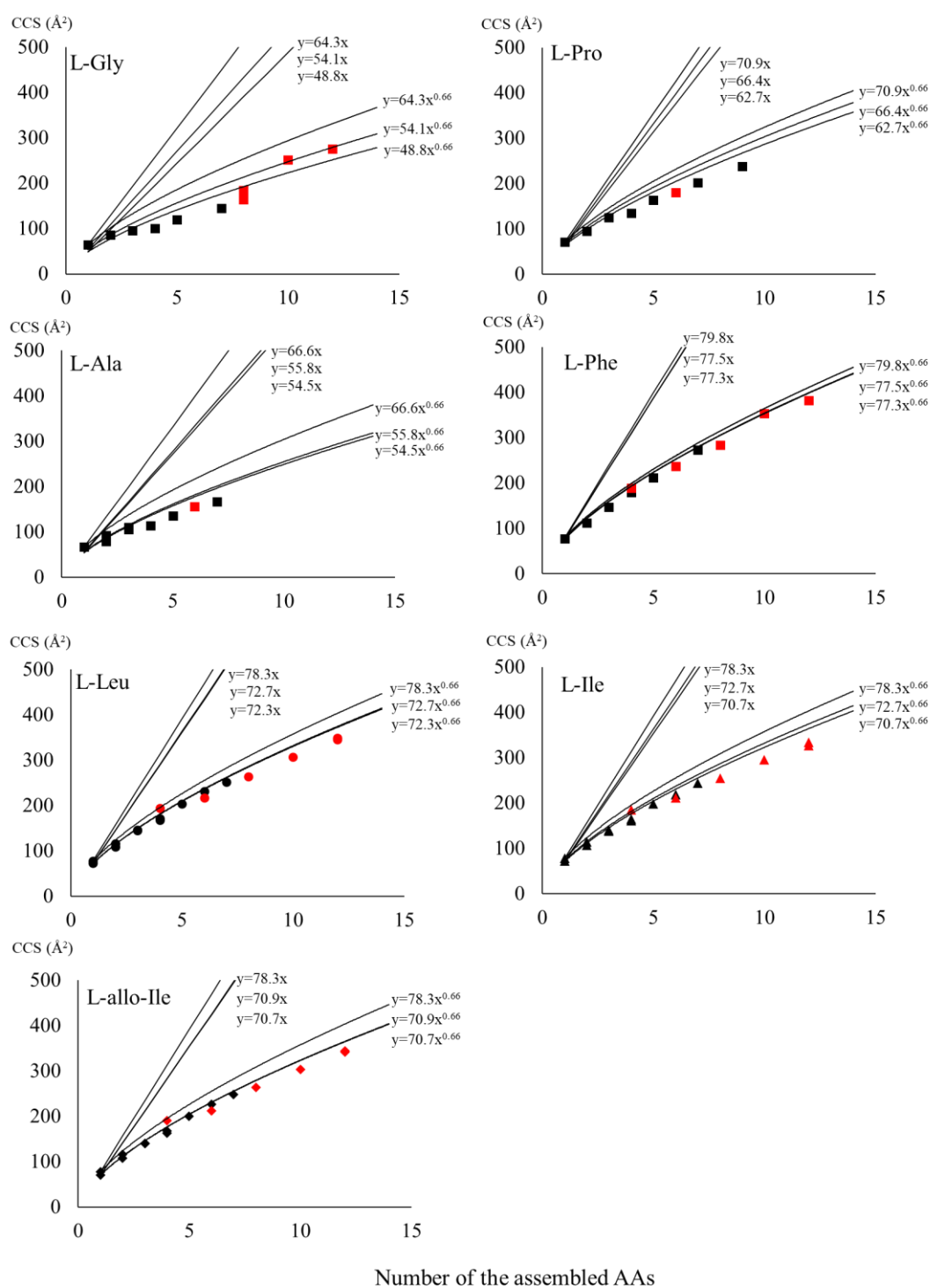


Figure 2-25. Shapes prediction model and plots of experimental CCS values versus assembly degree (n) for 7 amino acids. The determined CCS values were indicated in the dots for singly charged state (black) and doubly charged state (red). The standard linear and isotropic trends were plotted in solid lines.

2.3.4.2 The architectures of the assembly ions

There have been many studies on the architectures of lower assemblies of Cu ion and AA due to the important role of copper(II)-AA complexes in copper transportation,¹¹⁰ including cis- or trans-Cu(L-Ile)₂·H₂O, trans-Cu(L-Leu)₂·H₂O, cis-Cu(L-allo-Ile)₂·H₂O in crystal form, trans-Cu(L-Ile)₂ in solution or gas phase.¹¹¹⁻¹²⁰ The typical coordination sites are amino and carboxylate with bi or tridentate bridging modes (μ_n -N₁:O₁ or μ_n -N₁:O₁:O₂) Figure 2-26.¹²¹⁻¹²⁴ However, until now, there has been no study on the architectures of the higher assemblies of the copper bound Leu, Ile and allo-Ile.

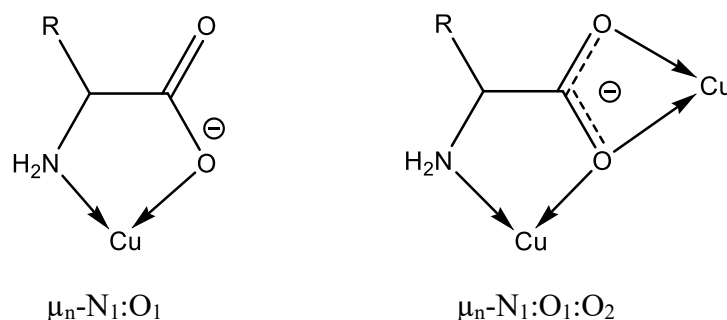


Figure 2-26. Typical coordination modes.

In this study, the structures were explored by computational molecular modeling with the assistance from structural interpretation by MS/MS dissociation. Figure 2-27 displays structures, such as $[(\text{L-Ile})_2 + \text{Cu}^{\text{II}} - \text{H}]^+$, $[(\text{L-Ile})_3 + 2\text{Cu}^{\text{II}} - 3\text{H}]^+$ and $[(\text{L-Ile})_4 + 2\text{Cu}^{\text{II}} - 3\text{H}]^+$.

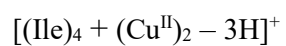
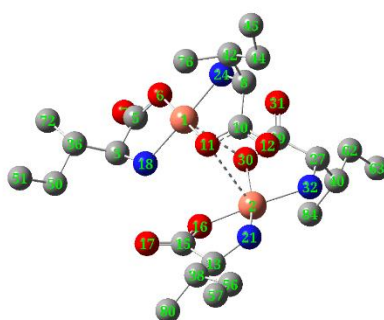
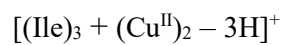
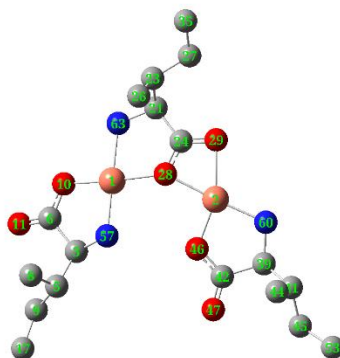
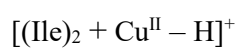
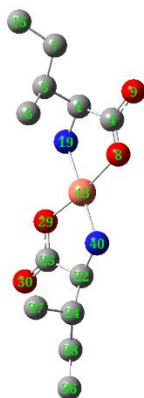


Figure 2-27. Simulated structures for assembly ions optimized at the B3LYP/6-31+G(d) level. Relative energies at 0 K, ΔH_0° , are in kJ mol^{-1} .

It could be observed that all the above structures have a common feature: the amino (–NH₂) and a carboxyl (–COOH) group with lone pair of electrons are attracted to the transition metal ions in the center and the alkyl chains are on the exterior of the structure acids protruded outwards as a hydrophobic shell. To explain why the metal ions are existing in the center, computational modelling was applied. In Figure 2-28, to exemplify, [(L-Ile)₄ + 2Cu^{II} – 3H]⁺, a complex ion that is distinguishable from the analog isomers [(L-Leu)₄ + 2Cu^{II} – 3H]⁺ and [(L-allo-Ile)₄ + 2Cu^{II} – 3H]⁺ with ΔCCS (Å²) of ≈ 4. the lower-energy geometry **1** (Δ*G*^o₂₉₈ = 0 kJ mol^{–1}; CCS_{EHSS} = 166 Å²) is more compact than the higher-energy geometry **2** (Δ*G*^o₂₉₈ = 60 kJ mol^{–1}; CCS_{EHSS} = 171 Å²). The two Cu²⁺ ions can be bridged closely together, e.g., Cu–Cu = 2.91 Å in **1**, supporting that, in this kind of constitutes, because of electrostatic and dispersion forces and hydrogen bonds, metal ions are likely to attract electron donors thus existing in the center. Therefore, we propose to regard the architecture characteristics as two parts: coordination core, comprised of the amino group, carboxyl group and metal ions, and the alkyl chains shell.

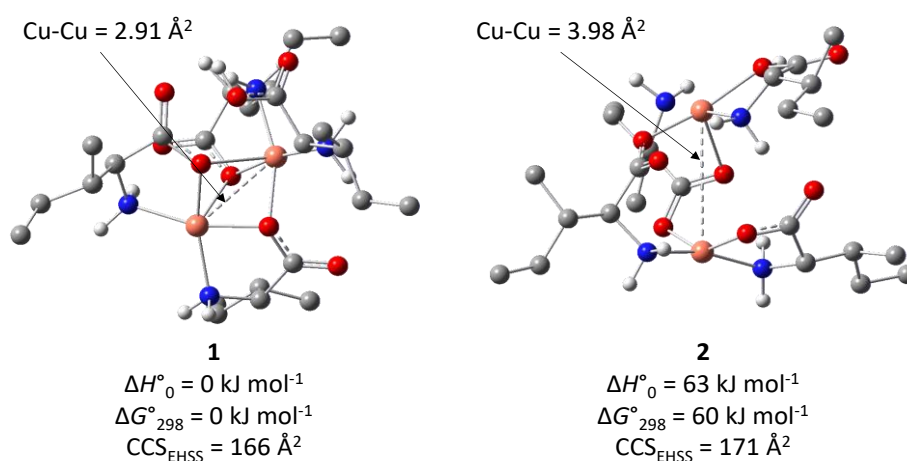


Figure 2-28. Simulated interaction between sub-assemblies.

Given that the rigid side chains predominate the exterior of the structure, we here assume that in this type of architectures the CCSs of assemblies would be minorly affected by coordination core but majorly affected by the covering chains. Several tests were designed to further confirm the assumption. First, we tested whether the concentration of metal and the type of the metal ions could affect the geometry, since the geometry of certain metallo-macromolecules (terpyridine-based) that are solely based on coordination bridges can be easily affected by the metal ion concentration and types.¹²⁵⁻¹²⁶ However, as shown in Figure 2-29, there were no drift time and CCS shifts when the concentration changed in our study. Further study with different metal ions revealed that the CCSs of the assemblies were slightly affected by the types of metal ions, as shown in Figure 2-30. The computational molecular modeling further supported our observations. As shown in Figure 2-31, the coordination of metal-AA in $[(M - H)_3 + (\text{Metal})_2]^+$ was optimized by replacing Ile to Gly and it could be observed that the 1a

geometry is with the lowest related energy after optimization with three types of metals including Ni, Zn and Cu. Under 1a geometry, the alkyl chains of Leu, Ile and allo-Ile were added back to the geometries and the theoretical CCS values were calculated and compared with measured CCSs from our experiments (Table 2-10). It could be found that, the different metal centers have little influence on the CCSs especially on the Δ CCSs between isomers in smaller complexes.

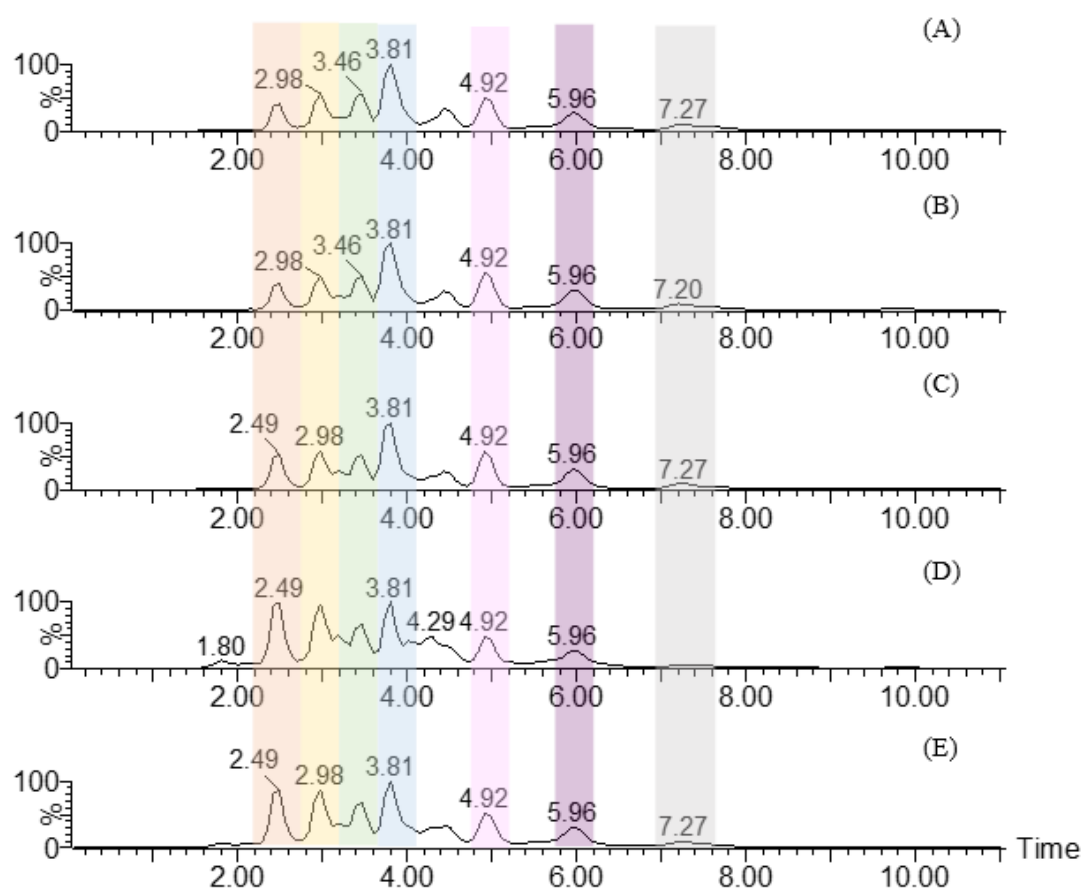


Figure 2-29. IM-MS spectra for L-Ile with addition of $\text{Cu}(\text{COOCH}_3)_2$ at concentration ratios of Ile: $\text{Cu}^{2+}=10:1$ (A) , Ile: $\text{Cu}^{2+}=5:1$ (B), Ile: $\text{Cu}^{2+}=3:1$ (C), Ile: $\text{Cu}^{2+}=2:1$ (D), Ile: $\text{Cu}^{2+}=1:1$ (E).

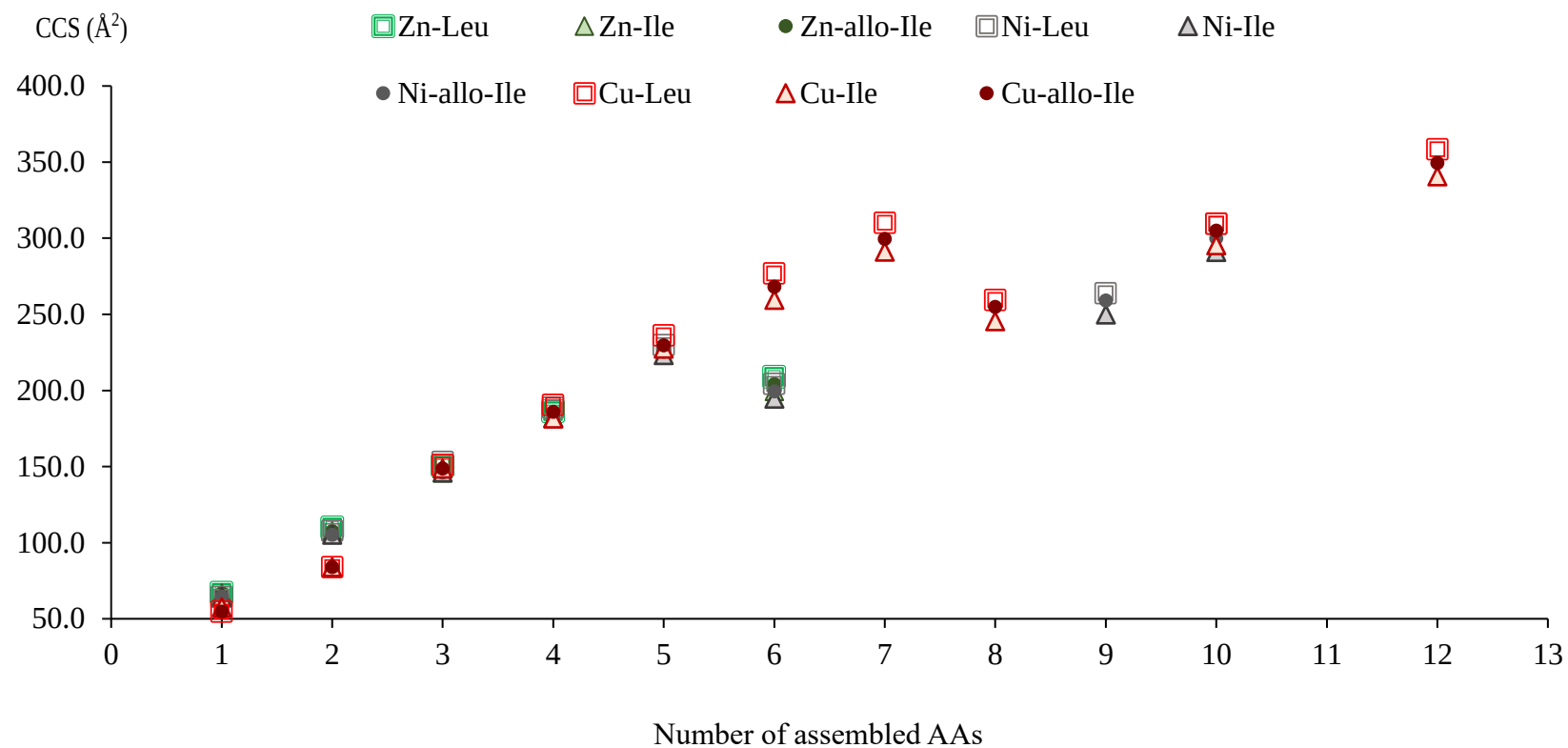


Figure 2-30. Plots of experimental CCSs vs. degree of multimerization from metal ion (Cu^{II} , Zn^{II} and Ni^{II})-AA (L-Leu, L-Ile and L-allo-Ile) complexes observed in IM-MS.

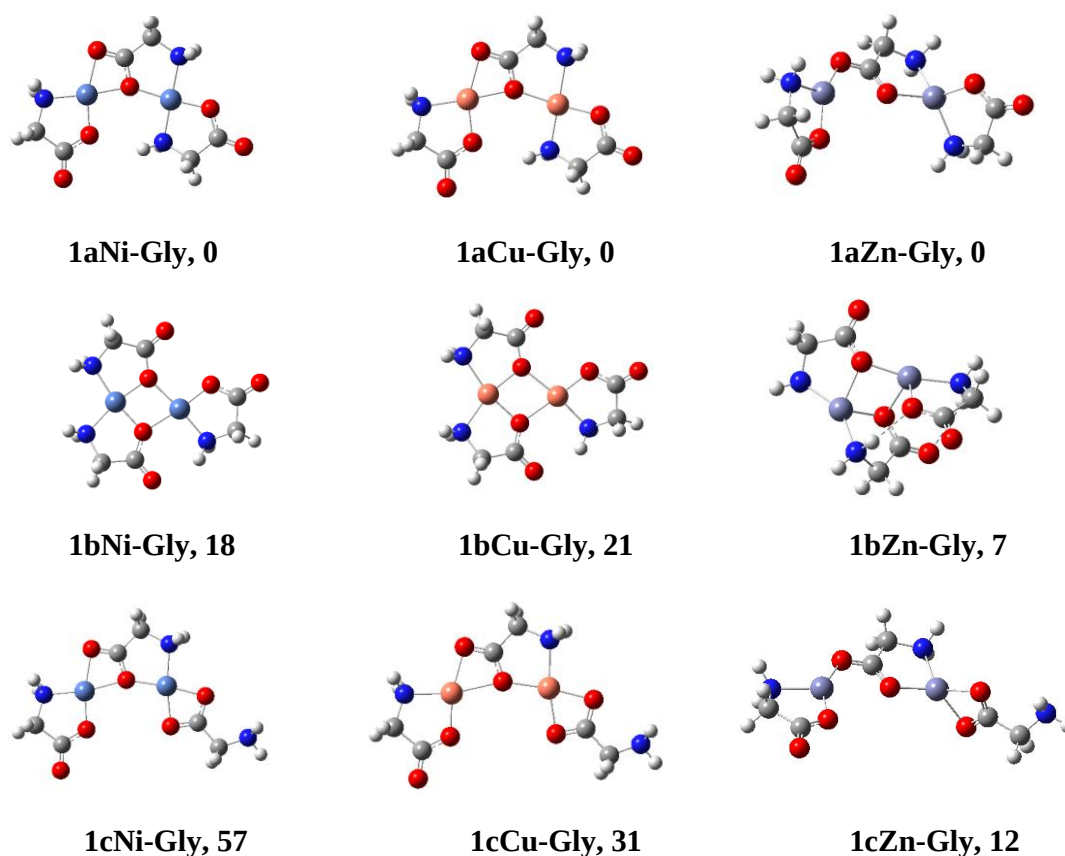


Figure 2-31. Geometries of $[(\text{Gly} - \text{H})_3 + (\text{Metal})_2]^+$ optimized at the B3LYP/6-31+G(d) level. Relative energies at 0 K ($\Delta H_0^\circ/\text{kJ mol}^{-1}$).

Table 2-10. Theoretical CCS and measured experimental CCS of complex ion $[(\text{M} - \text{H})_3 + (\text{Metal}^{\text{II}})_2]^+$ (Leu, Ile and allo-Ile) under 1a geometries.

Metals	CCS ($\text{\AA}^2$)	Leu	Ile	allo-Ile
Cu	Ω_{theo}	146.0	140.5	140.5
	Ω_{exp}	151.2	148.6	148.6
Ni	Ω_{theo}	152.9	146.0	146.0
	Ω_{exp}	150.2	147.5	148.9
Zn	Ω_{theo}	150.5	145.9	145.9
	Ω_{exp}	155.9	153.1	155.1

Second, linear peptides constituted with the same amino acids with a length of one to four of AAs (Phe, Pro, Ala, Gly) were then selected to interact with copper ion to generate new assemblies (Tables 2-11 to 2-26). According to the literature,¹²¹ the new assemblies could have different coordination geometries among AAs, dipeptides, tripeptides and tetrapeptides thus different coordination cores. However, due to the comparison is between assemblies from AA and assemblies from its own peptides, the exterior chains are the same. Figure 2-32 shows the plots of all the CCSs of the assemblies from AA and their peptides together as the z axis, with unit number as x axis and chain number as y axis. It could be clearly seen that for the plots of z versus y the CCSs trends of the AA and peptides are almost overlapping (except Gly and its peptides which have no side chain) as opposed to the plots of z versus x where the CCS trends are wide apart, indicating the number of side chains was the key factor on the shape or size of the assemblies. The results further verified our computationally predicted structures that the coordination was at the core while the side chains form the shell of the assemblies, and the measured CCS values are predominated by the shell-like alkyl chains.

Table 2-11. Summary of the measured CCSs and t_d for copper bound assembly ions of Phe (n = 3 intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	2.54	77.5
$[(M)_2 + Cu^{II} - H]^+$	4.41	112.7
$[(M)_2 + Cu^{II} + Cu^I - 2H]^+$	4.41	112.8
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	6.50	147.1
$[(M)_4 + (Cu^{II})_2 - 3H]^+$	8.60	179.1
$[(M)_4 + (Cu^{II})_2 + (Cu^I)_2 - 4H]^{2+}$	3.31±0.07	189.6±1.1
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	10.91	211.2
$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	4.52	236.8
$[(M)_7 + (Cu^{II})_4 - 7H]^+$	15.66	273.4
$[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$	5.73	283.5
$[(M)_{10} + (Cu^{II})_6 - 10H]^{2+}$	7.71	353.1
$[(M)_{12} + (Cu^{II})_7 - 12H]^{2+}$	8.60	382.6

Table 2-12. Summary of the measured CCSs and t_d for copper bound assembly ions of dipeptides Phe-Phe (n = 3 intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + H]^+$	4.30	111.6
$[(M) + Cu^I]^+$	4.08	106.7
$[(M)_2 + Cu^{II} - H]^+$	7.94	169.5
$[(M)_2 + Cu^{II} + H_2O - 2H]^+$	7.72	166.0
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	8.16	172.6
$[(M)_2 + Cu^I + (Cu^{II})_2 - 4H]^+$	7.94	169.0
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	12.57	234.2
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	13.56	246.7
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	18.30	305.2
$[(M)_4 + (Cu^{II})_4 - 6H]^{2+}$	5.73±0.11	283.8±1.1
$[(M)_5 + (Cu^{II})_5 - 8H]^{2+}$	6.84	323.2
$[(M)_6 + (Cu^{II})_6 - 10H]^{2+}$	8.38	375.5
$[(M)_7 + (Cu^{II})_6 - 10H]^{2+}$	9.70	419.5
$[(M)_7 + (Cu^{II})_7 - 12H]^{2+}$	9.48	411.2
$[(M)_8 + (Cu^{II})_8 - 14H]^{2+}$	11.25	469.1

Table 2-13. Summary of the measured CCSs and t_d for copper bound assembly ions of tripeptides Phe-Phe-Phe ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + H]^+$	5.73	135.4
$[(M) + Cu^{II} - H]^+$	5.84	136.7
$[(M)_2 + Cu^{II} - H]^+$	12.02	227.1
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	12.57	235.3
$[(M)_2 + Cu^I + (Cu^{II})_2 - 4H]^+$	12.35	231.1
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	19.29	318.3
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	18.63	310.8
$[(M)_4 + (Cu^{II})_5 - 8H]^{2+}$	8.38 $\pm$ 0.07	375.6 $\pm$ 0.9
$[(M)_5 + (Cu^{II})_6 - 10H]^{2+}$	10.14	432.3
$[(M)_6 + (Cu^{II})_7 - 10H]^{2+}$	12.35	500.2
$[(M)_6 + (Cu^{II})_8 - 12H]^{2+}$	12.35	500.2

Table 2-14. Summary of the measured CCSs and t_d for copper bound assembly ions of tetrapeptides Phe-Phe-Phe-Phe ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I - H]^+$	8.27	174.7
$[(M)_2 + (Cu^I)_6 + 2H_2O - 3H]^{2+}$	6.28	303.6

Table 2-15. Summary of the measured CCSs and t_d for copper bound assembly ions of Pro ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	2.20	71.0
$[(M)_2 + Cu^{II} - H]^+$	3.42	94.9
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	5.18	125.8
$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	5.18	125.3
$[(M)_4 + (Cu^{II})_2 - 3H]^+$	5.73	134.5
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	7.61	164.1
$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	3.09	180.0
$[(M)_6 + (Cu^{II})_3 - 5H]^+$	8.71	180.4
$[(M)_7 + (Cu^{II})_4 - 7H]^+$	10.25	202.2
$[(M)_9 + (Cu^{II})_5 - 9H]^+$	12.90	237.9

Table 2-16. Summary of the measured CCSs and t_d for copper bound assembly ions of dipeptides Pro-Pro ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	3.20	90.7
$[(M)_2 + Cu^{II} - H]^+$	5.73	135.1
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	5.73 $\pm$ 0.11	134.7 $\pm$ 1.2
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	7.94	169.2
$[(M)_4 + (Cu^{II})_3 + H_2O - 7H]^+$	10.25	202.8

Table 2-17. Summary of the measured CCSs and t_d for copper bound assembly ions of tripeptides Pro-Pro-Pro ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	4.08	106.7
$[(M)_2 + Cu^{II} - H]^+$	7.94	169.5
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	11.25	216.3
$[(M)_5 + (Cu^{II})_3 - 7H]^+$	13.78	249.8

Table 2-18. Summary of the measured CCSs and t_d for copper bound assembly ions of tetrapeptides Pro-Pro-Pro-Pro ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I - H]^+$	5.40	129.6

Table 2-19. Summary of the measured CCSs and t_d for copper bound assembly ions of Ala ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	1.98	66.6
$[(M)_2 + Cu^I]^+$	2.65	79.7
$[(M)_2 + H_2O + Cu^I]^+$	2.87	84.1
$[(M)_2 + (Cu^I)_2 - H]^+$	3.31	92.5
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	4.30 $\pm$ 0.07	110.6 $\pm$ 0.9
$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	4.08	106.0
$[(M)_4 + (Cu^{II})_2 - 3H]^+$	4.52	113.9
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	5.84	136.1
$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	2.54	155.9
$[(M)_7 + (Cu^{II})_4 - 7H]^+$	7.83	167.1

Table 2-20. Summary of the measured CCSs and t_d for copper bound assembly ions of dipeptides Ala-Ala($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	2.54	77.6
$[(M)_2 + Cu^{II} - H]^+$	4.52	114.8
$[(M)_2 + Cu^{II} + H_2O - 2H]^+$	4.08	106.4
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	6.06 $\pm$ 0.11	139.9 $\pm$ 1.1
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	6.84	152.3
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	8.60	178.7

Table 2-21. Summary of the measured CCSs and t_d for copper bound assembly ions of tripeptides Ala-Ala-Ala ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^{II} - H]^+$	3.20	90.4
$[(M)_2 + Cu^{II} - H]^+$	5.62	133.0
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	6.28	143.7
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	8.49	177.3
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	9.26 $\pm$ 0.07	188.5 $\pm$ 1.2
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	11.91	225.5

Table 2-22. Summary of the measured CCSs and t_d for copper bound assembly ions of tetrapeptides Ala-Ala-Ala-Ala ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I - H]^+$	4.3 $\pm$ 0.07	110.9 $\pm$ 0.8
$[(M)_2 + Cu^{II} - H]^+$	7.72	166.2
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	7.28	159.0
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	11.25	216.3
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	12.02	226.2

Table 2-23. Summary of the measured CCSs and t_d for copper bound assembly ions of Gly ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	1.87	64.4
$[(M)_2 + (Cu^I)_2 - H]^+$	2.98	86.1
$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	3.53	95.8
$[(M)_4 + (Cu^{II})_2 - 3H]^+$	3.75	99.9
$[(M)_8 + (Cu^{II})_4 + (Cu^I)_4 - 8H]^{2+}$	3.20±0.07	184.3±1.3
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	4.85	119.4
$[(M)_7 + (Cu^{II})_4 - 7H]^+$	6.39	144.6
$[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$	2.76	163.8
$[(M)_{10} + (Cu^{II})_6 - 10H]^{2+}$	4.85	251.5
$[(M)_{12} + (Cu^{II})_7 - 21H]^{2+}$	5.51	276.0

Table 2-24. Summary of the measured CCSs and t_d for copper bound assembly ions of dipeptides Gly-Gly ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M)_3 + (Cu^{II})_3 + H_2O - 11H]^+$	5.73	134.4
$[(M)_3 + (Cu^{II})_4 + H_2O - 12H]^+$	5.95	137.8
$[(M)_4 + (Cu^{II})_5 + H_2O - 10H]^+$	8.27	173.8
$[(M)_8 + (Cu^{II})_6 + 2H_2O - 26H]^{2+}$	4.30±0.07	229.0±0.9
$[(M)_8 + (Cu^{II})_8 + 2H_2O - 20H]^{2+}$	4.41	233.1
$[(M)_8 + (Cu^{II})_9 + 2H_2O - 20H]^{2+}$	4.74	246.0

Table 2-25. Summary of the measured CCSs and t_d for copper bound assembly ions of tripeptides Gly-Gly-Gly ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^{II} - H]^+$	2.76	81.9
$[(M)_2 + Cu^{II} - H]^+$	4.41	112.2
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	5.07 ± 0.11	123.6 ± 1.2
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	6.95	153.7
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	8.82	182.2

Table 2-26. Summary of the measured CCSs and t_d for copper bound assembly ions of tetrapeptides Gly-Gly-Gly-Gly ($n = 3$ intraday measurements).

Composition	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I - H]^+$	3.53	96.9
$[(M)_2 + Cu^{II} - H]^+$	5.73	134.7
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	5.40 ± 0.11	128.7 ± 1.1
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	7.94	168.8
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	8.27	173.6

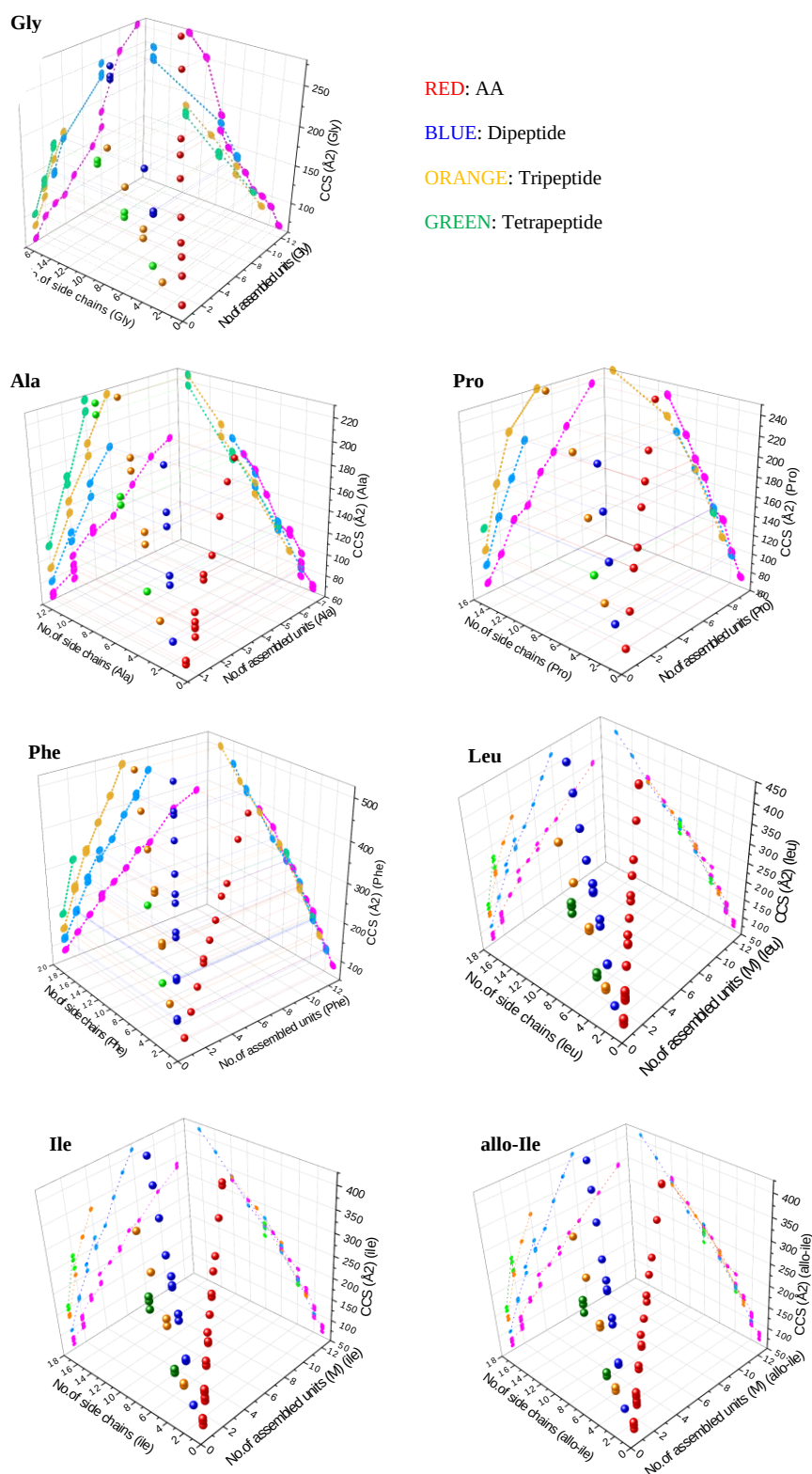


Figure 2-32. Plots of experimental CCSs vs. number of units and chains from amino acids, dipeptides, tripeptides and tetrapeptides coordinated with copper ion.

2.3.4.3 Relevance of assembly, structures and enhanced differentiation

As discussed previously, the alkyl side chains of Leu, Ile and allo-Ile are exposed to the exteriors of the assemblies. Since the alkyl chains could not form any hydrogen bonds or coordinate with metal ions, the distribution of alkyl chains is the main factor deciding the shape, size and even the concaveness of the assemblies,⁴⁹ which could be described by CCSs in IM-MS. In lower degrees of assemblies such as $[(M)_3 + (Cu^{II})_2 - 3H]^+$, alkyl side chains are farther away from each other, while in the higher degrees of assemblies $[(M)_4 + (Cu^{II})_2 - 3H]^+$, they are closer with each other.

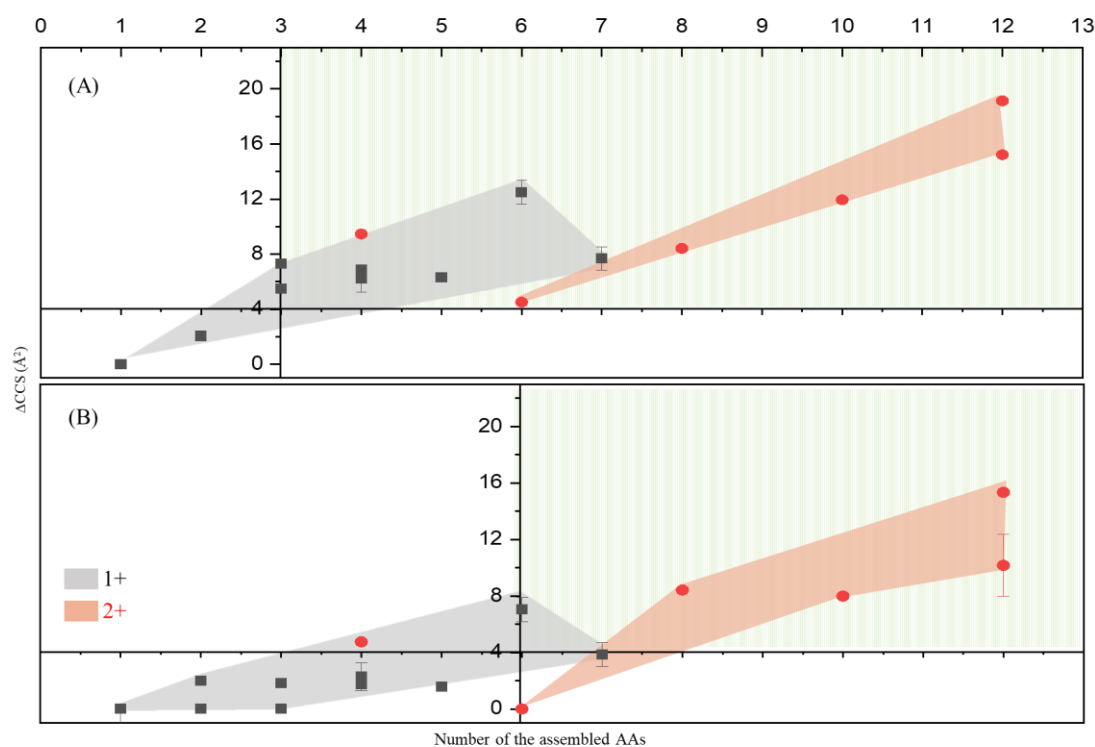


Figure 2-33. The effect of self-assembly on the ΔCCS between L-Leu and L-Ile (A) and between L-allo-Ile and L-Ile (B). Black dots are for singly charged ions and red dots for doubly charged ions.

Because most of the higher assemblies were doubly charged with shorter drift times and the lower assemblies were singly charged with longer drift times, for easier classification, we tentatively classified the singly charged ions and doubly charged ions as lower assembly group and higher assembly group, respectively. Since most of the assemblies were found to be isotropic shapes, all the experimental CCSs of the assemblies from AAs (Ile, Leu, allo-Ile, Phe, Pro, Ala and Gly) in our study were exclusively fitted into isotropic curves (Table 2-27).

Table 2-27. Fitting curves upon experimental CCSs of the assemblies of 7 amino acids.

AAs	CCS_{theo}	CCS_{exp} for [M+H]⁺	All ions	Group⁺	Group²⁺
Gly	40.36	54.12	$y=47.9*x^{0.66}$	$y=42.79*x^{0.66}$	$y=50.2*x^{0.66}$
Ala	45.41	55.78	$y=48.97*x^{0.66}$	$y=49.26*x^{0.66}$	/
Pro	51.50	66.36	$y=56.48*x^{0.66}$	$y=56.69*x^{0.66}$	/
Phe	68.76	77.32	$y=74.13*x^{0.66}$	$y=73.42*x^{0.66}$	$y=74.47*x^{0.66}$
Leu	62.01	72.97	$y=68.71*x^{0.66}$	$y=70.25*x^{0.66}$	$y=67.85*x^{0.66}$
Ile	60.70	72.41	$y=65.98*x^{0.66}$	$y=67.71*x^{0.66}$	$y=65.00*x^{0.66}$
Allo	60.61	71.2	$y=67.60*x^{0.66}$	$y=68.65*x^{0.66}$	$y=67.00*x^{0.66}$

The fitting curves were isolated and re-plotted as two groups based on the assembly level (Figure 2-34). The obtained c_{fitting} in each fitting equation was the average CCS of the additive unit. Notably, the curve for allo-Ile was very close to Ile's curve, and both of them were below the curve for Leu in the lower assembly group, which was consistent with the CCS order of the monomer units ($\text{Leu} > \text{Ile} \approx \text{allo-Ile}$). If the assembly trends were following the curves of the lower assembly group, $n \geq 2$ for Leu vs. Ile and $n \geq 9$ for allo-Ile vs. Ile would be required for achieving the significant difference (at least $4 \text{ \AA}^2$) between the isomers. However, for the higher assembly group, the order was found to be: $\text{Leu} \approx \text{allo-Ile} > \text{Ile}$, and if the assembly trends were following these curves, the required assembly level for achieving the significant difference was still $n \geq 2$ for Leu vs. Ile but only $n \geq 3$ for allo-Ile vs. Ile. Obviously, the difference between Ile with *sec*-butyl side chain and Leu with isobutyl side chain could be quickly accumulated starting from the lower assemblies, while the significant difference between Ile with (S)-amino-(S)-*sec*-butyl chain and allo-Ile with (S)-amino-(R)-*sec*-butyl chain was only appeared in higher assemblies where they were distributed with higher density.

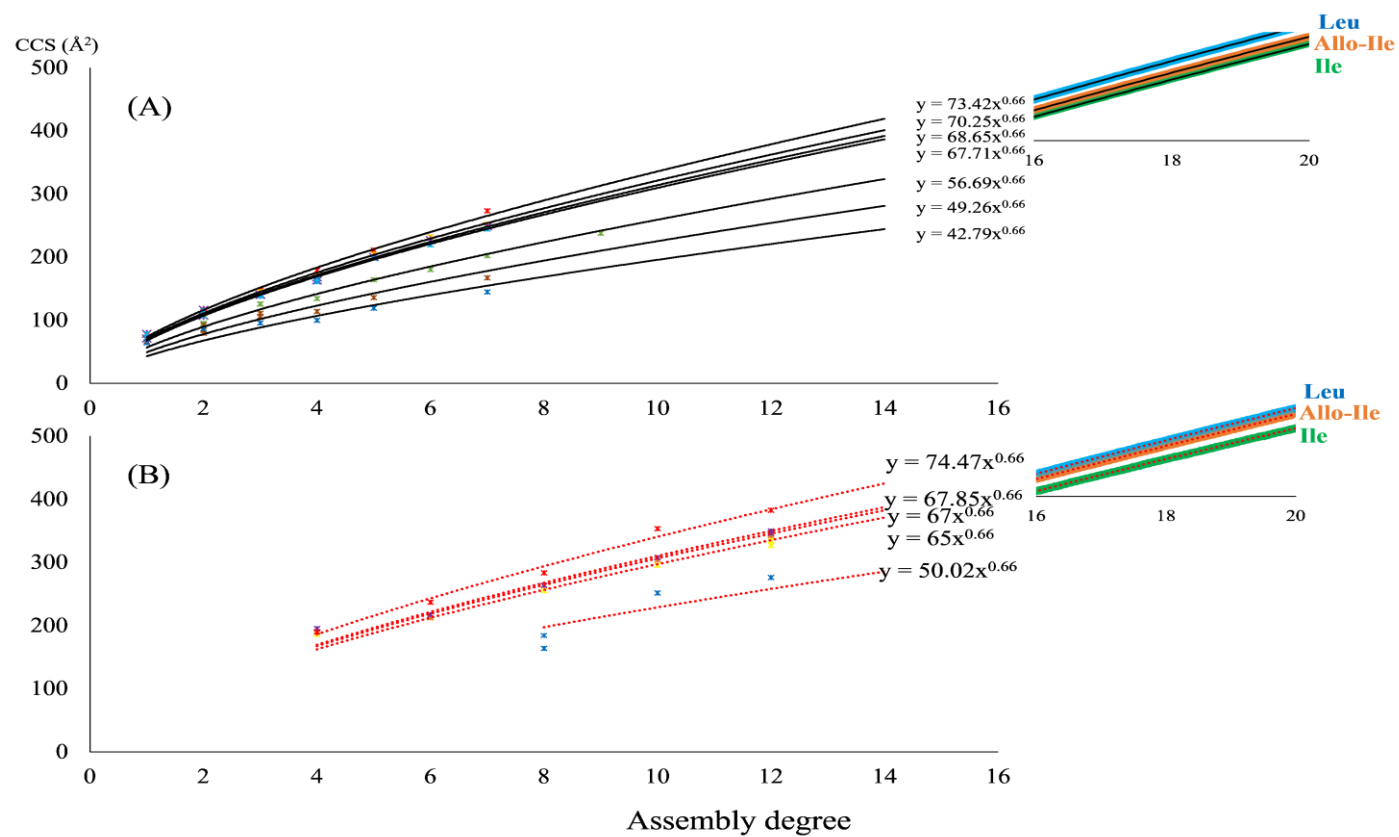


Figure 2-34. Isolated fitting curves of the assembly ions of 7 amino acids. Lower assembly group (A) and higher assembly group (B) are plotted in black solid lines and red dashed lines, respectively.

In higher assembly, distribution of the alkyl side chains in the assemblies may differ between isomers due to their differences in intrinsic properties and interactions and is thus necessary to be investigated. A new term, CCS_{unit} , the CCS increment of an additional monomer unit in the assembly, which could be obtained by subtracting the CCS of $(M)_n$ from that of $(M)_{n+1}$, was employed for the discussion. As shown in Table 2-27, the $c_{fitting}$ values, i.e., the average CCS_{unit} of AAs in the assemblies, were always smaller than the experimental or theoretical CCSs which were detected or calculated for the monomers alone. This was because that in the assemblies, these monomer units would be only partially exposed and not detected as a whole one in IM. Thus, the CCS_{unit} is related with the unit location, interface area or embed depth, etc., and could be applied to reveal distribution of monomer units in the assemblies.

Additive density (AD) was defined as additive mass divided by CCS_{unit} and were calculated based on the data and shown in Figure 2-35. A larger AD indicates that the unit is less exposed in the assemblies, while a smaller AD for more exposed. For examples, one unit M in $Cu_3(M)_5^+$ showed larger AD value with Ile than that with Leu and allo-Ile, indicating that this unit was less exposed with Ile but more exposed with Leu and allo-Ile; the unit $Cu(M)_2$ in $Cu_5(M)_8^{2+}$ showed smaller AD value with allo-Ile than with Leu and Ile, indicating that this unit was more exposed with allo-Ile but less exposed with Leu and Ile. The isomers had the same coordination functional groups to

form the assemblies but different alkyl side chains to have different exposure in the assemblies of isomers, which caused discrepancy on their sizes, shapes and IM behaviors.

It could be observed that, for lower assembly group with lower density of side chains, similar *AD* values were obtained for the isomers, while the adding positions of the units exhibited more uncertainties as the size of assemblies increased; for the higher assembly group with more neighboring side chains, different *AD* values were obtained for the isomers, which demonstrated more position diversities of the units among the isomers. This result explained why the isomeric difference could be accumulated thus caused better differentiation of isomers after the assembly.

The computational modeling results further supported the idea of chains distribution. As shown in Figure 2-36, the chains distances in dimers or trimers of Leu, Ile and allo-Ile were compared through the measurement between each pair of beta, gamma and furthest carbon atoms. The measured distances in dimers, trimers and tetramers were listed in Table 2-28. It could be observed that, generally, the dimer of allo-Ile had smallest diameter, the chains distribution in trimer of Leu was more averaged-distributed than that in trimer of Ile and allo-Ile, while the chains distribution in tetramer of Leu was less averaged-distributed than that in tetramer of Ile and allo-Ile. Although

in such lower assemblies, the CCSs measured and calculated are closed to each isomer. The results indicated that due to the different steric hindrances on the isomeric alkyl chains, the chains distribution in assemblies was affected. Because the CCS is also related with the surface density of the molecular, in such type of assemblies with alkyl chains as exterior, the chains distribution will affect the final detected CCS values. Van der Waals volumes (V_w) of assemblies listed in Table 2-35 was also measured and compared. The V_w of dimers of Leu, Ile and allo-Ile were 386.8, 335.6 and 351.7 Å³, the V_w of trimers of Leu, Ile and allo-Ile were 549.2, 550.0 and 578.6 Å³, the V_w of tetramers of Leu, Ile and allo-Ile were 727.1, 775.7 and 742.9 Å³, respectively. Interestingly, the dimer of Leu showed largest V_w than that of allo-Ile then Ile, in contrast, the trimer of allo-Ile showed largest V_w than that of Leu and Ile, and the tetramer of Ile showed largest V_w than that of allo-Ile then Leu. Because V_w is related with several factors such as rotational freedom or polarization. The results need to be further interpreted to explain the detail information.

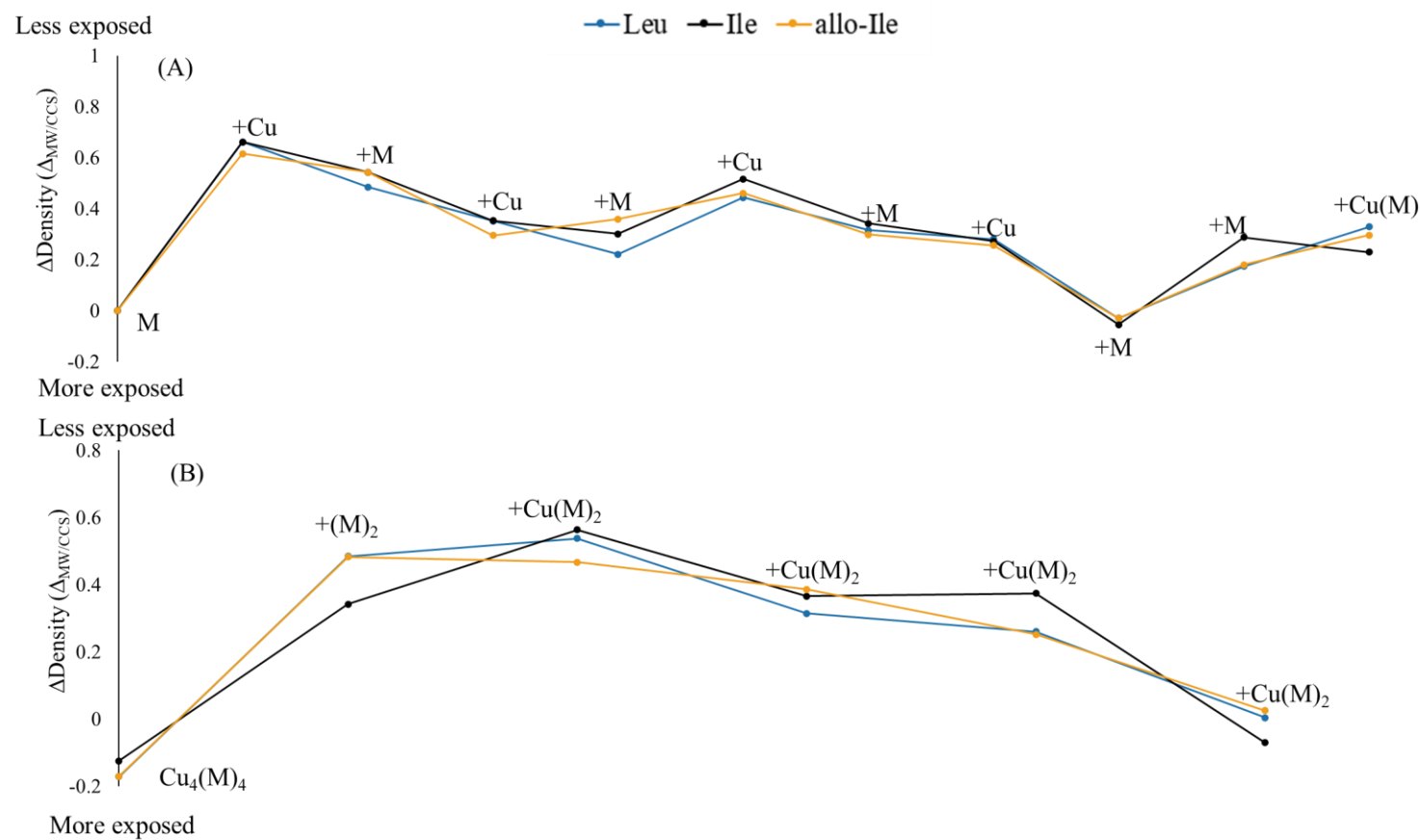


Figure 2-35. An assembly plot showing the distribution of AD values for the singly charged group (A) and doubly charged group (B).

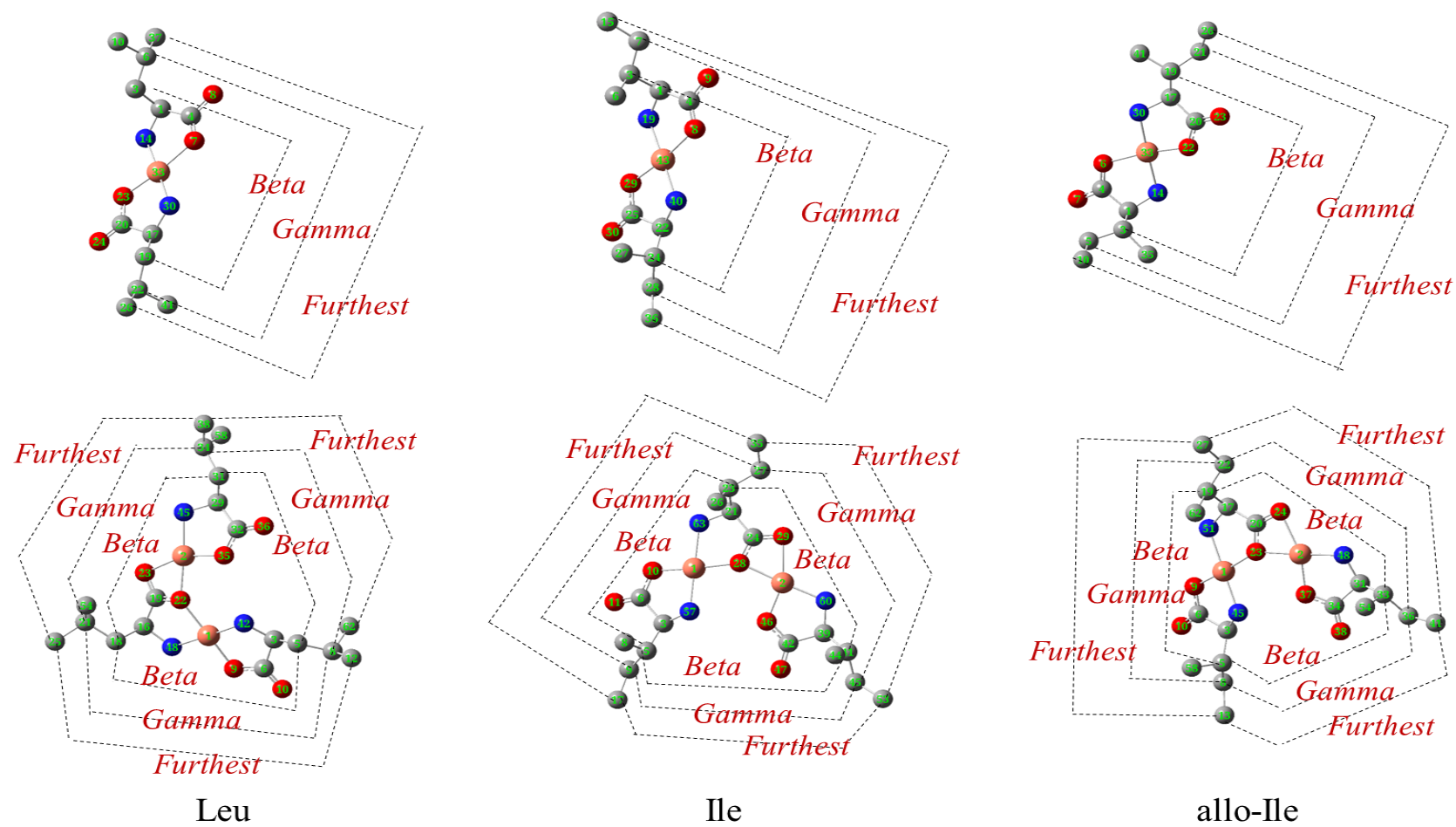


Figure 2-36. The comparison between pairs of beta, gamma, and furthest carbon atoms in simulated copper bound-dimers and -trimers of Leu, Ile and allo-Ile.

Table 2-28. Chains distances in assemblies of Leu, Ile and allo-Ile.

	Distances between carbon atoms on alkyl chains [Å]								
Units (M)	Leu			Ile			allo-Ile		
Carbon atoms	<i>Beta</i>	<i>Gamma</i>	<i>Furthest</i>	<i>Beta</i>	<i>Gamma</i>	<i>Furthest</i>	<i>Beta</i>	<i>Gamma</i>	<i>Furthest</i>
$[(M)_2 + Cu^{II} - H]^+$	7.68	10.37	12.55	7.81	10.51	12.70	7.67	9.92	11.77
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	8.24	10.67	12.73	8.45	9.56	11.18	8.45	9.56	11.17
	8.90	9.44	11.94	8.90	10.35	12.69	8.90	10.35	13.24
	7.92	10.51	12.24	8.18	10.77	13.24	8.18	10.77	12.69
$[(M)_4 + (Cu^{II})_2 - 3H]^+$	7.60	9.11	12.03	7.05	8.44	10.34	7.65	9.01	10.73
	7.11	9.01	11.81	7.47	10.29	10.13	7.42	8.51	11.81
	7.55	9.70	9.60	7.41	9.58	11.33	7.66	9.50	10.21
	8.14	7.57	11.81	8.93	7.63	12.87	8.05	7.80	12.01
	7.65	8.25	11.92	7.80	8.26	11.09	7.58	8.20	11.29
	7.31	9.51	9.70	7.31	9.16	12.10	7.44	9.29	10.93

2.3.5 Extension study on analogue isomers and peptides

Leu, Ile and allo-Ile differ on their alkyl side chains. To explore how assembly enhances the differentiation, other isomers including amino alcohols, keto acids and fatty acids, which have the same alkyl chains with Leu, Ile and allo-Ile, were selected for investigation. As shown in Table 2-29, (S)-isoleucinol, (R)-isoleucinol and (S)-leucinol are isomers of each other, (S)-3-methyl-2-oxovaleric acid, (R)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid are isomers of each other, and (S)-3-methyl-pentanoic acid, (R)-3-methyl-pentanoic acid, 4-methylvaleric acid and 2-methylvaleric acid are isomers of each other. (S)-Isoleucinol and (S)-3-methyl-2-oxovaleric acid have the same alkyl side chain with L-Ile and similar alkyl side chain with (S)-3-methyl-pentanoic acid; (R)-isoleucinol and (R)-3-methyl-2-oxovaleric acid have the same alkyl side chain with L-allo-Ile and similar alkyl side chain with (R)-3-methyl-pentanoic acid; (S)-leucinol and 4-methyl-2-oxovaleric acid have the same alkyl side chain with L-Leu and similar alkyl side chain with 4-methylvaleric acid. As discussed previously, not all these molecules could form higher assemblies under the optimized experimental conditions, probably due to their differences in available binding sites and coordination ability.

Table 2-29. Comparison of Leu, Ile, allo-Ile and analogue isomers.

Compounds	Side chains			
	<i>sec</i> -butyl / active pentyl		Isobutyl / isopentyl	<i>sec</i> -pentyl
Amino acids	Ile	allo-Ile	Leu	/
Amino alcohols	(S)-Isoleucinol	(R)-Isoleucinol	(S)-Leucinol	/
Keto acids	(S)-3-Methyl-2-oxovaleric acid	(R)-3-Methyl-2-oxovaleric acid	4-Methyl-2-oxovaleric acid	/
Fatty acids	(S)-3-Methyl-pentanoic acid	(R)-3-Methyl-pentanoic acid	4-Methylvaleric acid	2-Methylvaleric acid

For amino alcohols, as shown in Table 2-30, increased drift time differences between isomers with the assembly were observed. The drift time differences between the assemblies of (S)-isoleucinol and (R)-isoleucinol, the assemblies of (S)-isoleucinol and (S)-leucinol, and the assemblies of (R)-isoleucinol and (S)-leucinol were increased from 0 to 0.49 ms, 0.07 to 0.21 ms, and 0.07 to 0.42 ms, respectively. From $n = 1$ to 2, the ΔCCS between the assemblies of (S)-isoleucinol and (R)-isoleucinol was increased from 0 to $8.8 \text{ \AA}^2$, the ΔCCS between the assemblies of (S)-isoleucinol and (S)-leucinol was increased from 1.8 to $3.8 \text{ \AA}^2$, and the ΔCCS between the assemblies of (R)-isoleucinol and (S)-leucinol was increased from 1.8 to $7.5 \text{ \AA}^2$. Interestingly, although the assembly level of amino alcohols was only $n=2$, the difference between the isomers was significantly enhanced, and the water molecule also showed effect on the differentiation which was probably due to its involvement in the bonding of the assemblies.

For keto acids, the drift time difference of assemblies between (S)-3-methyl-2-oxovaleric acid and (R)-3-methyl-2-oxovaleric acid was observed as 3.4 ms when $n=8$, while the difference between the assemblies of (S)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid was increased from 0.07 to 0.76 ms. As shown in Table 2-31, the assembly ΔCCS s between (S)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid, and between (R)-3-methyl-2-oxovaleric acid and 4-methyl-2-

oxovaleric acid were increased from 0 to 18.7 Å², and 1.6 to 29.8 Å², respectively. The ΔCCS of the assemblies of (S)-3-methyl-2-oxovaleric acid and (R)-3-methyl-2-oxovaleric acid was 0 Å² when $n \leq 4$, and was increased to 11.1 Å² with the assembly level up to 8 with doubly charged. This result was similar with that for Ile and allo-Ile which have the same enantiomeric side chains but different numbers of chiral centers. As shown in Figure 2-2, these two AAs have one more chiral center than keto acids, thus the coordination part in AA assemblies should have more stereo-centers than that of keto acids assemblies, indicating the assembly-enhanced differentiation might be related with the assembly level and the assembly mode.

For fatty acids, as shown in Table 2-32, assembly could provide enhanced differentiation if alkyl chains on the fatty acids are the constitutional isomers such as 3-methyl-pentanoic acid, 4-methylvaleric acid and 2-methylvaleric acid. For example, the ΔCCS of assemblies was increased from 1.5 Å² in monomer form 4.3 Å² in trimer form between (S)-3-methyl-pentanoic acid and 4-methylvaleric acid, and from 1.5 Å² in monomer form to 10.0 Å² in pentamer form between 2-methyl-pentanoic acid and 4-methylvaleric acid. It could be found that for the enantiomers, such as (S)-3-methyl-pentanoic acid and (R)-3-methyl-pentanoic acid, no accumulated differences were observed for the assembly.

Table 2-30. Summary of the measured CCSs and t_d for copper bound assembly ions of (S)-isoleucinol, (R)-isoleucinol and (S)-leucinol (n = 3 intraday measurements).

Composition	(S)-Isoleucinol		(R)-Isoleucinol		(S)-Leucinol	
	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
[(M) + H] ⁺	2.01	69.4	2.01	69.4	2.08	71.2
[(M) + 2H ₂ O + (Cu ^I) ₂ – H] ⁺	3.53	97.7	3.39	94.9	3.67	100.4
[(M) + 3H ₂ O + (Cu ^I) ₂ – H] ⁺	4.16	109.5	4.36	113.2	4.22	110.6
[(M) ₂ + Cu ^{II} + 2H ₂ O – 5H] ⁺	4.57	116.5	4.64	117.8	4.78	120.3
[(M) ₂ + H ₂ O + Cu ^{II} + Cu ^I – 2H] ⁺	4.43	113.4	4.92	122.2	4.5	114.7

Table 2-31. Summary of the measured CCSs and t_d for copper bound assembly ions of (S)-3-methyl-2-oxovaleric acid, (R)-3-methyl-2-oxovaleric acid and 4-methyl-2-oxovaleric acid (n = 3 intraday measurements).

Composition	(S)-3-Methyl-2-oxovaleric acid		(R)-3-Methyl-2-oxovaleric acid		4-Methyl-2-oxovaleric acid	
	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	2.49	77.7	2.56	79.3	2.49	77.7
$[(M)_2 + (Cu^I)_3 - 2H]^+$	4.92	121.5	4.92	121.5	5.06	123.9
$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	6.51	147.2	6.51	147.2	6.37	145.0
$[(M)_3 + (Cu^I)_4 - 3H]^+$	6.58	148.0	6.58	148.0	6.51	146.9
$[(M)_4 + (Cu^{II})_2 + (Cu^I)_2 - 4H]^{2+}$	5.33	254.1	5.33	254.1	4.57	227.7
$[(M)_4 + (Cu^{II})_2 + Cu^I - 4H]^+$	9.14	186.6	9.14	186.6	7.89	168.0
$[(M)_8 + (Cu^{II})_4 + (Cu^I)_2 - 8H]^{2+}$	5.75	265.7	6.09	276.8	5.19	247.0

Table 2-32. Summary of the measured CCSs and t_d for copper bound assembly ions of (S)-3-methyl-pentanoic acid, (R)-3-methyl-pentanoic acid, 4-Methylvaleric acid and 2-Methylvaleric acid (M) (n = 3 intraday measurements).

Composition	(S)-3-Methyl-pentanoic acid		(R)-3-Methyl-pentanoic acid		4-Methylvaleric acid		2-Methylvaleric acid	
	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	2.42	76.6	2.42	76.6	2.49	78.1	2.42	76.6
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	3.81	102.1	3.88	103.4	3.95	104.7	4.02	106.0
$[(M)_3 + (Cu^{II})_3 + H_2O - 5H]^+$	6.65±0.11	149.7±1.2	6.65	149.7	6.86	153.0	6.65	149.7
$[(M)_5 + (Cu^{II})_3 - 5H]^+$	9.28	188.3	9.28	188.3	9.42±0.11	190.3±1.6	8.73	180.3

Dipeptides, tripeptides, tetrapeptides constituted with L-Leu, L-Ile and L-allo-Ile were prepared and studied with the assembly-based approach. The obtained drift time and calculated CCSs were shown in Tables 2-33 to 2-35. It could be observed that only dipeptides could form higher level of assemblies and the tripeptide and tetrapeptides could only form smaller assemblies such as dimeric, trimeric or tetrameric ions. For dipeptides, higher assembly still enabled larger drift time differenced, and with the assembly, the drift time difference between Leu-Leu and Ile-Ile, Leu-Leu and allo-Ile-allo-Ile, Ile-Ile and allo-Ile-allo-Ile, was increased from 0 to 4.4 ms, 0.11 to 0.55 ms and 0.11 to 0.22 ms, respectively. However, the enhanced difference between Ile-Ile and allo-Ile-allo-Ile was less significant than between Leu-Leu and Ile-Ile. For example, the assembly Δ CCSs were gradually increased from 0 to 14.1 Å² from n=1 to 8 between Leu-Leu and Ile-Ile, but did not exceed the system error until n=7 between Ile-Ile and allo-Ile-allo-Ile.

For tripeptides, slight differences could be observed even for their monomers, e.g., a Δ CCS of 4.2 Å² between Ile-Ile-Ile and allo-Ile-allo-Ile-allo-Ile. With the assembly, the Δ CCSs were increased from 1.0 to 3.9 Å², 3.2 to 8.2 Å², between Leu-Leu-Leu and Ile-Ile-Ile, and between Leu-Leu-Leu and allo-Ile-allo-Ile-allo-Ile, respectively. The differences between assemblies of Ile-Ile-Ile and allo-Ile-allo-Ile-allo-Ile were not significantly enhanced. For tetrapeptides, the Δ CCSs of assemblies between Leu-Leu-

Leu-Leu and Ile-Ile-Ile-Ile were increased from 1.9 to 12.8 Å². Similar to the case for the tripeptides, the differences between Ile tetrapeptide and allo-Ile tetrapeptide were not significantly enhanced with the assembly, which was probably also related to the low assembly level.

Although the peptides could not form complexes as large as for the amino acids, it was interesting to observe that with the same number of alkyl side chains, the assemblies show close CCS values. For example, the CCS of [(Leu-Leu)₄ + (Cu^{II})₅ – 6H]²⁺ with eight alkyl chains was 259.5 Å² and the CCS of [(Leu-Leu-Leu-Leu)₂ + (Cu^{II})₂ – 3H]⁺ also with eight alkyl chains was 261.6 Å², the CCS of [(Leu-Leu)₃ + (Cu^{II})₃ – 5H]⁺ with six alkyl chains was 231.8 Å² and the CCS of [(Leu-Leu-Leu)₂ + (Cu^{II})₂ – 3H]⁺ also with six alkyl chains was 231.9 Å². CCS is an average value of all orientations, and is related with the shape and size of the ion flying in the drift tube. The measured CCS values of these assemblies could be more related to the exposed alkyl chains. This will be further discussed in the next section.

Table 2-33. Summary of the measured CCSs and t_d for copper bound assembly ions of dipeptides (Leu-Leu, Ile-Ile and allo-Ile-allo-Ile) (n = 3 intraday measurements).

Composition	Leu-Leu		Ile-Ile		allo-Ile-allo-Ile	
	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + Cu^I]^+$	3.64	99.0	3.64	99.0	3.53	96.9
$[(M)_2 + Cu^{II} - H]^+$	7.39	161.8	7.17	158.3	7.17	158.3
$[(M)_2 + Cu^{II} + H_2O - 2H]^+$	7.28	159.9	6.95	154.7	7.06±0.07	156.4±1.1
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	7.28	159.7	7.39	161.4	7.39	161.4
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	11.02	213.8	10.58	207.6	10.69	209.1
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	12.35	231.8	11.69	222.8	11.69	222.8
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	16.1	279.4	15.44	271.1	15.44	271.1
$[(M)_4 + (Cu^{II})_4 - 6H]^{2+}$	4.96	255.4	4.74	246.9	4.74	246.9
$[(M)_4 + (Cu^{II})_5 - 6H]^{2+}$	5.07	259.5	4.85	251.0	4.85	251.0
$[(M)_5 + (Cu^{II})_5 - 8H]^{2+}$	6.06	295.8	5.84	287.8	5.73	283.7
$[(M)_6 + (Cu^{II})_6 - 10H]^{2+}$	7.17±0.11	334.8±1.8	7.06	331.0	7.17	334.8
$[(M)_7 + (Cu^{II})_7 - 12H]^{2+}$	8.71	386.6	8.60	383.0	8.38	375.6
$[(M)_8 + (Cu^{II})_8 - 14H]^{2+}$	10.03	429.1	9.59	415.0	9.48	411.4

Table 2-34. Summary of the measured CCSs and t_d for copper bound assembly ions of tripeptides (Leu-Leu-Leu, Ile-Ile-Ile and allo-Ile-allo-Ile-allo-Ile) ($n = 3$ intraday measurements).

Composition	Leu-Leu-Leu		Ile-Ile-Ile		allo-Ile-allo-Ile-allo-Ile	
	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + H]^+$	5.33	129.5	5.39	130.5	5.14	126.3
$[(M) + Cu^{II} - H]^+$	5.84	137.6	5.73 $\pm$ 0.07	135.7 $\pm$ 0.8	5.66	134.4
$[(M)_2 + Cu^{II} - H]^+$	11.85	224.0	11.11	214.0	11.45	218.6
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	12.47	231.9	12.29	229.5	12.04	226.2
$[(M)_3 + (Cu^{II})_3 - 5H]^+$	17.14	289.0	16.87	285.8	16.55	281.9
$[(M)_4 + (Cu^{II})_4 - 7H]^+$	8.07	337.7	7.93	333.8	7.79	329.5

Table 2-35. Summary of the measured CCSs and t_d for copper bound assembly ions of tetrapeptides (Leu-Leu-Leu-Leu, Ile-Ile-Ile-Ile and allo-Ile-allo-Ile-allo-Ile-allo-Ile) (n = 3 intraday measurements).

Composition	Leu-Leu-Leu-Leu		Ile-Ile-Ile-Ile		allo-Ile-allo-Ile-allo-Ile-allo-Ile	
	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + H]^+$	6.61	149.9	6.50	148.1	6.50	148.1
$[(M) + Cu^I - H]^+$	7.39	161.4	7.06	156.7	7.06	156.7
$[(M)_2 + Cu^{II} - H]^+$	15.44	270.8	14.22	255.7	14.33	258.1
$[(M)_2 + (Cu^{II})_2 - 2H]^{2+}$	4.63±0.11	243.1±1.2	4.41	234.4	4.30	230.0
$[(M)_2 + (Cu^{II})_2 - 3H]^+$	14.66	261.6	13.67	248.8	13.67	248.3

2.4 Conclusions

In this chapter, by optimizing the ESI conditions, additive metal ions and their concentration ratios with the amino acids, we have been able to observe higher degrees of assemblies of Leu, Ile or allo-Ile in the presence of copper ion. Interestingly, it was found that the differences of the isomer assemblies in drift time and CCSs in IM-MS became larger with the increased number of the assembled amino acids. The Δ CCS slope between assemblies of Leu and assemblies of Ile was larger than the Δ CCS slope between assemblies of Ile and assemblies of allo-Ile, which is in the same order as the Δ CCS among the individual Leu, Ile and allo-Ile, suggesting that the enhanced differences were originated from the accumulation of initial differences of the isomers during the assembly. Further investigation with the analogue compounds (e.g., amino alcohols, keto acids and fatty acids) and peptides by IM-MS supported the enhanced differentiation of isomers in the assemblies.

The combination of IM-MS, MS/MS and computational modelling have been employed to obtain the structure information of the assemblies and understand their gradually increased differentiation. The structures were revealed as near-sphere shape, with the coordination core and the alkyl side chains of amino acids protruded outwards. The lower assemblies had farther away side chains distribution and the higher assemblies had higher density neighboring side chains. The *AD* results showed that the

differences between alkyl side chains in isomers had a significant effect on the side chain distribution especially in the assemblies with higher density neighboring side chains. The difference between Ile and Leu could be accumulated from lower assemblies, while the difference between Ile and allo-Ile were accumulated in higher assemblies. Hence, the differences between those aliphatic side chain isomers could be accumulated after assembly and eventually caused significant differences. This work proposes a unique insight into the effect of side chains on assembly of amino acids as well as analogue compounds and provides a theoretical basis for differentiation of isomers by assembly-based approaches.

Chapter 3: Differentiation of saccharide isomers by IM-MS

3.1 Introduction

Saccharides are the unit structure of carbohydrates, which have a general chemical formula of $C_n(H_2O)_n$ and are important biological molecules that play vital roles in living organisms, e.g., as the structural component of cells and source of energy.¹²⁷ According to the number of monomeric units, saccharides can be classified into monosaccharide, disaccharide, oligosaccharide and polysaccharide.¹²⁸ Unlike oligonucleotides or proteins, saccharides do not have a linear sequence but branch, with diverse regio- and even stereochemistry in them, leading to a vast number of possible isomers.^{29, 50} Even monosaccharides, the basic building blocks, are often stereoisomers because the carbon atoms along the chain can exist in different spatial arrangements.¹²⁹ The small structural differences of isomers in saccharides can lead to significant differences in their physical and chemical properties. For example, the α - and β -configurations of glycosidic bonds of glucose cause the very different properties and functions of cellulose and starch.¹⁶ The disaccharides with different linkages or monosaccharide compositions are associated with their different properties such as the selectivity of microbial fermentation.⁵ Thus, for better understanding the biological roles of saccharides in living organisms, the detailed knowledge of their structures is essential.

To describe the structure of a saccharide, three structural features, composition, connectivity and configuration, need to be expounded. The composition is defined by its type of monosaccharide building blocks, such as Glc, Fru, Man and Gal (Figure 3-1). The connectivity means the position of the glycosidic bond and branching, as in the case of Mal with 1→4 linkage and IsoMal with 1→6 linkage (Figure 3-1). The configuration is defined by the stereo-center when a glycosidic bond is formed in two different configurations, such as α -anomer (Mal) and β -anomer (Cel) as shown in Figure 3-2.

In the last decade, IM–MS emerged as a very promising tool for saccharide analysis.^{41, 65, 130} However, due to the inherent complexity on the structures of saccharides, the characterization of saccharides especially for their isomers, is still challenging. Efforts have been made to enhance the differentiation of saccharide isomers, e.g., addition of metal ions, however, they did not universally work.¹³¹⁻¹³²

Based on our previous observation with amino acids that assembly might enhance the differentiation of isomers in IM-MS, in this chapter, we would like to extend the approach to analysis of saccharides. Typical monosaccharides including Glc, Fru, Man, Gal, and disaccharides including Mal, Cel, IsoMal, Lac and Suc with different compositions, connectivity and configurations (see Figure 3-1 and 3-2) were selected and investigated with addition of metal ions, and the effect of assembly on isomer

differentiation was studied.

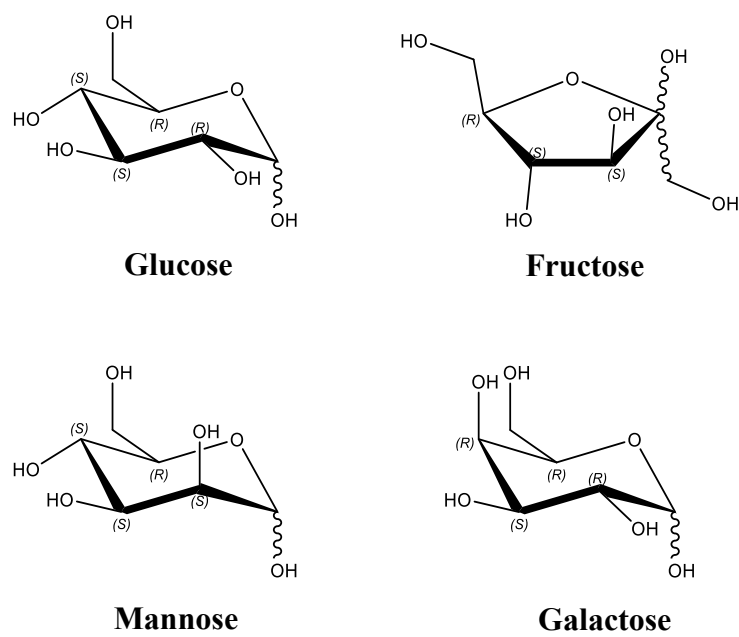
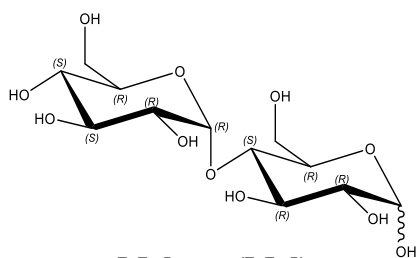
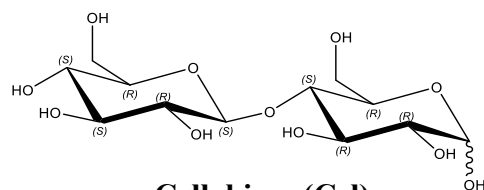


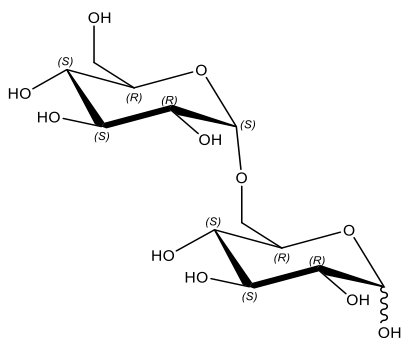
Figure 3-1. Structures of monosaccharides.



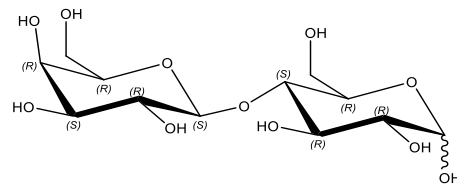
Maltose (Mal)
 α -D-Glc-[1→4]-D-Glc



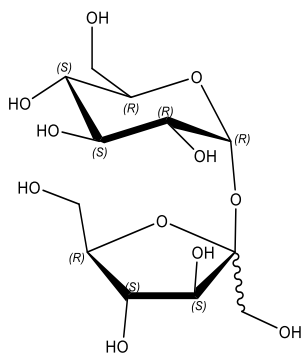
Cellobiose (Cel)
 β -D-Glc-[1→4]-D-Glc



Isomaltose (IsoMal)
 α -D-Glc-[1→6]-D-Glc



Lactose (Lac)
 β -D-Gal-[1→4]-D-Glc



Sucrose (Suc)
 α -D-Glc-[1→2]-β-D-Fru

Figure 3-2. Structures of disaccharides.

3.2 Experimental

3.2.1 Materials

The optically pure monosaccharide (D-glucose, D-mannose, D-fructose and D-galactose) and disaccharide (maltose, cellobiose, isomaltose, lactose and sucrose) samples were purchased from Sigma-Aldrich Co (St. Louis, MO). The metal salts ($\text{Cu}(\text{CH}_3\text{COOH})_2$, CuCl_2 , ZnCl_2 , NiCl_2 , MgCl_2 and LiNO_3 , NaNO_3 , KNO_3) were purchased from Aladdin (Shanghai, China). The HPLC grade methanol was purchased from BCR International Trading Co Ltd. (California, USA). Ultrapure water was produced by a MilliQ device (Millipore, Milford, MA, USA). All the chemicals were used directly without further purification.

3.2.2 Sample preparation

Monosaccharides (D-glucose, D-mannose, D-fructose and D-galactose) and disaccharides (maltose, cellobiose, isomaltose, lactose and sucrose) and metal salts were dissolved separately in water to achieve a concentration of 5 mM. The sample solutions used for the MS analysis were prepared by mixing the stock solutions, and then diluting with 50:50 water/methanol, to obtain a series of final solutions each containing 50 μM metal salts with 5 μM , 25 μM , 50 μM , 100 μM , 250 μM , 500 μM or 1000 μM monosaccharides or disaccharides.

3.2.3 Mass Spectrometry

IM-MS experiments were performed using a Synapt G2 mass spectrometer (Waters, Manchester, UK). ESI conditions were as follows: capillary voltage, 3 kV; sampling cone, 40 V; extraction cone, 5 V; source temperature, 100 °C; desolvation temperature, 350°C; cone gas, 50 L h⁻¹; desolvation gas, 500 L h⁻¹. All the experiments were performed in the positive ion mode. The sample solutions were directly introduced to the mass spectrometer through the ESI source at a flow rate of 10 μ L min⁻¹. All the spectral data represented the combined 50 scans in the direct infusion experiments. For TWIM experiments, the drift gas was N₂, the traveling wave height was set at 40 V, and traveling wave velocity was chosen as 800 m s⁻¹. The mass spectra acquired were analyzed by the Driftscope program. The CCS values of the investigated complexes were calculated by using tetraalkylammonium ions (TAA-n, n=1-8) with a concentration of 10 mg mL⁻¹ to plot a calibration curve as described by the literature,⁴⁵ and the CCS values of polyalanine were obtained from the literature.¹³³ The mass spectra acquired were analyzed by the Driftscope program.

3.3 Results and discussion

3.3.1 Optimization for detection of assembly ions

Addition of metal salts, including LiNO₃, KNO₃, MgCl₂, CuCl₂, NiCl₂ and ZnCl₂, and different concentration ratios of saccharide to metal ion, i.e., 10:1 or 1:1 or 1:10, were

attempted for optimal detection of the self-assembled saccharide ions. For monosaccharides, it was found that only up to dimeric or trimeric complex ions could be detected under the various testing conditions (Table 3-1). For example, for Glc, the largest complex ions detected were trimeric ions such as $[(M)_3 - H]^+$ (m/z 539.2500) with the addition of KNO_3 , and dimeric ions such as $[(M)_2 + Li]^+$ (m/z 367.1480), $[(M)_2 + Zn^{II} + 2H_2O - H]^+$ (m/z 459.0345), $[(M)_2 + Cu^{II} + 2H_2O - H]^+$ (m/z 458.0440), $[(M)_2 + Mg^{II} + 2H_2O - H]^+$ (m/z 419.0977) and $[(M)_2 + (Ni^{II})_3 + 3H_2O - 7H]^+$ (m/z 582.9111) with $LiNO_3$, $ZnCl_2$, $CuCl_2$, $MgCl_2$ and $NiCl_2$, respectively (Figures 3-3). For disaccharides, as shown in Figure 3-4, it was found that only Ni^{2+} and Zn^{2+} could allow better detection of assemblies of disaccharides. With Zn^{2+} at a 1:10 molar ratio to the maltose, higher degree assembly ions of disaccharides, such as at m/z 545.1104, 716.1688, 887.2247, 1058.2683, 1230.3264 and 1400.3959, corresponding to $[(M)_3 + Zn^{II}]^{2+}$, $[(M)_4 + Zn^{II}]^{2+}$, $[(M)_5 + Zn^{II}]^{2+}$, $[(M)_6 + Zn^{II}]^{2+}$, $[(M)_7 + Zn^{II}]^{2+}$ and $[(M)_8 + Zn^{II}]^{2+}$ respectively, were detected. With Ni^{2+} at a 1:10 ratio to maltose, complex ions at m/z 542.1178, 713.1694, 884.2234, 1055.2739, 1227.3270 and 1397.3752, corresponding to $[(M)_3 + Ni^{II}]^{2+}$, $[(M)_4 + Ni^{II}]^{2+}$, $[(M)_5 + Ni^{II}]^{2+}$, $[(M)_6 + Ni^{II}]^{2+}$, $[(M)_7 + Ni^{II}]^{2+}$ and $[(M)_8 + Ni^{II}]^{2+}$ respectively, could be detected. The detected assembled ions of monosaccharides and disaccharides were identified through analyses of accurate monoisotopic masses, isotopic patterns and charge states, and the results are summarized in Tables 3-1 and 3-2.

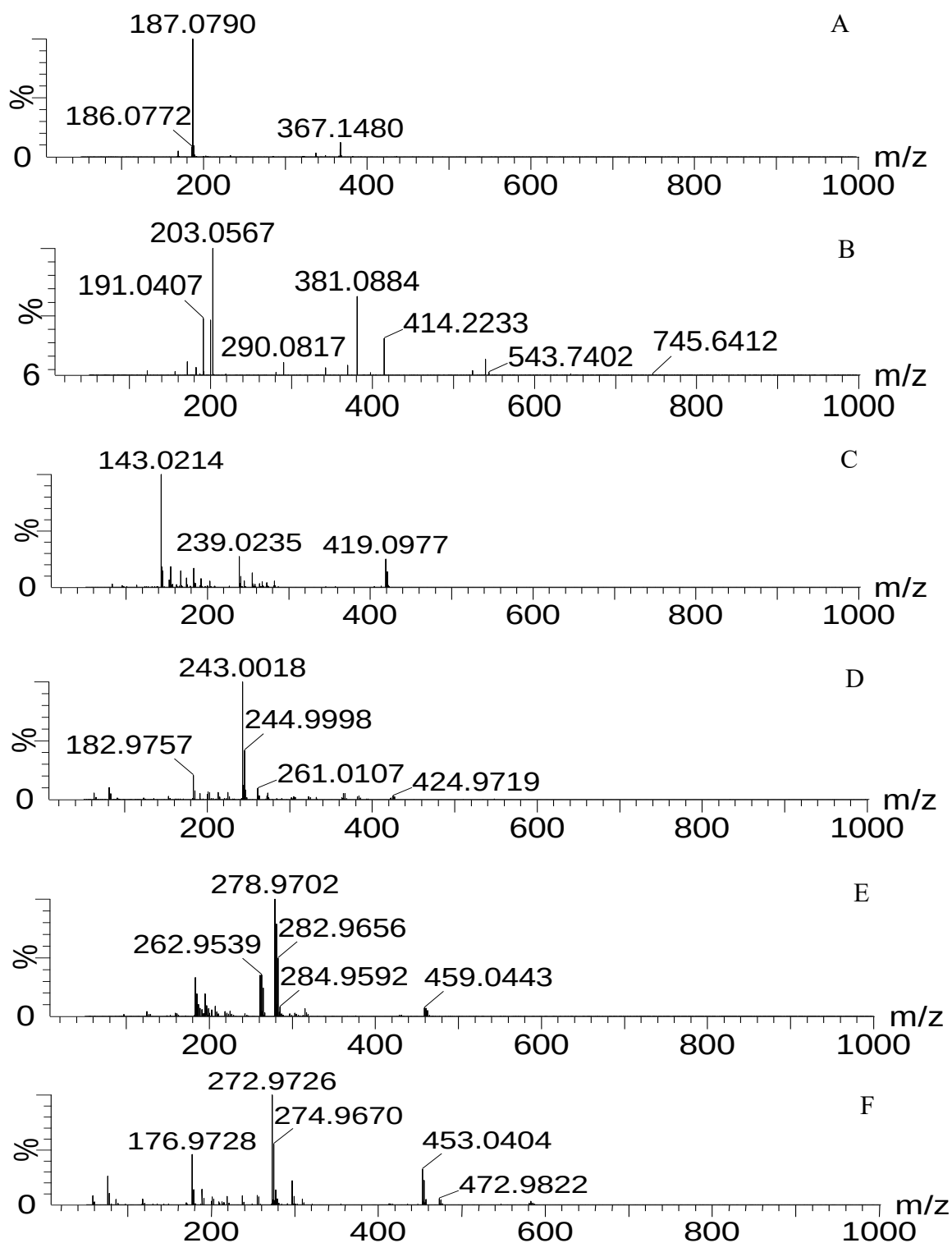


Figure 3-3. ESI-MS spectra for glucose with addition of LiNO₃ (A), KNO₃ (B), MgCl₂ (C), CuCl₂ (D), NiCl₂ (E), and ZnCl₂ (F) at a concentration ratio of Glc: Metal=1:10. The assignments of major peaks are summarized in Table 3-1.

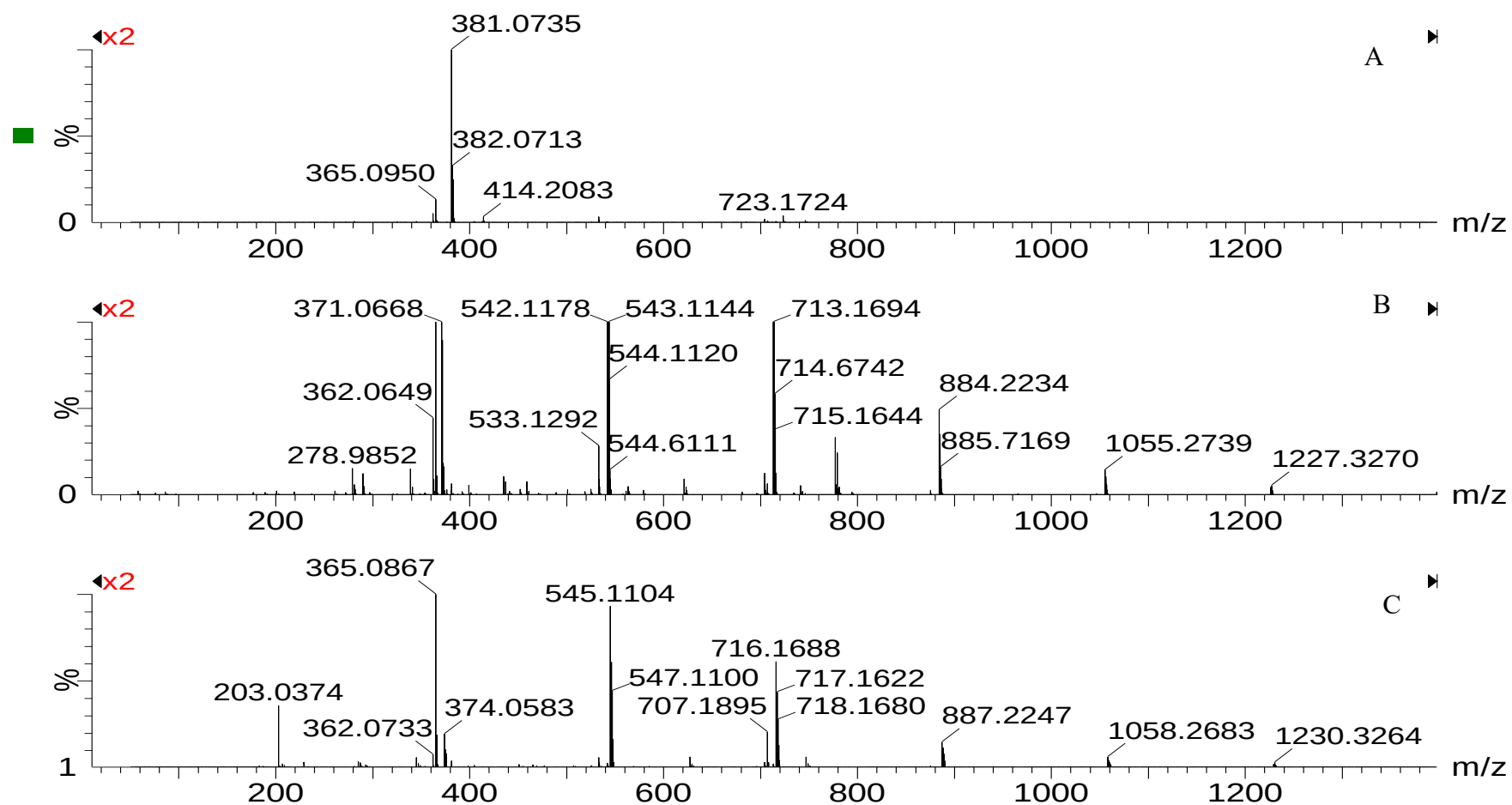


Figure 3-4. ESI-MS spectra for maltose with addition of KNO₃ (A), NiCl₂ (B), and ZnCl₂ (C), at a concentration ratio of Mal: metal=10:1. The assignments of major peaks are summarized in Table 3-2.

Table 3-1. Results for exact mass measurements of the metal ion bound assembled ions of monosaccharides (M: glucose, fructose, galactose and mannose).

Metal ion	Composition	Formula	Monoisotopic mass (Da)		
			Theoretical mass	Measured mass	Error
Na ⁺	[(M) + Na] ⁺	C ₆ H ₁₂ O ₆ Na	203.0532	203.0567	0.0035
	[(M) ₂ + Na] ⁺	C ₁₂ H ₂₄ O ₁₂ Na	383.1165	383.1194	0.0029
	[(M) ₂ +Na – H ₂ O] ⁺	C ₁₂ H ₂₂ O ₁₁ Na	365.1060	365.1089	0.0029
Li ⁺	[(M) + Li] ⁺	C ₆ H ₁₂ O ₆ Li	186.0785	186.0772	-0.0013
	[(M) ₂ + Li] ⁺	C ₁₂ H ₂₄ O ₁₂ Li	366.1419	366.1449	0.0030
K ⁺	[(M) + K] ⁺	C ₆ H ₁₂ O ₆ K	219.0271	219.0296	0.0025
	[(M) ₂ + K – H ₂ O] ⁺	C ₁₂ H ₂₂ O ₁₁ K	381.0799	381.0884	0.0085
	[(M) ₃ – H] ⁺	C ₁₈ H ₃₅ O ₁₈	539.1823	539.2500	0.0677
	[(M) ₃ – OH] ⁺	C ₁₈ H ₃₅ O ₁₇	523.1874	523.2516	0.0642
Ni ²⁺	[(M) + Ni ^{II} – H] ⁺	C ₆ H ₁₁ O ₆ Ni	236.9909	236.9962	0.0053
	[(M) + Ni ^{II} + H ₂ O – H] ⁺	C ₆ H ₁₃ O ₇ Ni	255.0015	254.9595	-0.0420
	[(M) + Ni ^{II} + 2H ₂ O – H] ⁺	C ₆ H ₁₅ O ₈ Ni	273.0120	272.9726	-0.0394
	[(M) ₂ + Ni ^{II} + 2H ₂ O – H] ⁺	C ₁₂ H ₂₇ O ₁₄ Ni	453.0754	453.0404	-0.0350
	[(M) ₂ + (Ni ^{II}) ₂ – 3H] ⁺	C ₁₂ H ₂₁ O ₁₂ Ni ₂	472.974	472.9822	0.0082
	[(M) ₂ + (Ni ^{II}) ₃ + 3H ₂ O – 7H] ⁺	C ₁₂ H ₂₃ O ₁₅ Ni ₃	580.9097	580.9207	0.0110
Zn ²⁺	[(M) + Zn ^{II} – H] ⁺	C ₆ H ₁₁ O ₆ Zn	242.9847	242.995	0.0103
	[(M) + Zn ^{II} + H ₂ O – H] ⁺	C ₆ H ₁₃ O ₇ Zn	260.9953	260.9614	-0.0339
	[(M) + Zn ^{II} + 2H ₂ O – H] ⁺	C ₆ H ₁₅ O ₈ Zn	279.0058	278.9702	-0.0356
	[(M) ₂ + Zn ^{II} + 2H ₂ O – H] ⁺	C ₁₂ H ₂₇ O ₁₄ Zn	459.0692	459.0443	-0.0249
Mg ²⁺	[(M) + Mg ^{II} -H] ⁺	C ₆ H ₁₁ O ₆ Mg	203.0406	203.0567	0.0161
	[(M) + Mg ^{II} + 2H ₂ O – H] ⁺	C ₆ H ₁₅ O ₈ Mg	239.0617	239.0235	-0.0382
	[(M) ₂ + 2Mg ^{II} + 6H ₂ O – 6H] ²⁺	C ₁₂ H ₃₀ O ₁₈ Mg ₂	255.0567	255.078	0.0213
	[(M) ₂ + 2Mg ^{II} + 6H ₂ O – 2H] ²⁺	C ₁₂ H ₃₄ O ₁₈ Mg ₂	258.0810	258.1621	0.0811
	[(M) ₂ + 2Mg ^{II} + 7H ₂ O – 6H] ²⁺	C ₁₂ H ₃₂ O ₁₉ Mg ₂	264.0619	264.0878	0.0259
	[(M) ₂ + 2Mg ^{II} + 8H ₂ O – 6H] ²⁺	C ₁₂ H ₃₄ O ₂₀ Mg ₂	273.0672	273.0880	0.0208
	[(M) ₂ + 2Mg ^{II} + 9H ₂ O – 6H] ²⁺	C ₁₂ H ₃₆ O ₂₁ Mg ₂	282.0725	282.0997	0.0272
	[(M) ₂ + Mg ^{II} + 2H ₂ O – H] ⁺	C ₁₂ H ₂₇ O ₁₄ Mg	419.1251	419.0977	-0.0274
Cu ²⁺	[(M) + Cu ^I] ⁺	C ₆ H ₁₂ O ₆ Cu	242.9930	243.0018	0.0088
	[(M) + Cu ^{II} + H ₂ O] ⁺	C ₆ H ₁₄ O ₇ Cu	261.0036	261.0107	0.0071
	[(M) ₂ + 3Cu ^{II} – 4H] ²⁺	C ₁₂ H ₂₀ O ₁₂ Cu ₃	272.4421	272.4535	0.0114
	[(M) ₂ + Cu ^{II} – H] ⁺	C ₁₂ H ₂₃ O ₁₂ Cu	422.0486	422.0611	0.0125
	[(M) ₂ + Cu ^{II} + 2H ₂ O – H] ⁺	C ₁₂ H ₂₇ O ₁₄ Cu	458.0697	458.0440	-0.0257

Table 3-2. Results for exact mass measurements of the metal ion bound assembled ions of disaccharides (M: maltose, cellobiose, isomaltose, lactose and sucrose).

Metal ion	Composition	Formula	Monoisotopic mass (Da)		
			Theoretical Mass	Measured Mass	Error
K ⁺	[(M) + K] ⁺	C ₁₂ H ₂₂ O ₁₁ K	381.0799	381.0735	-0.0064
Zn ²⁺	[(M) + Zn ^{II} - H] ⁺	C ₁₂ H ₂₁ O ₁₁ Zn	405.0375	405.0194	-0.0181
	[(M) + Zn ^{II} + 2H ₂ O - H] ⁺	C ₁₂ H ₂₅ O ₁₃ Zn	441.0587	440.9896	-0.0691
	[(M) ₂ + Zn ^{II}] ²⁺	C ₂₄ H ₄₄ O ₂₂ Zn	374.0808	374.0583	-0.0225
	[(M) ₂ + Zn ^{II} - H] ⁺	C ₂₄ H ₄₃ O ₂₂ Zn	747.1537	747.1177	-0.036
	[(M) ₂ + Zn ^{II} + 2H ₂ O - H] ⁺	C ₂₄ H ₄₇ O ₂₄ Zn	783.1749	783.1010	-0.0739
	[(M) ₃ + Zn ^{II}] ²⁺	C ₃₆ H ₆₆ O ₃₃ Zn	545.1389	545.1104	-0.0285
	[(M) ₄ + Zn ^{II}] ²⁺	C ₄₈ H ₈₈ O ₄₄ Zn	716.1970	716.1688	-0.0282
	[(M) ₅ + Zn ^{II}] ²⁺	C ₆₀ H ₁₁₀ O ₅₅ Zn	887.2551	887.2247	-0.0304
	[(M) ₆ + Zn ^{II}] ²⁺	C ₇₂ H ₁₃₂ O ₆₆ Zn	1058.3132	1058.2683	-0.0449
	[(M) ₇ + Zn ^{II}] ²⁺	C ₈₄ H ₁₅₄ O ₇₇ Zn	1229.3713	1229.3314	-0.0399
	[(M) ₈ + Zn ^{II}] ²⁺	C ₉₆ H ₁₇₆ O ₈₈ Zn	1400.4294	1400.3959	-0.0335
Ni ²⁺	[(M) + Ni ^{II} -H] ⁺	C ₁₂ H ₂₁ O ₁₁ Ni	399.0437	399.0234	-0.0203
	[(M) + Ni ^{II} + 2H ₂ O - H] ⁺	C ₁₂ H ₂₅ O ₁₃ Ni	435.0649	434.9960	-0.0689
	[(M) ₂ + Ni ^{II}] ²⁺	C ₂₄ H ₄₄ O ₂₂ Ni	371.0839	371.0668	-0.0171
	[(M) ₂ + Ni ^{II} - H] ⁺	C ₂₄ H ₄₃ O ₂₂ Ni	741.1599	741.1274	-0.0325
	[(M) ₂ + Ni ^{II} + 2H ₂ O - H] ⁺	C ₂₄ H ₄₇ O ₂₄ Ni	777.1811	777.1017	-0.0794
	[(M) ₃ + Ni ^{II}] ²⁺	C ₃₆ H ₆₆ O ₃₃ Ni	542.1420	542.1178	-0.0242
	[(M) ₄ + Ni ^{II}] ²⁺	C ₄₈ H ₈₈ O ₄₄ Ni	713.2001	713.1694	-0.0307
	[(M) ₅ + Ni ^{II}] ²⁺	C ₆₀ H ₁₁₀ O ₅₅ Ni	884.2583	884.2234	-0.0349
	[(M) ₆ + Ni ^{II}] ²⁺	C ₇₂ H ₁₃₂ O ₆₆ Ni	1055.3163	1055.2739	-0.0424
	[(M) ₇ + Ni ^{II}] ²⁺	C ₈₄ H ₁₅₄ O ₇₇ Ni	1226.3740	1226.3177	-0.0563
	[(M) ₈ + Ni ^{II}] ²⁺	C ₉₆ H ₁₇₆ O ₈₈ Ni	1397.4325	1397.3752	-0.0573

3.3.2 The effect of assembly on differentiation of saccharides

The ion mobility separation of isomers based on their assembled ions was investigated at different multimerization stages, with the results summarized in Tables 3-4 and 3-5.

In monosaccharides case, the complex of monosaccharides only recruited one or two sugar molecules, there is no clear enhanced differentiation trend observed after assembly. Because the system error of IM is $\Delta t_d \approx \pm 0.11 \text{ ms}$, thus $\Delta \text{CCS} \approx \pm 2 \text{ \AA}^2$, we tentatively regarded when the ΔCCS between isomers is larger $4 \text{ \AA}^2$ means the difference do exist. In Table 3-4, it could be observed that Glc, Man and Gal are positional isomers of each other with six carbons in the ring, whereas Fru has five carbons in the ring showed compact structures than others. The CCS difference between Glc and Man, Glc and Gal, Gal and Man could be increased from 0 to $5.6 \text{ \AA}^2$, 0 to $7.4 \text{ \AA}^2$ in Zn bound assemblies, 0 to 7.2 in Ni bound assemblies, respectively. The difference between Fru and Man, Fru and Glc was 2.1 and 0 $\text{ \AA}^2$ in the monomeric complex with Ni and Zn ions, and was increased to 7.2 and $7.4 \text{ \AA}^2$ for the dimeric complex ion $[(\text{M})_2 + (\text{Ni}^{\text{II}})_3 + 3\text{H}_2\text{O} - 7\text{H}]^+$ and $[(\text{M})_2 + \text{Zn}^{\text{II}} + 2\text{H}_2\text{O} - \text{H}]^+$, respectively. The reason probably due to the monosaccharides not always as ring form but easily be conversed to the open chain form. Because the monosaccharides applied in our study are all hexose, containing six carbons, the difference between them is chiral carbons at different positions. The results we obtained in this study provide an insight that assembly could

help to amplify the difference between monosaccharide isomers.

Table 3-3. Comparison of disaccharides.

	Maltose	Cellobiose	Isomaltose	Lactose	Sucrose
Composition	2Glc	2Glc	2Glc	Gal+Glc	Glc+Fru
Connectivity	1→4	1→4	1→6	1→4	1→2
Configuration	α	β	α	β	α and β
Molecular formula	$C_{12}H_{22}O_{11}$				

In disaccharides isomers, higher assemblies were observed with addition of Ni or Zn ions. The comparison was carried out between composition, connectivity and configuration isomers as shown in Table 3-3. The general CCS trends of assemblies of disaccharides were shown in Figure 3-5, as could be observed, the CCS growing trends of Mal, Cel, Isomal, Lac and Suc were similar and almost overlapped. The Mal showed larger slope than others indicated that the assemblies of Mas are with larger size, or shape or less compact than the assemblies from other disaccharides. Sur, one monomer is Fru with five carbons in the ring, showed clearly lower CCS growing slope than others, indicating the assemblies of Sur is smaller than the assemblies of other disaccharides whose units are all having six carbons in the rings. To further investigate the enhanced difference, the comparison was carried out one pair isomer by one pair isomer.

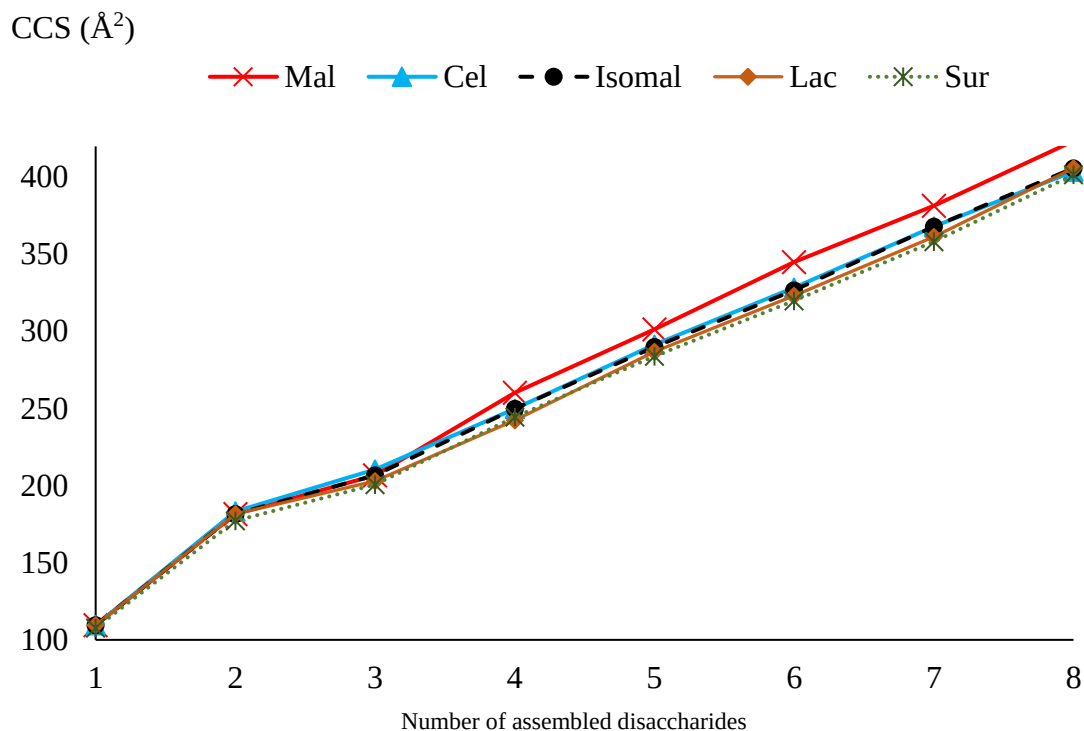


Figure 3-5. Plots of experimental CCSs vs. number of assembled disaccharides.

Lac and Cel are composition isomers of disaccharides. Both of them are β -configuration with 1 $\rightarrow$ 4 linkage for the connectivity, but Cel consists of two Glc while Lac consists of Glc and Gal. As shown in Figure 3-6, a general trend of larger Δ CCS with higher assembly of Cel and Lac could be observed with both of the metal ions. It can be seen from Table 3-5 and Figure 3-6 that the drift time difference was increased from 0 to 0.28 ms, corresponding to Δ CCS from 0 to 7.6 Å², with the zinc-bound assembled molecules from 1 to 3.

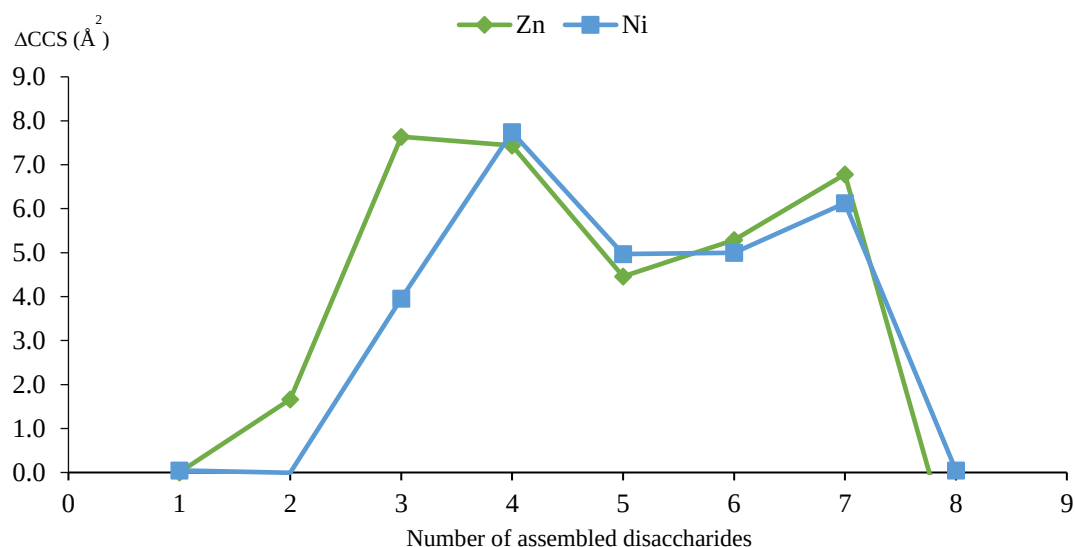


Figure 3-6. The CCS differences between Cel assembly ions and Lac assembly ions at different assembly states, with NiCl_2 (blue) and ZnCl_2 (green).

Mal and IsoMal are connectivity isomers of disaccharides. Both of them consist of two Glcs with α -configuration, but the connectivity between the blocks is 1 $\rightarrow$ 4 linkage for Mal and 1 $\rightarrow$ 6 linkage for IsoMal. As shown in Figure 3-7 and Table 3-5, a general trend of larger ΔCCS with higher assembly of Mal and IsoMal were also observed with both of the metal ions. With the zinc-bound assembled molecules increased from 1 to 6, the drift time between the Mal and IsoMal assemblies was increased from 0 to 0.34 ms, corresponding to ΔCCS increased from 0 to $18.2 \text{ \AA}^2$ and 8. It is interesting to observe that IsoMal assemblies showed earlier shorter drift time and smaller CCS values than those of Mal, indicating that 1 $\rightarrow$ 6 linkage disaccharides are more compact than 1 $\rightarrow$ 4 linkage ones.

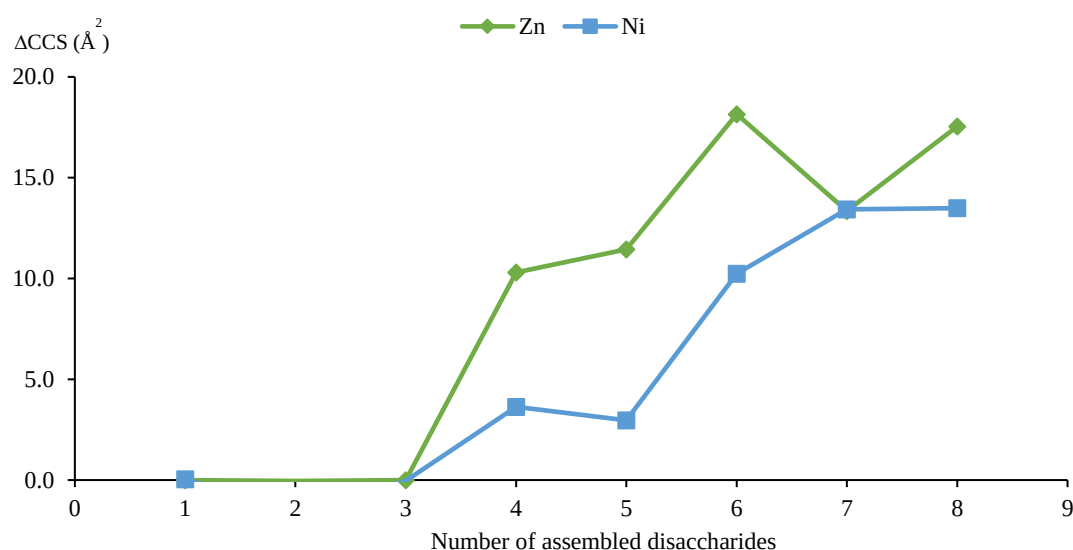


Figure 3-7. The CCS differences between Mal assembly ions and IsoMal assembly ions at different assembly states, with NiCl_2 (blue) and ZnCl_2 (green).

Maltose and cellobiose are configuration isomers of disaccharides. They have the same monomer (Glc) and connectivity (1→4 linkage), but maltose belongs to α -anomers while cellobiose belongs to β -anomers. According to the results, the separation in IM spectra and CCSs between the assemblies of maltose and cellobiose were also enhanced upon the assembly level. As shown in Figure 3-8 and Table 3-5, the IM separations of the assemblies were found to be enhanced with the assembled molecules from 1 to 8, allowing the ΔCCS between Cel and Mal assemblies to increase from 1.8 to 19.7 $\text{\AA}^2$ and from 1.8 to 13.5 $\text{\AA}^2$ with Zn and Ni ions, respectively. Similarly, comparing with the α - configuration of Mal, the β - configuration of Cel made it drift even slower, representing a less compact structure.

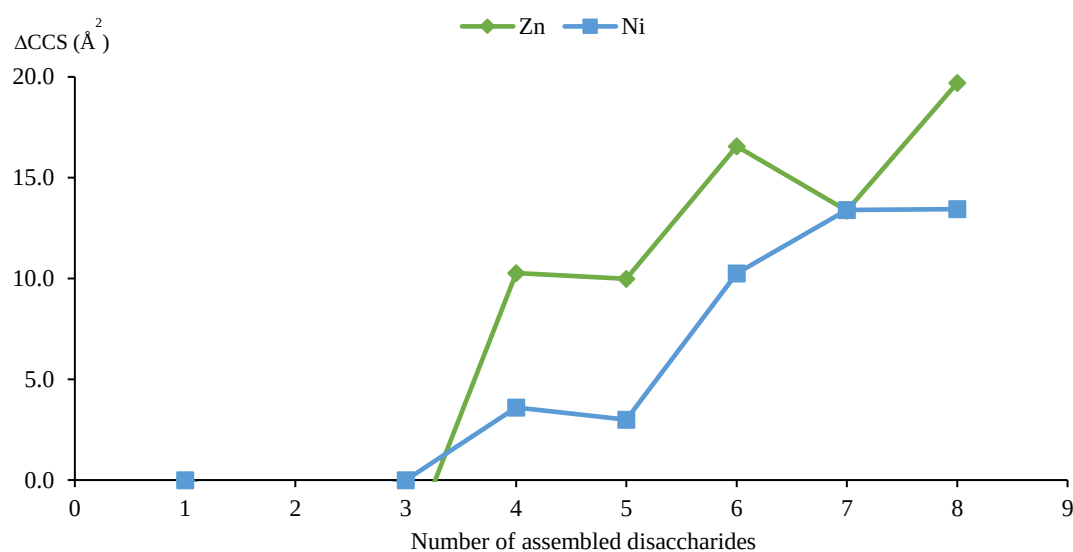


Figure 3-8. The effect of assembly on CCS differences among disaccharide isomers (Mal vs. Cel) with addition of NiCl_2 (blue) or ZnCl_2 (green).

Isomaltose and sucrose are isomers with different linkage and composition as listed in Table 3-3. Isomal contains two Glc with 1→6 linkage, Suc contains one Glc and Fru with 1→2 linkage. As shown in Figure 3-9 and Table 3-5, before assembly, their ΔCCS is $1.8 \text{ \AA}^2$, with the assembled molecules increased to 8, the ΔCCS s between Isomal and Suc assemblies were only increased to $4.3 \text{ \AA}^2$ and $2.6 \text{ \AA}^2$ with Zn and Ni ions. The results indicated that not all the isomers could be better differentiated after assembly. However, when $n=7$, the ΔCCS s between Isomal and Suc assemblies could be increased to $9.8 \text{ \AA}^2$ and $9.6 \text{ \AA}^2$ with Zn and Ni ions. Further exploration and interpretation on the structure of assemblies are needed to explain this phenomenon.

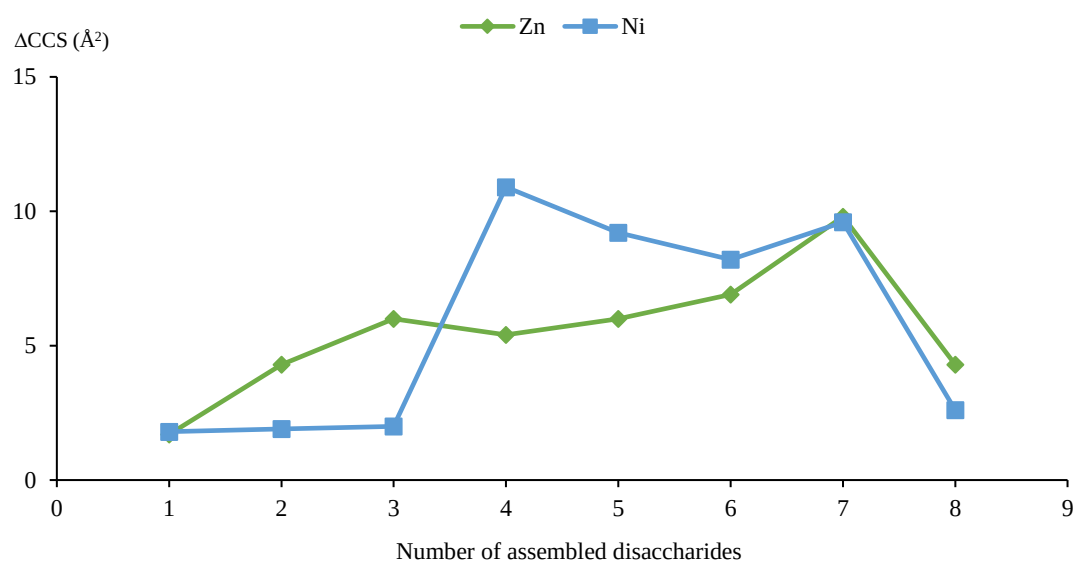


Figure 3-9. The effect of assembly on CCS differences among disaccharide isomers (Isomal vs. Suc) with addition of NiCl_2 (blue) or ZnCl_2 (green).

All the enhanced ΔCCS trends was compared in Figure 3-10. The results demonstrated that assembly of the disaccharide isomers could induce larger differences for their differentiation by IM-MS, particularly for the composition, connectivity and configuration isomers. The difference between α -anomers and β -anomers, 1 $\rightarrow$ 4 linkage and 1 $\rightarrow$ 6 linkage could be amplified better than others. This study extended the assembly approach from amino acids to saccharides and suggested its potential applications for differentiation of other isomers.

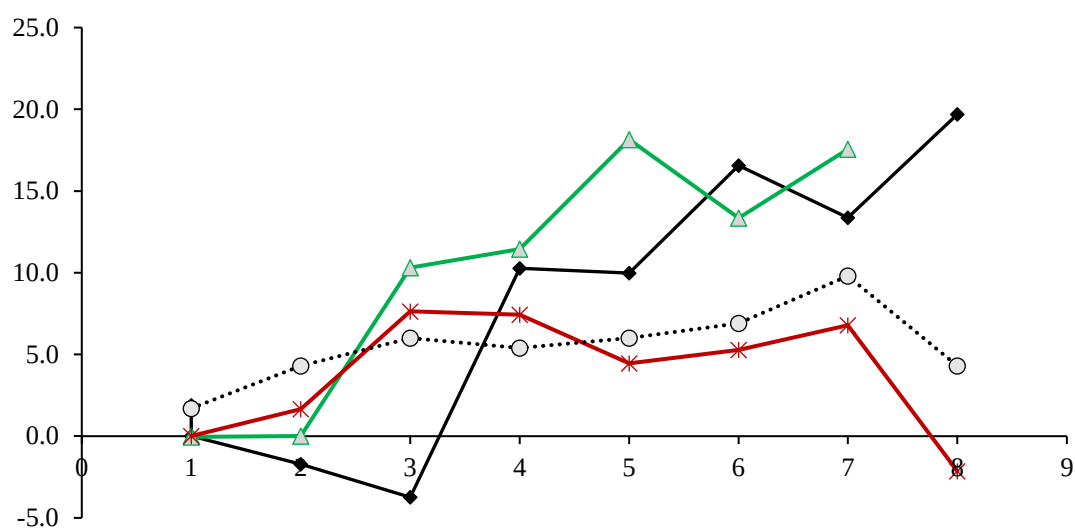


Figure 3-10. The effect of self-assembly on ΔCCS between Mal vs. Cel (black), Mal vs. Isomal (green), Cel vs. Lac (red) and Isomas vs. Suc (black dot) with addition of Zn ion.

According to some studies on the architectures of Ni- disaccharides,¹³⁴⁻¹³⁵ the structure of transition metal and saccharides are folding structure or metal-central saccharides surrounding. Because the main difference between of Cel and Mal, Mal and Isomal is angle as discussed above, herein we proposed that the angle will affect the shape such as concaveness as mentioned in previous chapter of the assemblies thus enhanced the differentiation in IM. Further modeling is still ongoing.

Table 3-4. Summary of the t_d and CCSs measured for the metal bound assemble ions ($n = 2$ measurements) of monosaccharides (glucose, fructose, galactose and mannose).

Composition	m/z	Glucose		Fructose		Galactose		Mannose	
		t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]	t_d [ms]	CCS _{exp} (Ω_{He}) [$\text{\AA}^2$]
$[(M) + K]^+$	219.0296	2.29±0.11	79.1±0.3	2.22	77.0	2.22	77.0	2.15	75.0
$[(M)_2 + K - H_2O]^+$	381.0884	3.60	113.5	3.46	109.7	3.53	111.6	3.53	111.6
$[(M) + Ni^{II} - H]^+$	236.9962	2.22	76.7	2.29	78.8	2.22	76.7	2.22	76.7
$[(M) + Ni^{II} + H_2O - H]^+$	254.9595	2.29	78.4	2.29	78.4	2.29	78.4	2.29	78.4
$[(M) + Ni^{II} + 2H_2O - H]^+$	272.9726	2.42	82.0	2.42	82.0	2.42	82.0	2.42	82.0
$[(M)_2 + Ni^{II} + 2H_2O - H]^+$	453.0404	3.95	122.2	3.81±0.11	118.5±1.2	3.88	120.3	3.88	120.3
$[(M)_2 + (Ni^{II})_2 - 3H]^+$	472.9822	3.60	112.7	3.60	112.7	N/A	N/A	3.60	112.7
$[(M)_2 + (Ni^{II})_3 + 3H_2O - 7H]^+$	580.9207	4.71	141.0	4.85	144.6	4.85	144.6	4.57	137.4
$[(M) + Zn^{II} - H]^+$	242.995	2.22	76.6	2.22	76.6	2.22	76.6	2.22	76.6
$[(M) + Zn^{II} + H_2O - H]^+$	260.9614	2.35	80.1	2.35	80.1	2.29	78.3	2.29	78.3
$[(M) + Zn^{II} + 2H_2O - H]^+$	278.9702	2.42	81.9	2.49	83.9	2.42±0.11	81.9±1.1	2.42	81.9
$[(M)_2 + Zn^{II} + 2H_2O - H]^+$	459.0443	4.16	127.7	3.88	120.3	3.88	120.3	3.95	122.1

Table 3-5. Summary of the t_d and CCSs measured for the metal bound assemble ions ($n = 2$ measurements) of disaccharides (maltose, cellobiose, isomaltose, lactose and sucrose).

Composition	m/z	Maltose		Cellobiose		Isomaltose		Lactose		Sucrose	
		t_d	CCS _{exp}	t_d	CCS _{exp}	t_d	CCS _{exp}	t_d	CCS _{exp}	t_d	CCS _{exp}
		[ms]	(Ω_{He}) [$\text{\AA}^2$]	[ms]	(Ω_{He}) [$\text{\AA}^2$]	[ms]	(Ω_{He}) [$\text{\AA}^2$]	[ms]	(Ω_{He}) [$\text{\AA}^2$]	[ms]	(Ω_{He}) [$\text{\AA}^2$]
$[(M) + Zn^{II} - H]^+$	405.0194	3.46	109.5	3.46	109.5	3.46	109.5	3.46	109.5	3.37	107.8
$[(M)_2 + Zn^{II}]^{2+}$	374.0583	2.35	154.8	2.42	158.8	2.35	154.8	2.35	154.8	2.29	150.8
$[(M)_2 + Zn^{II} - H]^+$	747.1177	6.37	181.6	6.44	183.3	6.37	181.6	6.37±0.11	181.6±1.1	6.16	177.3
$[(M)_3 + Zn^{II}]^{2+}$	545.1104	3.32	206.8	3.39	210.5	3.32	206.8	3.19	202.9	3.12	200.8
$[(M)_4 + Zn^{II}]^{2+}$	716.1688	4.29	260.2	4.16	249.9	4.16	249.9	3.88	242.5	3.88	244.5
$[(M)_5 + Zn^{II}]^{2+}$	887.2247	5.12	301.3	4.99	291.4	4.92	289.9	4.78	286.9	4.64	283.9
$[(M)_6 + Zn^{II}]^{2+}$	1058.2683	6.02	345.0	5.75	328.4	5.68	326.8	5.54	323.1	5.40	319.9
$[(M)_7 + Zn^{II}]^{2+}$	1229.3314	6.79	381.5	6.51±0.11	368.1±1.8	6.51	368.1	6.37	361.3	6.30	358.3
$[(M)_8 + Zn^{II}]^{2+}$	1400.3959	7.76	423.5	7.27	403.8	7.34	405.9	7.34	405.9	7.20	401.6
$[(M) + Ni^{II} - H]^+$	399.0234	3.32	105.7	3.32	105.7	3.32	105.7	3.32	105.7	3.24	103.9
$[(M)_2 + Ni^{II}]^{2+}$	371.0668	2.35	154.8	2.42	158.8	2.42	158.8	2.42	158.8	2.35	156.8
$[(M)_2 + Ni^{II} - H]^+$	741.1274	6.44±0.11	183.3±1.3	6.51	185.0	6.51	185.0	6.51	185.0	6.44	183.1

$[(M)_3 + Ni^{II}]^{2+}$	542.1178	3.39	210.6	3.39	210.6	3.39	210.6	3.25	206.6	3.32	208.6
$[(M)_4 + Ni^{II}]^{2+}$	713.1694	4.29	256.6	4.22	253.0	4.22	253.0	3.94	245.3	3.80	242.1
$[(M)_5 + Ni^{II}]^{2+}$	884.2234	5.12	297.9	5.06	294.9	5.06	294.9	4.92	289.9	4.78	285.7
$[(M)_6 + Ni^{II}]^{2+}$	1055.2739	5.96	338.6	5.75	328.4	5.75	328.4	5.61±0.11	323.4±1.9	5.48	320.2
$[(M)_7 + Ni^{II}]^{2+}$	1226.3177	6.79	378.1	6.51	364.7	6.51	364.7	6.37	358.6	6.30	355.1
$[(M)_8 + Ni^{II}]^{2+}$	1397.3752	6.62	369.6	6.34	356.1	6.34	356.1	6.34	356.1	6.20	353.5

3.4 Conclusions

The current study employed TWIM-MS to analyze different types of saccharides by forming complexes with metal ions in the gas phase. Among the tested alkali metal ions and transition metal ions, Zn and Ni ions were found to be more suitable to form complexes with saccharides, allowing complex ions up to metal bound dimers and metal bound octamers of monosaccharides and disaccharides were detected. It was observed that the assemblies tended to enlarge the differences of the isomers in IM-MS and allowed their better differentiations. The increasing CCSs difference of assemblies between Lac and Gal who are composition isomers of disaccharide indicating that although the monosaccharide isomers could not be separated well by IM, when they are the monomeric unit in disaccharides, the differences could be accumulated after assembly thus enhanced discrimination in IM-MS. The connectivity difference between 1→4 linkages in IsoMal and 1→6 linkage in Mal also could be accumulated after assembly, and the former structures of assemblies were more compact than the latter ones. The α -configuration in Mal is reported more compact than the β -configuration in Cel, however the assemblies of Mal showed fewer compact structures than the assemblies of Cel, indicating the configuration difference probably be transferred during assembly. Suc constituted with Fru who has five carbons in the ring showed compact structures always whatever before or after assembly than others. The reported study suggested that the structures of the metal-disaccharide assemblies might be metal-centered with the disaccharides surrounded and the difference of the configuration, linkage and monomer unit might cause the diameter or surface concaveness differences and thus their enhanced differentiations.

Chapter 4: Chiral recognition of propranolol by IM-MS

4.1 Introduction

Chirality is a common phenomenon in nature and chemical synthesis.¹³⁶ A chiral molecule has two mirror-image forms called enantiomers. Many basic biomolecules such as amino acids, saccharides and nucleotides are chiral, causing the chiral environment in living systems. The two enantiomers of chiral drugs can have different metabolic rates and biological effects, e.g., one enantiomer may have treatable effects, while the other may have no effect or even side-effects.^{22, 137} Because the pair of enantiomers are commonly coexisting in isolated and synthesized products, chiral analysis is essential and commonly applied in various fields including pharmaceutical science, organic synthesis, natural product chemistry, biochemistry, etc.

Propranolol, a chiral drug (Figure 4-1), belongs to beta blockers, which are nonselective competitive antagonists that block the receptor sites for the endogenous catecholamines epinephrine and norepinephrine on adrenergic beta receptors.¹³⁸ It is commonly used to treat hypertension, angina, atrial fibrillation, myocardial infarction, migraine and some other heart diseases. The adrenergic beta receptors have strong enantioselectivity on beta blocker drugs in human body because of the different conformational states of the receptors when the chiral drugs are binding in the pocket.¹³⁹ Reported protein-propranolol docking revealed that the binding energy of beta-2 receptor and S-propranolol is lower than that of R-propranolol.¹⁴⁰⁻¹⁴¹ It is reported that the S(-)-enantiomer has approximately 100 times affinity in blocking beta adrenergic receptors than the R(+)-enantiomer.¹³⁸ In

normal subjects receiving oral doses of racemic propranolol, S(-)-enantiomer concentrations in blood exceeded those of the R(+)-enantiomer by 40 to 90% due to the stereoselective hepatic metabolism. Clearance of the pharmacologically active S(-)-propranolol is lower than R(+)-propranolol after intravenous and oral doses.¹⁴² Due to the costs and challenging of production of S(-)-propranolol, it is sold as a racemic mixture with equal ratio enantiomers. Development of approaches for chiral differentiation of propranolol is thus necessary.

Chiral differentiation by MS has attracted significant research interest in past years.^{29-30, 33, 41} Especially with the development of MS/MS where fragmentation patterns enabled the study of energy differences between the pair of complex ions formed from the two enantiomers and chiral selectors, the discrimination of enantiomers could be achieved.¹⁴³ The two most common methods are the kinetic method (KM) and the chiral recognition ratio (CR) method, in the former method is achieved by the comparison on intensities of two product ions, while the latter method is achieved by the abundance ratio of product ion to precursor ion. The new increasingly popular tool for differentiation of enantiomers was IM-MS,³⁸⁻³⁹ as the coupling of ion mobility with MS provides an additional dimension to resolve enantiomer ions indistinguishable by MS alone. There were two reports about the chiral recognition of propranolol by MS, in which the propranolol was studied by KM and CR methods through copper bound trimers.¹⁴⁴⁻¹⁴⁵ As our mentioned in previous chapter, assembly could enhance the differentiation by MS. However, the largest complex ion of propranolol found is trimer and no exploration on the IM study. According to the discussion in previous chapters, that assembly might enhance the differentiation in MS, in this chapter, we would like to extend the method to enantiomers.

However, for isomers such as AAs and saccharides mentioned in previous chapter, we are isomers and can self-assemble to enlarge the differences of their monomers for differentiation by IM-MS. But for enantiomers, their monomeric molecules have the same IM-MS behaviours, so we need to introduce chiral selectors to cause their differences and further use assembly to enlarge these differences for the differentiation. In this chapter, several types and sizes of chiral selectors, including cyclodextrins ($\alpha/\beta/\gamma$ -CD), π -complex selector (R-N-(3,5-dinitrobenzoyl)- α -phenylglycine), crown ether (18-crown-6), cinchona alkaloids (quinine/quinidine) and metal-AAs, were selected and evaluated for assembly. The effect of assembly on discrimination and differentiation of propranolol by MS/MS and IM-MS was studied.

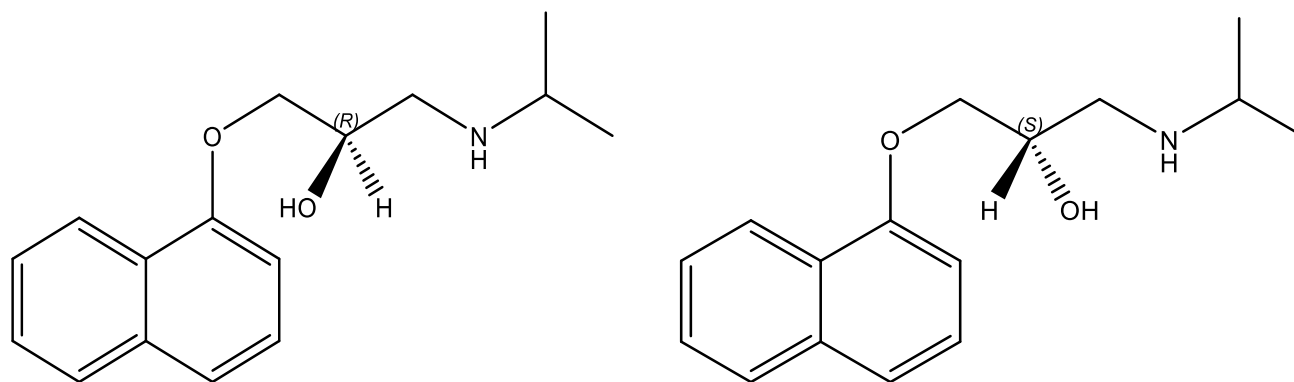


Figure 4-1. Structures of (S) and (R)-propranolol.

4.2 Experimental

4.2.1 Materials

The optically pure propranolol (S-propranolol and R-propranolol) samples were purchased from Sigma-Aldrich Co (St. Louis, MO). The chiral selectors (α -CD, β -CD, γ -CD, R-N-(3,5-dinitrobenzoyl)- α -phenylglycine, 18-crown-6, quinine, quinidine, L-Phe, *N*-Boc-L-Phe, L-His, L-Asn, L-Pro, R-epinephrine and S-naproxen), and the metal salts ($\text{Cu}(\text{CH}_3\text{COOH})_2$, CuCl_2 , ZnCl_2 , NiCl_2 , MgCl_2 and LiNO_3 , NaNO_3 , KNO_3) were purchased from Aladdin (Shanghai, China). The HPLC grade methanol was purchased from BCR International Trading Ltd. Co (California, USA). Ultrapure water was produced by a MilliQ device (Millipore, Milford, MA, USA). All the chemicals were used directly without further purification.

4.2.2 Sample preparation

S-propranolol, R-propranolol, chiral selectors and metal salts were dissolved separately in water to achieve a concentration of 5 mM. The sample solution used for MS analysis was prepared by mixing the stock solutions, and then diluting with 50:50 water/methanol, to obtain a series of final solutions each containing 50 μM metal salts with 5 μM , 25 μM , 50 μM , 100 μM , 250 μM , 500 μM or 1000 μM S-propranolol or R-propranolol, and 50 μM , 250 μM , 500 μM or 1000 μM chiral selectors.

4.2.3 Mass Spectrometry

MS/MS and IM-MS experiments were performed using a Synapt G2 HDMS mass

spectrometer (Waters, Manchester, UK). ESI conditions were as follows: capillary voltage, 3 kV; sampling cone, 40 V; extraction cone, 5 V; source temperature, 100 °C; desolvation temperature, 350°C; cone gas, 50 L h⁻¹; desolvation gas, 500 L h⁻¹. All the experiments were performed in the positive ion mode. The sample solutions were directly introduced to the mass spectrometer through the ESI source at a flow rate of 10 µL min⁻¹. All the spectral data represented the combined 50 scans in the direct infusion experiments. For the traveling wave ion mobility (TWIM) experiments, the drift gas was N₂, the traveling wave height was set at 40 V, and traveling wave velocity was chosen as 800 m s⁻¹. The mass spectra acquired were analyzed by the Driftscope program. The CCS values of the investigated complexes were calculated by using tetraalkylammonium ions (TAA-n, n=1-8) with a concentration of 10 mg mL⁻¹ to plot a calibration curve as described by the literature,⁴⁵ and the CCS values of polyalanine were obtained from the literature.¹³³ The mass spectra acquired were analyzed by the Driftscope program.

4.3 Results and discussion

4.3.1 Optimization for detection of complex ions

Different types and sizes of chiral selectors, as shown in Table 4-1, were chosen for chiral differentiation of the analyte propranolol (indicated as A in the formula). Among all these selectors, metal-AAs or -analogues showed stronger binding affinities than the others. The detection of the complex ions of propranolol was found to be more related with the interactions between propranolol and the chiral selectors. For example, when cyclodextrins were applied as the selectors, no complex ions could be detected even with the pH and solvents adjusted, probably because the cavity sizes of the cyclodextrins were

not matching the naphthyl group very well and the hydrogen bonds between propranolol and cyclodextrins were weak. Similar phenomena were observed for another host-guest chiral selector, 18-crown-6, whose cavity is smaller than those of cyclodextrins. When (R)-(-)-N-(3,5-dinitrobenzoyl)- α -phenylglycine, 1,2-diaminocyclohexane and quinine/quinidine were used as the chiral selectors, no significant signals of the complex ions containing the selectors and propranolol were observed, presumably due to the weak π - π interaction between propranolol and (R)-(-)-N-(3,5-Dinitrobenzoyl)- α -phenylglycine, and weak hydrogen bonding between propranolol and quinine/quinidine or 1,2-diaminocyclohexane.

For the metal-AAs or analogues, metal ions including Zn, Ni, Mg, Cu, K, Na ions, AAs and analogues including chiral Phe, *N*-Boc-Phe, Pro, Asn, His, naproxen (nap) and epinephrine were used to interact with propranolol. As discussed in the previous chapter, the concentration ratio of metal ion and ligand was crucial for the detection of more metal-AAs complex ions. When the pair of metal ion and AAs or analogues (M) was at a proper concentration ratio, different sizes of complex ions could be detected at once, as shown in Table 4-2, all of them could interact with the propranolol in MS to form assemblies. Among all the added metal ions, Cu²⁺ could allow better detection of complex ions for selection. For example, when concentration ratio of Phe and copper ion was over 10:1, complex ions such as [Cu₄(M)₆]²⁺, [Cu₄(M)₇]⁺, [Cu₅(M)₈]²⁺, [Cu₆(M)₁₀]²⁺, [Cu₇(M)₁₂]²⁺ could be observed in the spectrum. However, no complex ion was observed in the spectrum for copper and epinephrine, probably due to their weaker interactions.

Table 4-1. Chiral selectors investigated for chiral recognition of propranolol.

Types	Chiral selectors
Metal-AAs and analogues	Metals: Zn, Ni, Mg, Cu, K, Na AAs: L-Phe, L-Pro, L-Asn, L-His, <i>N</i> -Boc-L-Phe Analogues: S-naproxen and R-epinephrine.
π -Complexes	(R)-(-)- <i>N</i> -(3,5-Dinitrobenzoyl)- α -phenylglycine
Crown ethers	18-Crown-6
Cyclodextrins	Heptakis(2,3,6-tri- <i>O</i> -methyl)- β -cyclodextrin (2-Hydroxypropyl)- α -cyclodextrin (2-Hydroxypropyl)- β -cyclodextrin (2-Hydroxypropyl)- γ -cyclodextrin 3A-Amino-3A-deoxy-(2AS, 3AS)- β -cyclodextrin hydrate
Cinchona alkaloids	Quinine and quinidine
Others	(1R,2R)-(-)-1,2-Diaminocyclohexane (1S,2S)-(+)-1,2-Diaminocyclohexane

Table 4-2. Typical copper-bound assembled ions observed for the chiral selectors.

Phe	<i>N</i>-Boc- Phe	His	Pro	Asn	Naproxen
$[(M) + Cu^I]^+$	$[(M) + Cu^I]^+$	$[(M) + Cu^I]^+$	$[(M) + Cu^I]^+$	$[(M) + Cu^I]^+$	$[(M) + Cu^I]^+$
$[(M)_2 + Cu^{II} - H]^+$	$[(M)_2 + Cu^{II} - H]^+$	$[(M)_2 + Cu^{II} - 2H + H_2O]^+$	$[(M)_2 + Cu^{II} - H]^+$	$[(M)_2 + Cu^{II} - H]^+$	$[(M)_2 + Cu^I]^+$
$[(M)_2 + Cu^{II} + Cu^I - 2H]^+$	$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$[(M)_2 + Cu^{II} + Cu^I - 2H]^+$	$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$[(M)_4 + (Cu^{II})_2 - 3H]^+$
$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$[(M)_4 + (Cu^{II})_2 - H]^+$	$[(M)_3 + (Cu^{II})_2 - 3H]^+$	$[(M)_3 + Cu^{II} + (Cu^I)_2 - 3H]^+$	$[(M)_4 + (Cu^{II})_2 - 3H]^+$	
$[(M)_4 + (Cu^{II})_2 - 3H]^+$		$[(M)_4 + (Cu^{II})_2 - 3H]^+$	$[(M)_4 + (Cu^{II})_2 - 3H]^+$	$[(M)_5 + (Cu^{II})_3 - 5H]^+$	
$[(M)_4 + (Cu^{II})_2 + (Cu^I)_2 - 4H]^{2+}$		$[(M)_5 + (Cu^{II})_3 - 5H]^+$	$[(M)_5 + (Cu^{II})_3 - 5H]^+$	$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	
$[(M)_5 + (Cu^{II})_3 - 5H]^+$		$[(M)_6 + (Cu^{II})_2 - 2H]^+$	$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	$[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$	
$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$		$[(M)_6 + (Cu^{II})_4 - 5H]^+$	$[(M)_6 + (Cu^{II})_3 - 5H]^+$	$[(M)_{10} + (Cu^{II})_6 - 10H]^{2+}$	
$[(M)_7 + (Cu^{II})_4 - 7H]^+$		$[(M)_6 + (Cu^{II})_4 - 6H]^{2+}$	$[(M)_7 + (Cu^{II})_4 - 7H]^+$		
$[(M)_8 + (Cu^{II})_5 - 8H]^{2+}$		$[(M)_8 + (Cu^{II})_5 - 6H]^{2+}$	$[(M)_9 + (Cu^{II})_5 - 9H]^+$		
$[(M)_{10} + (Cu^{II})_6 - 10H]^{2+}$		$[(M)_8 + (Cu^{II})_6 - 10H]^{2+}$			
$[(M)_{12} + (Cu^{II})_7 - 12H]^{2+}$		$[(M)_{10} + (Cu^{II})_7 - 10H]^{2+}$			

The additional ratio of propranolol (Prp) was also important for detection of the complex ions. As shown in Figure 4-2, with a ratio of 1:1:1 or 1:3:3 for propranolol: Cu : Phe, complex ions at m/z 486.1877, 580.2681, 745.3472 and 817.3083, corresponding to $[\text{Cu}^{\text{II}} + (\text{Phe}) + (\text{Prp}) - \text{H}]^+$, $[\text{Cu}^{\text{II}} + (\text{Prp})_2 - \text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) + (\text{H}_2\text{O})_2 - 6\text{H}]^+$ and $[\text{Cu}^{\text{II}} + (\text{Phe})_3 + (\text{Prp}) - \text{H}]^+$ respectively, could be detected. When the ratio of Prp: Cu : Phe was 1:1:10, complexes ions at m/z 486.1877, 651.2370, 712.1509, 817.3238, 878.2377, 1042.3089 and 1268.3018, corresponding to $[\text{Cu}^{\text{II}} + (\text{Phe}) + (\text{Prp}) - \text{H}]^+$, $[\text{Cu}^{\text{II}} + (\text{Phe})_2 + (\text{Prp}) - \text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$, $[\text{Cu}^{\text{II}} + (\text{Phe})_3 + (\text{Prp}) - \text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_3 + (\text{Prp}) - 3\text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_4 + (\text{Prp}) - 3\text{H}]^+$ and $[(\text{Cu}^{\text{II}})_3 + (\text{Phe})_5 + (\text{Prp}) - 5\text{H}]^+$ respectively, were detected. It could be observed that the increasing concentration ratio of ligand enabled detection of larger complex ions with more numbers of ligand, metal ion and propranolol.

Different AAs and analogues were investigated and compared with the concentration ratio of Prp: Cu : M at 1:1:10. The complexes ions were identified through analyses of accurate monoisotopic masses, isotope patterns and charge states (see Table 4-3). It could be observed that Phe and Pro allowed better detection of the complex ions with copper ion and Prp. The largest complex ion detected was $[(\text{Cu}^{\text{II}})_3 + (\text{M})_6 + (\text{Prp}) - 5\text{H}]^+$ with Pro as the ligand. Interestingly, *N*-Boc-Phe showed much lower assembly ability with copper ion and Prp, compared to Phe, and although high assembled ions were

observed for Asn and His alone with copper ion, their higher assembled ions with Prp were not observed for their solutions with propranolol. Further adjustment of the pH value and MS parameters such as temperature and cone energy did not lead to detection of other complex ions, indicating that the detectable assembled ions were more related to the structures and intrinsic properties of the molecules.

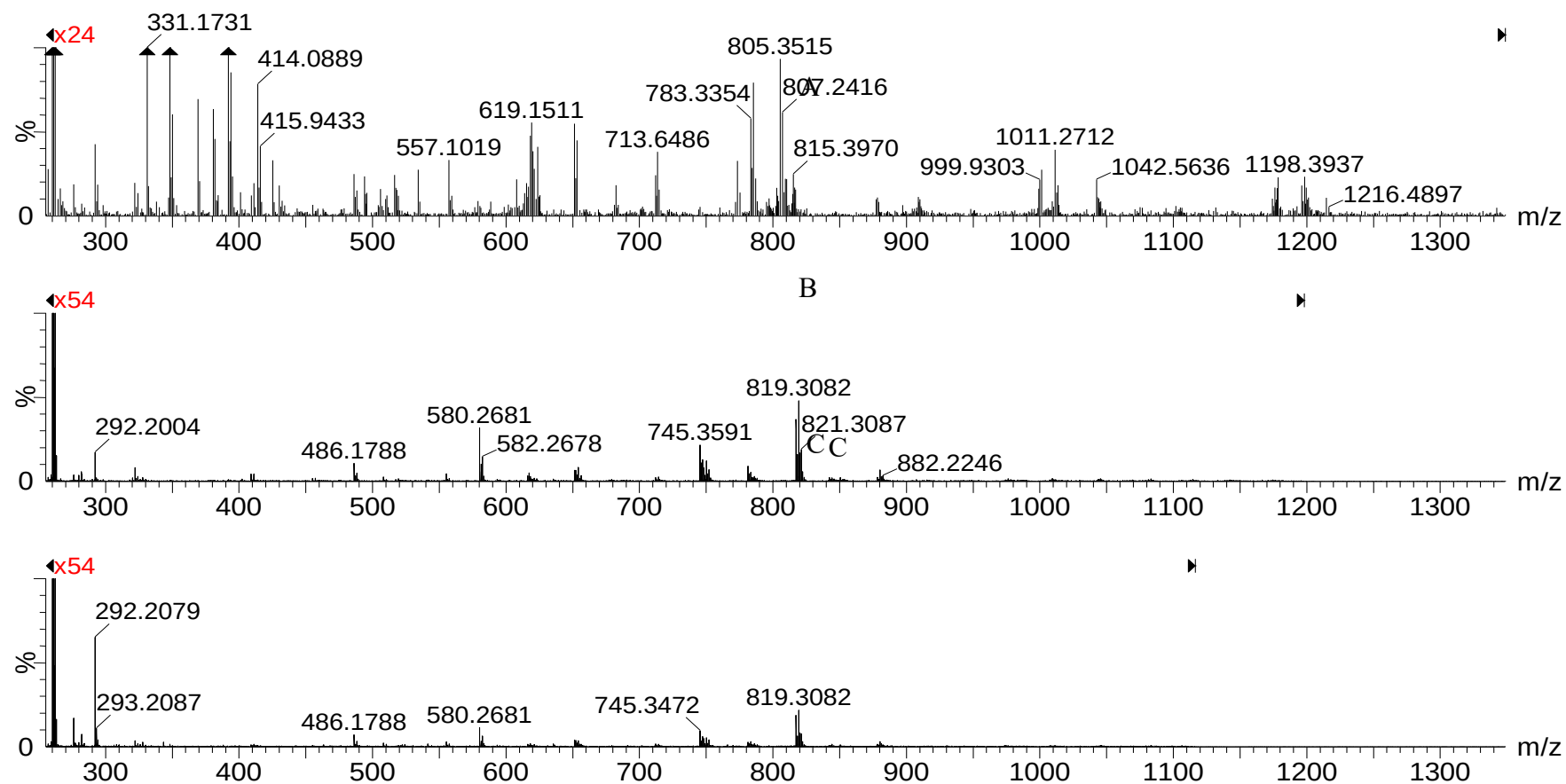


Figure 4-2. ESI-MS spectra for propranolol with addition of $\text{Cu}(\text{COOCH}_3)_2$ and Phe, at concentration ratio of Prp: copper: Phe (A) 1:1:10, (B) 1:3:3, (C) 1:1:1. The assignments of major peaks are summarized in Table 4-3.

Table 4-3. Copper bound complexes ions of propranolol observed with different selectors.

Phe	N-Boc- Phe	His	Pro	Asn	Naproxen
$[(M) + (Prp) + Cu^I - H]^+$ (<i>m/z</i> 486)	$[(M) + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 586)	$[(M) + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 476)	$[(M) + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 436)	$[(M) + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 453)	$[(M) + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 551)
$[(M)_2 + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 651)	$[(M)_2 + (Prp) + Cu^{II} - H_2O]^+$ (<i>m/z</i> 834)	$[(M)_2 + (Prp) + (Cu^I)_2 - 2H]^+$ (<i>m/z</i> 694)	$[(M)_2 + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 551)	$[(M)_2 + (Prp) + Cu^{II} - H]^+$ (<i>m/z</i> 585)	$[(M)_2 + (Prp) + Cu^{II} - 3H]^+$ (<i>m/z</i> 779)
$[(M)_2 + (Prp) + Cu^{II} + Cu^I - 3H]^+$ (<i>m/z</i> 712)		$[(M)_5 + (Prp) + (Cu^{II})_3 - 13H]^+$ (<i>m/z</i> 1209)	$[(M)_3 + (Prp) + (Cu^{II})_2 - 3H]^+$ (<i>m/z</i> 727)	$[(M)_4 + (Prp) + (Cu^{II})_2 - 3H]^+$ (<i>m/z</i> 910)	$[(M)_2 + (Prp) + (Cu^{II})_2 - 3H]^+$ (<i>m/z</i> 842)
$[(M)_3 + (Prp) + Cu^I]^+$ (<i>m/z</i> 817)			$[(M)_4 + (Prp) + (Cu^{II})_2 - 3H]^+$ (<i>m/z</i> 842)		$[(M)_3 + (Prp) + (Cu^{II})_2 - 3H]^+$ (<i>m/z</i> 1072)
$[(M)_3 + (Prp) + Cu^{II} + Cu^I - 2H]^+$ (<i>m/z</i> 878)			$[(M)_5 + (Prp) + (Cu^{II})_3 - 4H]^+$ (<i>m/z</i> 1019)		$[(M)_3 + (Prp) + (Cu^{II})_3 - 3H]^+$ (<i>m/z</i> 1135)
$[(M)_4 + (Prp) + (Cu^{II})_2 - 3H]^+$ (<i>m/z</i> 1042)			$[(M)_6 + (Prp) + (Cu^{II})_3 - 6H]^+$ (<i>m/z</i> 1132)		$[(M)_4 + (Prp) + (Cu^{II})_3 - 3H]^+$ (<i>m/z</i> 1365)
$[(M)_5 + (Prp) + (Cu^{II})_3 - 5H]^+$ (<i>m/z</i> 1268)					

4.3.2 Chiral differentiation of propranolol by IM-MS

The effect of assembly on differentiation of chiral propranolol by IM-MS was investigated in this project. The differentiation of S- and R-propranolol by the copper bound complexed ions was evaluated and compared using the drift time and CCSs, with the results summarized in Table 4-4. The differentiation by IM of enantiomers after assembly was shown in Figure 4-3, with Phe as the chiral selector, differentiation of the two enantiomers became more significant with the binuclear copper bound trimeric ions $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$, but less significant with $[\text{Cu}^{\text{II}} + (\text{Phe})_3 + (\text{Prp}) - \text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_3 + (\text{Prp}) - 3\text{H}]^+$ and $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_4 + (\text{Prp}) - 3\text{H}]^+$. We further compared the CCS difference between assemblies of enantiomers. As we could see, when Pro as the selector, the largest drift time difference was 0.20 and 0.21 with ΔCCS as 4.0 and 4.5 $\text{\AA}^2$ in the complex ions $[\text{Cu}^{\text{II}} + (\text{Pro}) + (\text{Prp}) - \text{H}]^+$ and $[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_4 + (\text{Prp}) - 3\text{H}]^+$. The largest ΔCCS we observed is 6.4 $\text{\AA}^2$ with Phe as selector in complex ion $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$. The results suggested an unclear trend for the effect of the assembly on the chiral differentiation. The reason probably due to all the assemblies we observed was containing only one chiral drug, thus the difference between enantiomers could not be accumulated during assembly thus could not reflect on the size or shape of the final assembly products. However, due to the ΔCCS between S-Prp and R-Prp actually is 0, we could confirm that with addition of chiral selectors, the difference of enantiomer was allowed to be detected.

Table 4-4. Summary of the chiral discrimination observed by IM-MS of propranolol (Prp) complexes ions

Complex ion	Δt_d [ms]	ΔCCS_{exp} (Ω_{He}) [$\text{\AA}^2$]
$[\text{Cu}^{\text{II}} + (\text{His}) + (\text{Prp}) - \text{H}]^+$	0.07 ± 0.07	0.5
$[(\text{Cu}^{\text{II}})_2 + (\text{His})_2 + (\text{Prp}) - 3\text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_3 + (\text{His})_5 + (\text{Prp}) - 13\text{H}]^+$	0.00	0.0
$[\text{Na}^{\text{I}} + (\text{Prp})_2]^+$	0.00	0.0
$[\text{Cu}^{\text{I}} + (\text{Phe}) + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{Prp})_2 - \text{H}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{Phe})_2 + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{Phe})_3 + (\text{Prp}) - \text{H}]^+$	0.07 ± 0.07	0.8 ± 0.8
$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$	0.35 ± 0.11	6.4 ± 1.5
$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_3 + (\text{Prp}) - 3\text{H}]^+$	0.14 ± 0.11	2.2 ± 1.7
$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_4 + (\text{Prp}) - 3\text{H}]^+$	0.07	0.9
$[(\text{Cu}^{\text{II}})_3 + (\text{Phe})_5 + (\text{Prp}) - 5\text{H}]^+$	--	--
$[\text{Cu}^{\text{II}} + (\text{Pro}) + (\text{Prp}) - \text{H}]^+$	0.21 ± 0.11	4.5 ± 1.1
$[\text{Cu}^{\text{II}} + (\text{Pro})_2 + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_3 + (\text{Prp}) - 3\text{H}]^+$	0.07	1.3
$[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_4 + (\text{Prp}) - 3\text{H}]^+$	0.20 ± 0.11	4.0 ± 1.5
$[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_5 + (\text{Prp}) - 4\text{H}]^+$	--	--
$[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_6 + (\text{Prp}) - 5\text{H}]^+$	--	--
$[\text{Cu}^{\text{II}} + (N\text{-Boc-Phe}) + (\text{Prp}) - \text{H}]^+$	0.28 ± 0.11	4.5 ± 1.3
$[\text{Cu}^{\text{II}} + (N\text{-Boc-Phe})_2 + (\text{Prp}) - \text{H}_2\text{O}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{Asn}) + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{Asn})_2 + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_2 + (\text{Asn})_4 + (\text{Prp}) - 3\text{H}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{nap}) + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[\text{Cu}^{\text{II}} + (\text{nap})_2 + (\text{Prp}) - \text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_2 + (\text{nap})_2 + (\text{Prp}) - 3\text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_2 + (\text{nap})_3 + (\text{Prp}) - 3\text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_3 + (\text{nap})_3 + (\text{Prp}) - 3\text{H}]^+$	0.00	0.0
$[(\text{Cu}^{\text{II}})_3 + (\text{nap})_4 + (\text{Prp}) - 3\text{H}]^+$	0.00	0.0

System error of IM is $\Delta t_d \approx \pm 0.11 \text{ ms}$.

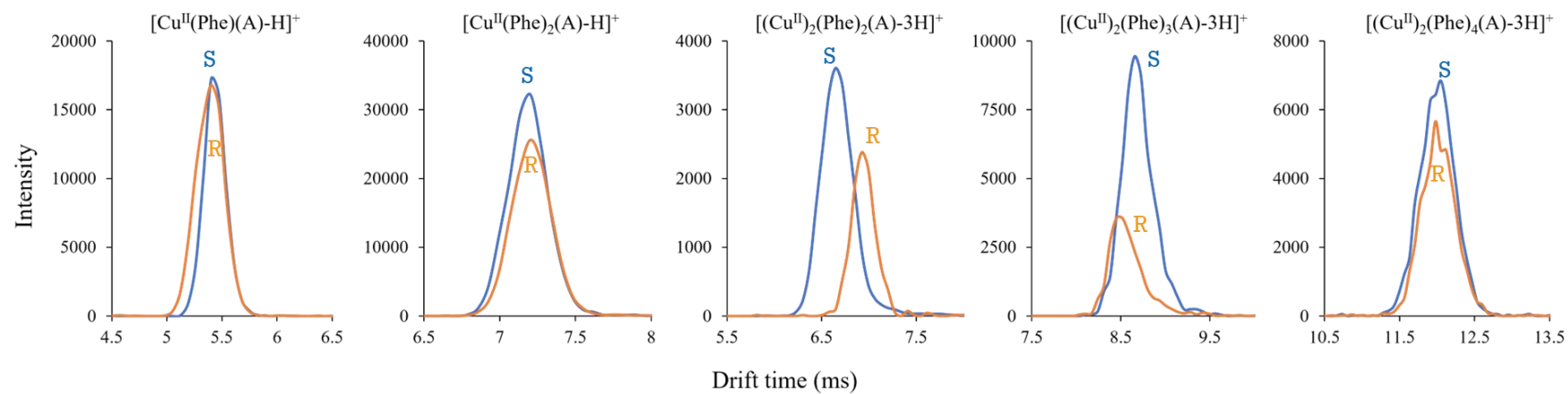


Figure 4-3. IM-MS spectra for $[\text{Cu}^{\text{II}} + (\text{Phe}) + (\text{A}) - \text{H}]^+$, $[\text{Cu}^{\text{II}} + (\text{Phe})_2 + (\text{A}) - \text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{A}) - 3\text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_3 + (\text{A}) - 3\text{H}]^+$ and $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_4 + (\text{A}) - 3\text{H}]^+$ of the propranolol enantiomers.

4.3.3 Chiral recognition of propranolol by MS/MS

The detected complex ions were also subjected to MS/MS with the optimized collision energy (CE) at 15 eV. Loss of Prp or chiral selectors was typically observed in the MS/MS spectra. For example, as shown in Figure 4-4, ion $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$ (m/z 712.1753) was dissociated to m/z 619.0896 and 488.1733, corresponding to loss of one Prp and $\text{Cu}(\text{Phe})$, formation of $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 - 3\text{H}]^+$ and $[\text{Cu}^{\text{II}} + (\text{Phe}) - \text{H}]^+$. However, the loss of one Prp is weakly observed in the spectrum and hard to be compared with that loss of chiral selectors.

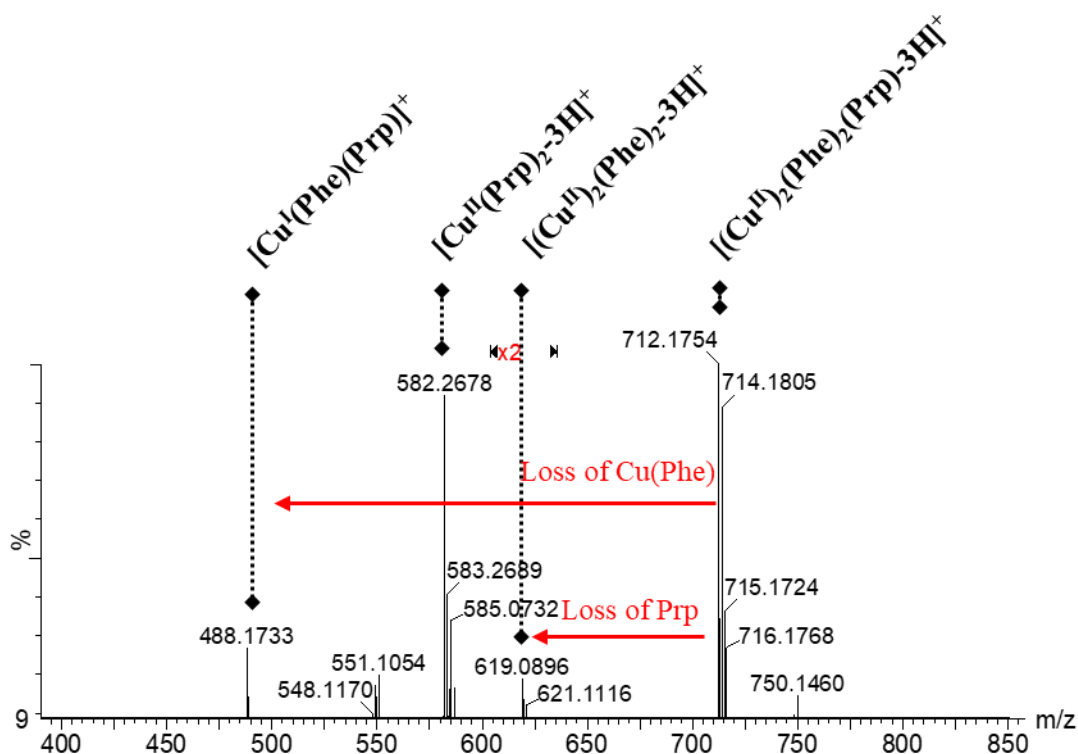


Figure 4-4. MS/MS spectrum of complex ion $[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$ obtained with a CE of 15 eV.

The chiral recognition (CR) values for $[(\text{Cu})_m(\text{Prp})(\text{M})_n - \text{H}]^+$ could be determined with Equation 1, where m and n are the number of the copper ion and chiral selectors (M), respectively.

$$\text{CR} = \frac{[(\text{Cu})_m(\text{Prp}_S)(\text{M})_{n-1} - \text{H}]^+ / [(\text{Cu})_m(\text{Prp}_S)(\text{M})_n - \text{H}]^+}{[(\text{Cu})_m(\text{Prp}_R)(\text{M})_{n-1} - \text{H}]^+ / [(\text{Cu})_m(\text{Prp}_R)(\text{M})_n - \text{H}]^+} \quad (1)$$

As shown in Table 4-5, significant chiral discrimination ($\text{CR} > 1.1$ or $\text{CR} < 0.9$) could be observed with His, Phe and Pro as the chiral selectors. For the same chiral selector, larger complex ions tended to show better chiral discrimination by MS/MS. For example, the CR values of 0.99, 1.05, 1.21, 2.13, 3.34 and 4.40 were obtained for $[\text{Cu}^{\text{II}} + (\text{Pro}) + (\text{Prp}) - \text{H}]^+$, $[\text{Cu}^{\text{II}} + (\text{Pro})_2 + (\text{Prp}) - \text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_3 + (\text{Prp}) - 2\text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_4 + (\text{Prp}) - 2\text{H}]^+$, $[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_5 + (\text{Prp}) - 3\text{H}]^+$ and $[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_6 + (\text{Prp}) - 5\text{H}]^+$, respectively, indicating a clear trend that the chiral discrimination could be enhanced with the assembly. This was probably due to the larger chiral differences caused by the higher degrees of assembly and was consistent with the in the preceding chapters. Compared with Phe, *N*-Boc-Phe with a bulky Boc group showed improvement for the chiral discrimination of propranolol for the same degrees of the assemblies. However, such a molecule with a bulky group could only form lower degrees of assemblies and lead to an overall lower discrimination towards the analyte, as compared with Phe, suggesting that a compromise between formation of assemblies

and distinguishable function groups might sometimes be considered when choosing chiral selectors for assembly-enhanced differentiation of isomers. Another interesting observation is that the assemblies formed always contained only one Prp, which is less loosed after dissociation. No clear evidence could be observed that which pathway, losing selectors or losing Prp, is better for recognition. Further study and the structural interpretation of the complexes for the enantiomer differentiation are still ongoing.

Table 4-5. CR values obtained by MS/MS study of the propranolol complexes ions with various chiral selectors.

Complex ion	Product ion	CR _{S/R}
$[\text{Cu}^{\text{II}} + (\text{His}) + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{His}) - \text{H}]^+$	1.032±0.089
$[(\text{Cu}^{\text{II}})_2 + (\text{His})_2 + (\text{Prp}) - 3\text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{His}) + (\text{Prp}) - \text{H}]^+$	1.113±0.104
$[(\text{Cu}^{\text{II}})_3(\text{His})_5(\text{Prp}) - 13\text{H}]^+$	$[(\text{Cu}^{\text{II}})_3 + (\text{His})_5 - 13\text{H}]^+$	2.314±0.093
$[\text{Cu}^{\text{I}} + (\text{Phe}) + (\text{Prp})]^+$	$[\text{Cu}^{\text{I}} + (\text{Phe})]^+$	1.001±0.113
$[\text{Cu}^{\text{II}} + (\text{Prp})_2 - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Prp}) - \text{H}]^+$	1.054±0.062
$[\text{Cu}^{\text{II}} + (\text{Phe})_2 + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Phe}) + (\text{Prp}) - \text{H}]^+$	1.021±0.021
$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_2 + (\text{Prp}) - 3\text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Phe}) + (\text{Prp}) - \text{H}]^+$	3.401±0.132
$[\text{Cu}^{\text{II}} + (\text{Phe})_3 + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Phe})_2 + (\text{Prp}) - \text{H}]^+$	2.106±0.046
$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_3 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_3 - 3\text{H}]^+$	2.893±0.131
$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_4 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{Phe})_4 - 3\text{H}]^+$	3.132±0.152
$[(\text{Cu}^{\text{II}})_3 + (\text{Phe})_5 + (\text{Prp}) - 5\text{H}]^+$	$[(\text{Cu}^{\text{II}})_3 + (\text{Phe})_5 - 3\text{H}]^+$	2.998±0.164
$[\text{Cu}^{\text{II}} + (\text{Pro}) + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Pro}) - \text{H}]^+$	0.993±0.032
$[\text{Cu}^{\text{II}} + (\text{Pro})_2 + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Prp}) + (\text{Pro}) - \text{H}]^+$	1.051±0.024
$[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_3 + (\text{Prp}) - 3\text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Prp}) + (\text{Pro})_2 - \text{H}]^+$	1.211±0.41
$[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_4 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_4 - 3\text{H}]^+$	2.130±0.076
$[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_5 + (\text{Prp}) - 4\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{Pro})_4 + (\text{Prp}) - 3\text{H}]^+$	3.341±0.092
$[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_6 + (\text{Prp}) - 5\text{H}]^+$	$[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_5 + (\text{Prp}) - 4\text{H}]^+$	4.401±0.159
$[\text{Cu}^{\text{II}} + (\text{N-Boc-Phe}) + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{N-Boc-Phe}) - \text{H}]^+$	1.035±0.042
$[\text{Cu}^{\text{II}} + (\text{N-Boc-Phe})_2 + (\text{Prp}) - \text{H}_2\text{O}]^+$	$[\text{Cu}^{\text{II}} + (\text{Prp}) + (\text{N-Boc-Phe}) - \text{H}]^+$	1.113±0.035
$[\text{Cu}^{\text{II}} + (\text{Asn}) + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Asn}) - \text{H}]^+$	0.967±0.098
$[\text{Cu}^{\text{II}} + (\text{Asn})_2 + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{Prp}) + (\text{Asn}) - \text{H}]^+$	0.998±0.107
$[(\text{Cu}^{\text{II}})_2 + (\text{Asn})_4 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{Asn})_4 - 3\text{H}]^+$	1.134±0.043
$[\text{Cu}^{\text{II}} + (\text{nap}) + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{nap}) - \text{H}]^+$	0.999±0.084
$[\text{Cu}^{\text{II}} + (\text{nap})_2 + (\text{Prp}) - \text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{nap})_2 - \text{H}]^+$	1.003±0.015
$[(\text{Cu}^{\text{II}})_2 + (\text{nap})_2 + (\text{Prp}) - 3\text{H}]^+$	$[\text{Cu}^{\text{II}} + (\text{nap})_2 - \text{H}]^+$	1.058±0.121
$[(\text{Cu}^{\text{II}})_2 + (\text{nap})_3 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{nap})_2 + (\text{Prp}) - 3\text{H}]^+$	1.109±0.052
$[(\text{Cu}^{\text{II}})_3 + (\text{nap})_3 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_2 + (\text{nap})_3 + (\text{Prp}) - 3\text{H}]^+$	1.032±0.069
$[(\text{Cu}^{\text{II}})_3 + (\text{nap})_4 + (\text{Prp}) - 3\text{H}]^+$	$[(\text{Cu}^{\text{II}})_3 + (\text{nap})_3 + (\text{Prp}) - 3\text{H}]^+$	1.045±0.108

4.4 Conclusions

In this chapter, we have investigated the chiral differentiation of propranolol by MS/MS and IM-MS with several potential chiral selectors, including cyclodextrins ($\alpha/\beta/\gamma$ -CD), π -complex selector (R-N-(3,5-dinitrobenzoyl)- α -phenylglycine), crown ether (18-crown-6), cinchona alkaloids (quinine/quinidine), and amino acids and analogues. Among all the tested chiral selectors, amino acids and analogues could allow better detection of the complex ions with propranolol, with complex ions such as $[(\text{Cu}^{\text{II}})_3 + (\text{Phe})_5 + (\text{Prp}) - 5\text{H}]^+$, $[(\text{Cu}^{\text{II}})_2 + (\text{Asn})_4 + (\text{Prp}) - 3\text{H}]^+$, $[(\text{Cu}^{\text{II}})_3 + (\text{Nap})_4 + (\text{Prp}) - 3\text{H}]^+$ and $[(\text{Cu}^{\text{II}})_3 + (\text{Pro})_6 + (\text{Prp}) - 5\text{H}]^+$ detected. For the IM-MS approach, binuclear copper bound trimeric ions with Phe showed larger drift time and higher CCS differences between the enantiomers than other detected complex ions, indicating the chiral selector with aromatic rings allowing better differentiation by IM-MS.

For the MS/MS approach, the amino acid chiral selectors His, Phe and Pro enabled significant chiral discrimination of propranolol, and generally larger copper bound complex selectors showed stronger enantioselectivity. This study demonstrated that chiral recognition of propranolol could be achieved by MS/MS and IM-MS of their copper bound complex ions with chiral selectors such as Phe, and further prove that assembly could provide more abundant chiral environment thus more possibilities on

chiral discrimination. Further experiments and modeling are needed to validate the observations, understand the discrimination mechanism, and extend the approaches for other applications.

Chapter 5: Overall conclusions and prospects

An assembly-based approach has been developed for differentiation of isomers by IM-MS in this study. Several important and representative isomers, including amino acids and relevant compounds such as amino alcohols, keto acids, fatty acids and peptides, saccharides, and chiral drugs which are constitutional, configurational, and chiral isomers, respectively, were investigated for their differentiations by IM-MS. With the presence of metal ions and optimized ESI conditions, various multimeric assemblies of isomers were observed in ESI-MS. IM-MS were employed to measure the CCSs values and determine their differences among different isomers at different assembly degrees. An important observation in our studies was that higher assembly of isomers tended to enable larger differences in drift time and CCS values in IM-MS and thus more significant discrimination.

For isomers Leu, Ile and allo-Ile, we have been able to observe assemblies up to hepta-copper bound dodecamers in MS. A general trend was observed that the difference of the isomer assemblies in drift time and CCSs in IM-MS became larger with the increased number of the assembled amino acids. The slopes of ΔCCS against the degree of assembly were found to be increased in the order: Leu vs. Ile > Ile vs. allo-Ile > L-form vs. D-form, which is the difference order of the individual isomers, suggesting

that the isomeric differences might be accumulated during the assembly to result in enhanced discrimination in IM-MS. Further investigation by IM-MS on di-, tri-, tetrapeptides and analogue isomers (e.g., amino alcohols, keto acids and fatty acids) with branched alkyl side chains supported the enhanced differentiation of isomers in the assemblies. The combination of IM-MS, MS/MS and computational modelling have been employed to obtain the structure information on the assemblies of Leu, Ile, allo-Ile and understand their gradually increased differentiation. The overall structures were revealed as a near-sphere shape, with a coordination core and the alkyl chains of amino acids protruded outwards, and the slight differences between the alkyl chains could be accumulated after assembly to eventually cause significant differences. This work provides a unique insight on improving isomer differentiation and may help explain the alkyl chains influence on membrane thickness of cell and material synthesis.

To further explore the assembly-based approach and to test whether assembly could also enhance the differentiation of saccharide isomers in IM-MS, monosaccharide and disaccharide isomers with different compositions, connectivity and configurations have been studied. Metal bound assemblies up to dimers of monosaccharides and octamers of disaccharides were detected using ESI-MS with addition of nickel or zinc ions at the optimal ratio. The IM separations of the assemblies were found to be enhanced with the assembly. The increased ΔCCS allowed differentiation between composition isomers Lac and Gal, between connectivity isomers IsoMal (1 $\rightarrow$ 4 linkage) and Mal (1 $\rightarrow$ 6 linkage), and between configuration isomers Mal (α -configuration) and Cel (β -

configuration). It was suggested that the structures of the metal-disaccharide assemblies might be metal-centered with the disaccharides surrounded and the difference of the configuration, linkage and monomer unit might cause the diameter or surface concaveness differences and thus their enhanced differentiations.

To extend the assembly-based approach to differentiation of enantiomers, propranolol, a chiral drug with its S-form more active than the R-form, has been selected for investigation. Several types of chiral selectors were evaluated for chiral discrimination by MS/MS and IM-MS, and amino acids or amino acids analogues with metal ions showed stronger binding affinities and discrimination potentials than the others. The complexes achieving better chiral discrimination of propranolol has been explored, and the results revealed that the larger copper bound complexes allowed better chiral recognition for propranolol in the MS/MS study. The results also revealed that the chiral selectors with aromatic rings tended to form complex ions with more drift time differences and larger CCS differences between the enantiomers, thus allowing better enantioselectivity in the IM-MS study.

Overall, the assembly-enhanced differentiation of isomers by IM-MS has been developed by testing several representative types of isomeric molecules. Amino acid isomers with different alkyl chains and saccharide isomers with different composition, connectivity and configuration could be better differentiated by IM-MS with the

assembly, revealing that the isomeric differences could be accumulated in the assembly to lead to significant differences in higher assemblies of the isomers. S- and R-Prp could be better recognized by MS/MS and separated by IM-MS after assembly with chiral selectors, indicating the effects of assembly and chiral selectors on the chiral MS analysis. Therefore, in general, higher assembly could enhance the differentiation of isomers by MS. However, more structural investigation is needed in later study to interpret how the various types of isomeric difference could be accumulated. The knowledge acquired is also needed to be summarized further and hopefully to propose a general strategy for differentiation of isomers even in complicated samples, and provide insights for applications of molecular assemblies in the gas phase or design of stereoselective materials such as chromatographic columns for isomer separation.

Appendix

Assembly in solution

In past decades, several studies about metal-Ile or Leu complexes in solid state or solution phase were reported. The complexes detected by X-ray, NMR and computational method were most commonly chelated dimers: five-membered chelate ring with one water molecule in axial positions $\text{Metal(Ile or Leu)}_2(\text{H}_2\text{O})$.¹¹¹⁻¹²⁰ Unlike other AAs such as asparagine or glutamic acid which were reported to easily coordinate with various transition metal ions and produce metal–organic frameworks (MOFs) in solution,¹²¹ Leu, Ile and allo-Ile were less prone to higher polymerize in solution.

To detect what complexes were already formed in our prepared solution, two approaches, Circular dichroism (CD) and Nuclear magnetic resonance (NMR), will be used to monitor the subunits formation in solution. Different concentration ratio of copper salt and AA mixture solution were tested. The UV–Vis and CD spectrum of Ile and copper- Ile water solutions are shown as an example in Figure A-1. Both spectra of Ile solution agree with the one previously reported.¹⁴⁶ In UV-Vis, the bands in the range 190-200 nm can be assigned as is the $\pi \rightarrow \pi^*$ transition of C=O bond in amino acid.¹⁴⁶ With the increased addition of copper ion, this band is shifted to lower energy around 210 nm in Figure 2-36B.¹⁴⁷⁻¹⁴⁸ The range of 230-240nm shown in Figure A-1

is contributed by C=O bond in copper acetate.⁷⁴ Whereas, the d-d transition specific for square planar bis(amino-acidato) copper (II) complexes appears around 550-650 nm haven't been found.^{147, 149-151} That might because the reaction condition is mild, only trace chelated Cu(AA)₂ could form in a such short time and it is hard to be detected by CD.

The CD positive signal in around wavelength 200-210nm is also from the C=O bond in amino acid affected by L-amine group.^{146, 152} In the range of 250-260nm, a ligand field negative cotton effect exhibit under the shifted n→π* band, suggests that there would be opposite optically active transitions which might be contributed from dissymmetric electron distributions around the coordinated.^{151, 153} A decrease intensity of band around 200nm and an increase intensity of band at 250nm-260nm are observed with the increasing concentration of added copper ions in two salts format. That indicate the coordination was happened and easily affected by the metal to ligand ratio in the mixture solution.¹⁵² The intensity difference among two kinds of copper salts could be explained by the pH influence on copper-amino acid formation. The higher pH value (Cu(COOCH₃)₂>CuCl₂) is, the easier the copper-AA complexes will form and their coordination geometry will change accordingly.¹⁵⁴

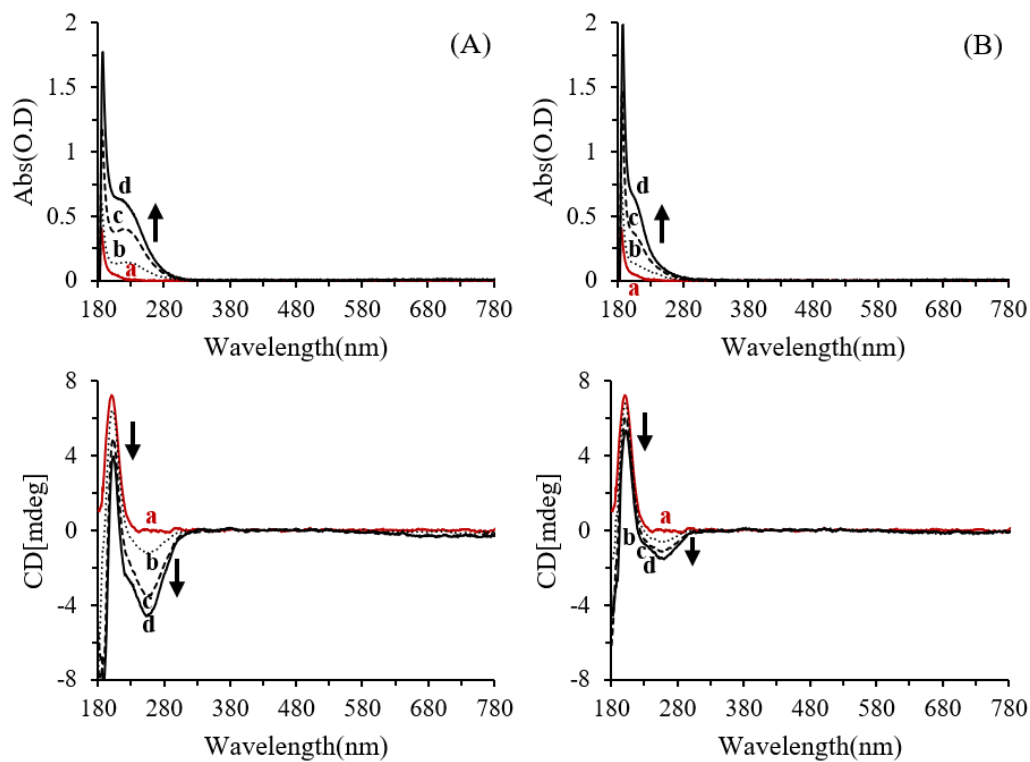


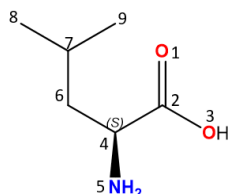
Figure A-1. UV Absorption (top) and CD spectra (bottom) for Ile with addition of $\text{Cu}(\text{COOCH}_3)_2$ (A), CuCl_2 (B) at concentration ratios of Ile: $\text{Cu}^{2+}=4:0$ (a), Ile: $\text{Cu}^{2+}=4:1$ (b), Ile: $\text{Cu}^{2+}=4:4$ (c), Ile: $\text{Cu}^{2+}=4:8$ (d).

The experimental NMR spectra of L-Ile, L-Leu, L-allo-Ile given below have an agreement with the reported data.¹⁵⁵⁻¹⁵⁶ As shown in Figure A-2, different transition metal salts were given in the L-Ile D₂O solution, because the paramagnetic of Cu(II) and Ni(II) nucleus will cause intramolecular electronic effect on other atoms, the broaden peak shown in Figure A-2-C and D could help to reveal whether the amino acids are involved in complex formation.¹⁵⁴ In L-Ile-Cu(II) solution, the chemical shifts difference was observed which could give more assignable signals to monitor the coordination process. In this spectra, ¹H chemical shifts of 4-CH group shifted from $\delta = 3.52$ ppm to $\delta = 3.72$ ppm, ¹H chemical shifts of 6-CH group shifted from $\delta = 1.83$ ppm to $\delta = 1.97$ ppm indicate that the coordination happened. These results were further investigated by replacing the CuCl₂ to Cu(COOCH₃)₂. As talked above, due to the higher PH, Cu(COOCH₃)₂ is easier to coordinate with amino acid.¹⁵⁴ Also Cu(COOCH₃)₂ in ¹H NMR could supply an extra signal come from the H on acetoxy group whose chemical shift is easily affect by whether C=O is binding with Cu(II). This is shown in Figure A-3, the chemical shift of H on CH₃ of copper acetate is $\delta = 5.83$ ppm, the H on CH₃ of acetic acid is $\delta = 1.48$ ppm. It is possible to assign the peaks from free and bound acetoxy group. The NMR spectra of L-Leu, L-Ile and L-allo-Ile containing a various concentration of Cu(COOCH₃)₂ are given in Figure A-4. Addition of Cu(COOCH₃)₂ to AA led to the appearance of new peaks in the ¹H-NMR spectrum between 5.83-1.48 ppm. As discussed above, the coordination has been happened, while

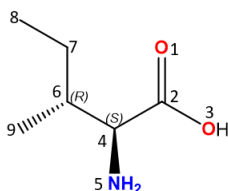
there are no separate peaks from free or bound acetoxy group observed indicating fast exchange on the NMR time scale. When the ratio of AA and Cu(II) is in a 1:1, the peaks appear at 2.65, 3.13 and 3.65 ppm, with the increasing ratio of AA, above peaks shift to 2.13, 2.54 and 2.56 ppm then 2.05, 2.07 and 2.07 ppm. That means the growth on the value of free acetoxy group present in solution gradually surmount the bound one. Until the molar ratio of the addition of AA is 5:1. The spin-spin splitting pattern could be clearly observed especially at high field which signal comes from AA. This is an assumption that there are redundant free ligand existing in the solution. As some research works mentioned the complexes species changes dynamically along with varying ratio of metal-ligand in solution.¹⁵⁷ Here, we propose the Cu(AA), Cu(AA)₂ or Cu₂(AA)₂ are complexes formed in solution.

In our experiments, Leu, Ile and allo-Ile form larger assemblies under ESI conditions, indicating that assembly occurred when the mixture was transposed into a non-solvent environment. That is likely because the gently desolvation without breaking coordinate bonds under the ESI condition enabled more exposure of binding sites which had been protected by solvent molecule spheres,¹²³ or because the ionization induced more interaction forces between molecules, and in turn initiated higher assembly level.

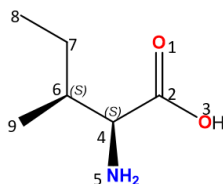
¹H NMR spectroscopic data for L-Leu, L-Ile and L-allo-Ile



L-Leu. ¹H NMR (495 MHz, D₂O) δ 3.65 – 3.57 (m, 1H), 1.58 (ddt, *J* = 23.2, 16.8, 7.6 Hz, 3H), 0.83 (t, *J* = 5.6 Hz, 6H).



L-Ile. ¹H NMR (495 MHz, D₂O) δ 3.53 (d, *J* = 4.0 Hz, 1H), 1.85 (dq, *J* = 14.0, 7.0, 3.6 Hz, 1H), 1.39 – 1.29 (m, 1H), 1.18 – 1.08 (m, 1H), 0.88 (d, *J* = 7.0 Hz, 3H), 0.80 (t, *J* = 7.4 Hz, 3H).



L-allo-Ile. ¹H NMR (495 MHz, D₂O) δ 3.61 (d, *J* = 3.6 Hz, 1H), 1.97 – 1.89 (m, 1H), 1.29 (dq, *J* = 14.2, 7.1 Hz, 1H), 1.24 – 1.15 (m, 1H), 0.84 – 0.77 (m, 6H).

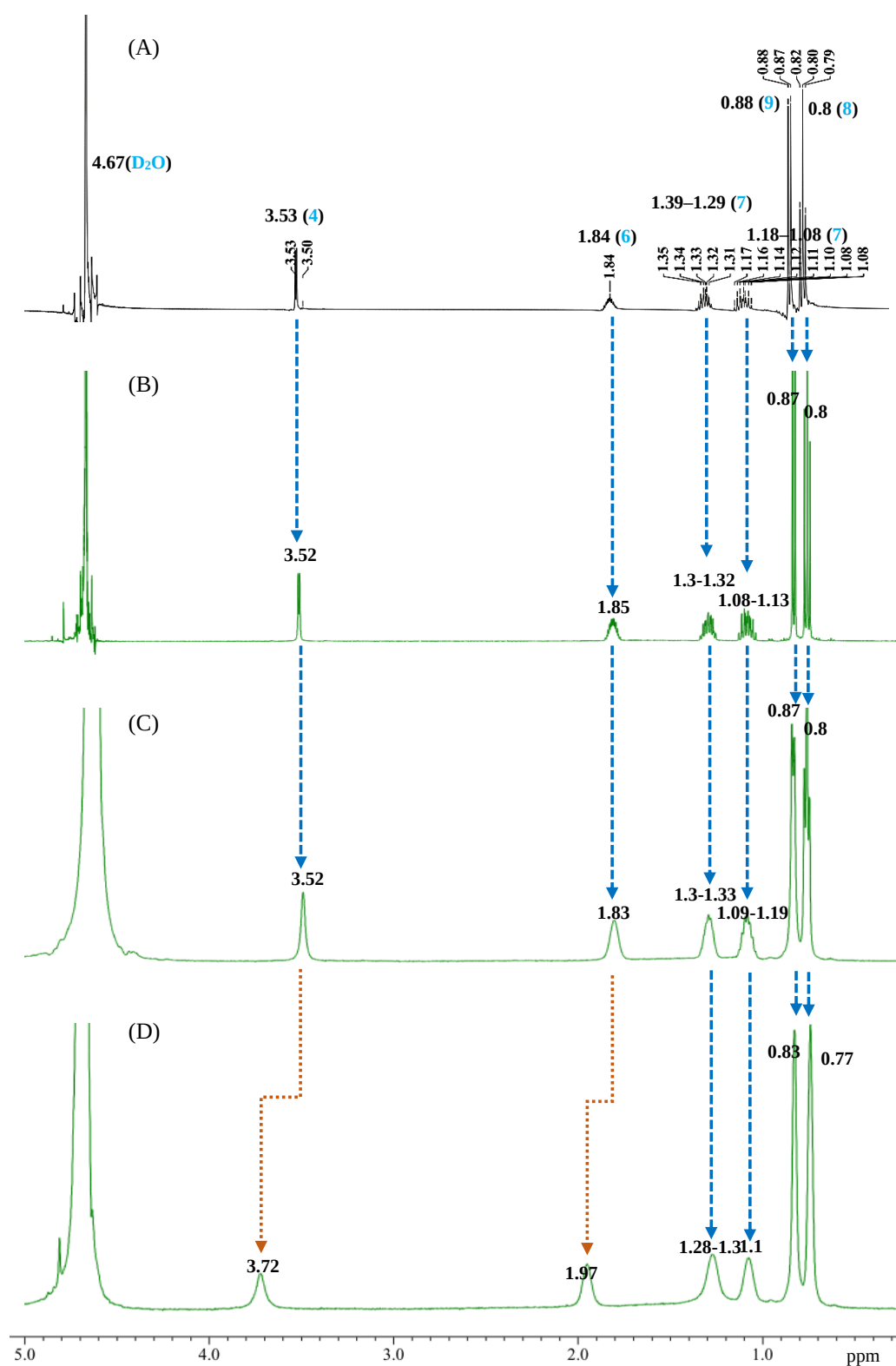


Figure A-2. ^1H NMR spectrum of L-Ile (A) with addition of ZnCl₂ (B), NiCl₂ (C), CuCl₂ (D) at a 1:1 molar ratio in D_2O .

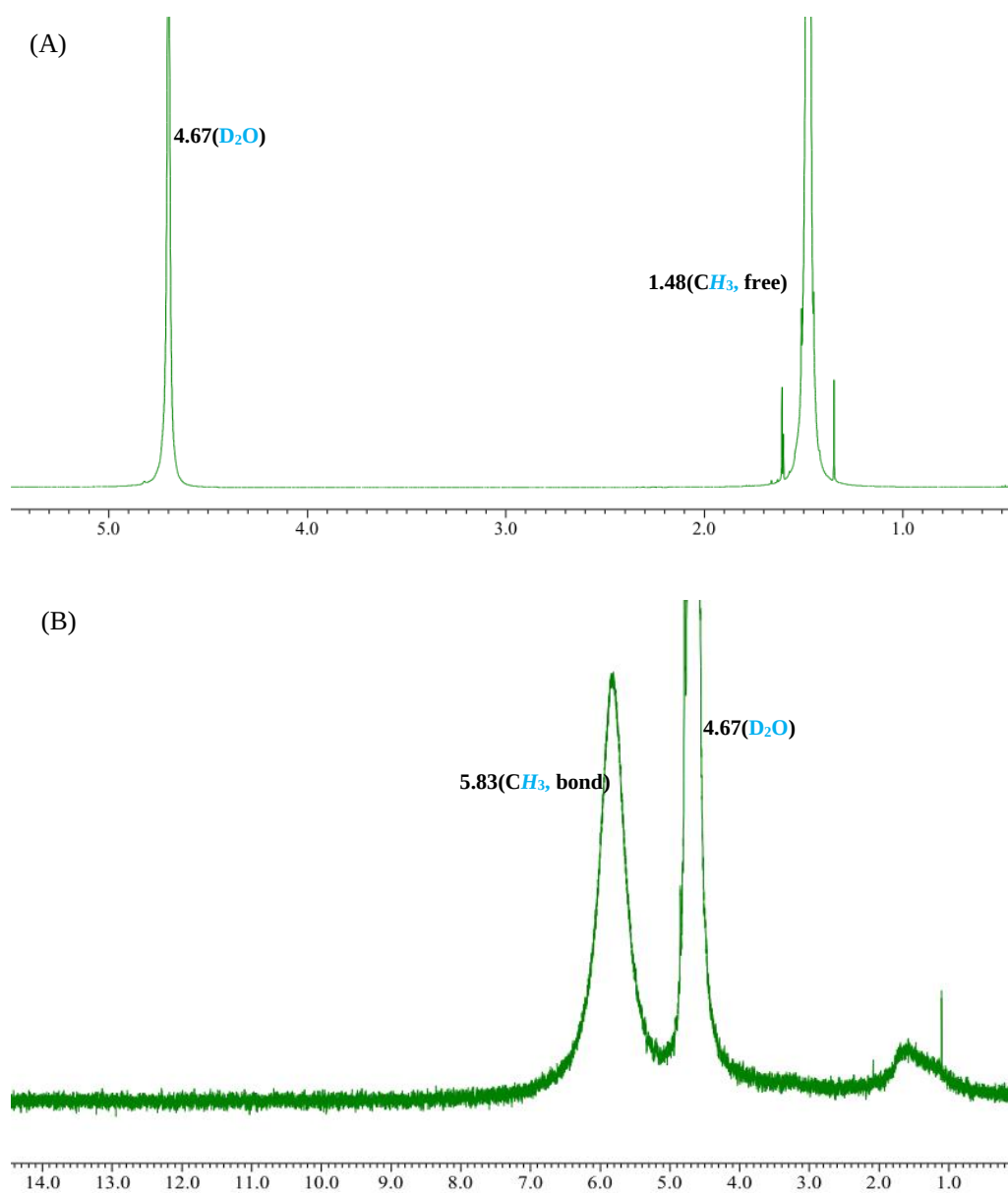


Figure A-3. ¹H NMR spectrum of CH₃COOH (A) and Cu(COOCH₃)₂ (B) in D₂O.

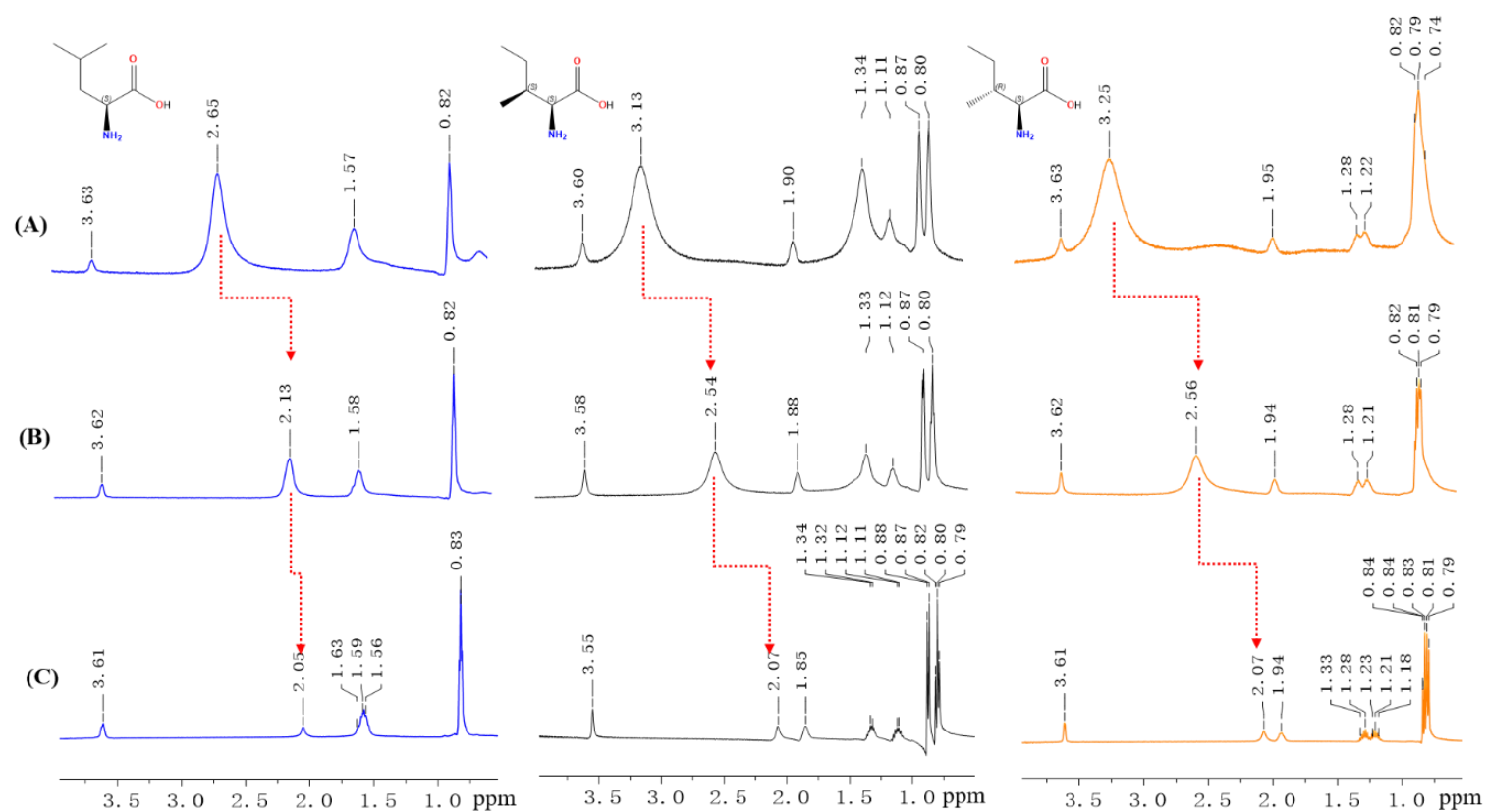


Figure A-4. ^1H NMR spectrum of L-Leu (blue), L-Ile (black) and L-allo-Ile (orange) addition $\text{Cu}(\text{COOCH}_3)_2$ at molar ratio 1:1 (A), 2:1 (B), 5:1 (C) in D_2O .

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